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ACIES – LLM powered SOC automation

A Thesis Submitted in
(Partial) Fulfillment of the
Requirements for the Degree of

Bachelor of Science in Information & Communication Technology

at

Bahrain Polytechnic January 2025

Title Page



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An abstract

My goal was to revolutionize SOC operations by developing an intelligent assistant powered by Large Language Models that addresses the fundamental challenges analysts face daily. Traditional SOC environments have many shortcomings when dealing with a large number of alerts and false positives, and the learning curve is extremely steep. Security analysts frequently experience alert fatigue, as they are

overwhelmed by the high volume of alerts generated by traditional SOC environments. The aim of my project was to create an automation solution that not only filters and prioritizes alerts intelligently but also acts as a knowledgeable assistant, providing context-aware recommendations and guidance to take measures according to incident response lifecycle.

The project presents an intelligent llm powered SOC designed to transform security operations through automated alert analysis and intelligent recommendations. The system leverages Large Language Models to provide context-aware assistance that helps analysts prioritize threats, reduce false positives, and make informed decisions during incident investigation. By automating routine alert triage and enrichment tasks, the assistant enables analysts to focus on genuine security threats while providing guided support for threat response activities.

The solution focuses on two main areas providing AI-powered assistance to our analysts during their daily workflows, delivering actionable recommendations for detected alerts to help prioritize genuine threats, and automating response actions to reduce manual workload. This makes it easier for new analysts to handle security incidents effectively while helping experienced analysts work more efficiently, leading to faster threat detection and improved incident response.

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List of Symbols

Symbol	Meaning
	Alert/Warning
	Critical/High Priority Alert
	Medium Priority Alert
	Low Priority/Resolved
	Real-time Detection
	Targeted Attack/Threat
	Investigation Required
	Verified/Confirmed Threat
	False Positive

List of Abbreviations:

Abbreviation	Full Form
LLM	Large Language Model
SOC	Security Operations Center
SIEM	Security Information and Event Management
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
MTTR	Mean Time To Respond
MTTD	Mean Time To Detect
SOAR	Security Orchestration, Automation and Response
IOC	Indicator of Compromise
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
EDR	Endpoint Detection and Response
API	Application Programming Interface
RAG	Retrieval Augmented Generation
TTP	Tactics, Techniques, and Procedures
VM	Virtual Machine
UML	Unified Modeling Language
XDR	Extended Detection and Response
MCP	Model Context Protocol

Introduction

Project Rational

Gulf security solution a Cybersecurity solution company operating in Bahrain and GCC countries, and mainly it provides typical solution for SOC but organizations relying on traditional Security Operations Centers (SOC) face critical operational challenges that significantly impact their security posture and incident response capabilities. Security analysts are overwhelmed with high volumes of security alerts generated by SIEM platforms like Splunk, where logs are presented in raw, technical formats that are difficult to interpret quickly. This creates severe alert fatigue, as analysts must manually go through hundreds of alerts daily, many of which are false positive or low-priority events.

Information security is a critical sector in today's digital landscape, where organizations face an ever-evolving threat whereas traditional SOC workflow is relatively slow and labor-intensive. Analysts spend valuable time analyzing complex log data, correlating events across multiple sources, and attempting to understand the context and severity of each alert. This manual process delays threat detection and response times, leaving organizations vulnerable to security breaches. Furthermore, the steep learning curve required to effectively read and analyze SIEM logs means that junior analysts struggle to contribute meaningfully, while senior analysts are bombarded with routine triage tasks.

Project Objectives

The objective of this project is to create an LLM (Large Language Model) powered SOC which will assist our analysts to engage with SIEM (Security information and Event Management) and to enhance threat detection, accelerate incident response and reduce analyst workload through Integration of SIEM tools, artificial intelligence, and automated workflow orchestration using natural-language interaction.

Technical objectives:

Integrate Splunk SIEM with Claude AI via MCP Server: Establish a robust connection between Splunk SIEM and Claude AI using the Model Context Protocol (MCP) server to enable seamless data exchange, automated log querying, and direct access to security event data for AI-powered analysis.

Deploy AI-Powered Human Assistance Interface: Implement Claude AI as an intelligent assistant that provides security analysts with interactive support and assistance for log interpretation, threat analysis, incident investigation guidance, and context-aware security recommendations in natural language.

Configure Multi-Platform Threat Detection: Deploy Splunk SIEM to monitor and detect various types of security attacks (malware, intrusion attempts, ransomware, unauthorized access, data exfiltration) across diverse operating system endpoints including Windows, Linux, and macOS environments.

Implement Automated Alert Processing Pipeline: Deploy an N8N workflow automation that receives security alerts from Splunk via webhooks, passes complex log data, and transforms raw technical alerts into human-readable formats.

Enable Real-Time Alert Distribution: Configure Slack integration to deliver processed, humanized security alerts directly to SOC analysts, ensuring immediate visibility and rapid response capabilities.

Create Scalable Automation Architecture: Design a modular, maintainable system architecture that can scale with growing alert volumes and adapt to evolving security requirements.

Create Scalable Automation Architecture: Design a modular, maintainable system architecture that can scale with growing alert volumes and adapt to evolving security requirements.

Business Objective:

Reduce Mean Time to Detect (MTTD) and Respond (MTTR): To Significantly decrease the time required to identify and respond to security threats through automated alert processing and AI-assisted analysis.

Improve Analyst Productivity and Efficiency: Free security analysts from repetitive, time-consuming tasks, allowing them to focus on high-value activities such as threat hunting, strategic security initiatives, and complex investigations.

Lower Operational Costs: Reduce the resource burden on SOC teams by automating routine tasks, minimizing the need for extensive manual log analysis, and improving overall team efficiency by leveraging Artificial Intelligence.

Enhance Security Posture: Improve the organization's ability to detect, analyze, and respond to security threats more effectively, reducing the risk of successful cyberattacks and potential business impact.

Accelerate Analyst Onboarding: Lower the barrier to entry for junior analysts by providing AI-assisted guidance and human-readable alert formats, reducing training time and enabling faster contribution to SOC operations.

Increase Alert Accuracy and Reduce Fatigue: Minimize false positives and alert overload through intelligent parsing and prioritization, improving analyst morale and decision-making quality.

Prior Work

As our project is or what we are creating a AI-assisted security operation center (SOC) which will use artificial Intelligence to detect ,investigate, analyze and help our Analyst by alerting about threat actors, So the most relevant work to our project is ORRYX AI by Beyon Cyber, ORRYX AI is an **AI-driven, autonomous SOC platform** designed to

automate and improve how cybersecurity operations handle threats. Built to replace many manual SOC tasks with intelligent AI automation, reducing alert fatigue and boosting operational speed. Positioned for **global deployment** through strategic partnerships — showing it's meant to be part of next-gen SOC automation ecosystems.

Hypothesis

This paper hypothesizes that the integration of Large Language model (LLMs) into Security operation center (SOC) workflows through Model Context Protocol (MCP) and multi layered automation will significantly enhance threat detection accuracy, reduce mean time to respond (MTTR) to security incidents, and alleviate analyst workload compared to traditional manual SOC operations. We are talking about Connecting our SIEM Splunk to extremely advanced AI model like Claude, these models are commonly called *generative AI* because they can generate text, images, and audio. With protocols like the Model Context Protocol (MCP), they can go even further by securely interacting with external applications, data sources, and tools enabling deeper integration with systems such as Splunk. MCP provides a standard way for Claude to access external tools, live data, and workflow actions rather than being limited to static training data, enabling richer integration with systems such as Splunk and other internal services. Then we use n8n to automate the workflow so that security alerts are received on a webhook then sent to an AI (like OpenAI) for initial sorting and labeling. After the AI classifies and prioritizes the alerts, the system sends the useful alerts directly to Slack channels. This shows how AI-powered security automation can handle and organize alerts faster and more consistently than if people did it all by hand.

Proposed Solution

Traditional Security Operations Centers often struggle with large number of alerts, most of which are false or low priority, forcing analysts to spend too much time on repetitive manual triage and investigation. This leads to alert fatigue, slower detection and delayed incident response increased mean time to detect (MTTD) and mean time to respond (MTTR), and inconsistent handling of threats, making it harder to keep up with sophisticated attacks. Additionally, relying heavily on human effort makes it difficult to

scale during high-alert periods and requires significant investment in staffing and training for 24/7 coverage.

To address these critical limitations, this project proposes an LLM-Powered SOC Automation. This solution uses advanced AI to change security operations from manual and reactive to automated and proactive. It uses two AI models working together, each handling different tasks, to create a smarter and more efficient security workflow.

This solution is proposed in 3-layer workflow, First, the Intelligence Layer uses Claude AI connected through the Model Context Protocol (MCP). This allows the AI to securely access context, tools, and data sources to perform deeper security reasoning and generate intelligent recommendations. Second, the Alert Processing Layer uses n8n workflow automation. Security alerts from sources such as Splunk SIEM, are sent to n8n via webhooks, where they are automatically routed to an AI model in a human readable form.

Third, the Communication Layer delivers the processed and prioritized alerts to Slack channels, including clear AI-generated insights and recommended next steps, allowing security teams to respond quickly and effectively.

Tasks
• Collect & Aggregate Security Logs from Endpoints via Splunk SIEM
• Automate Alert Processing through N8N Workflow with OpenAI Triage
• Integrate Claude AI Assistant with Splunk via MCP Server
• Generate Intelligent Security Recommendations & Threat Analysis
• Deliver Real-time Enriched Alerts & Insights to Slack Channels

Table 1 Tasks

Description of the Report

This report documents the complete development of the LLM-Powered SOC Automation project. Following this introduction, the **Background** section presents the related theory, project technologies, and related work that form the foundation of this project.

The **Design** section then details the system's technical architecture through UML diagrams and system architecture documentation it will explain how the different components of our project communicate with each other it's an essential phase since our design isn't easy to understand.

Next, the **Implementation** section describes how the system was built, deployed and configured demonstrating its development, maybe with the help of visuals too.

The **Testing** section represents one of the most critical components of this report, as it concludes all testing procedures and methodologies to validate the system's efficiency, effectiveness, and reliability in real-world SOC environments.

The **Discussion and Conclusion** section provides a comprehensive review of all accomplished objectives and the challenges encountered during development, along with recommendations for future enhancements and upgrades to the system. This section also incorporates the **LESPI** analysis, examining the legal, ethical, social, and professional implications of implementing AI-driven security automation, and concludes with a personal reflection on the knowledge and experience gained throughout this project journey.

The report concludes with **References** and **Appendices** containing system manuals, detailed design documentation, and implementation source code.

Background

Related Theory

A Security Operations Center is a centralized unit that monitors, detects, analyzes, and responds to cybersecurity threats and incidents across an organization's IT infrastructure. Traditional SOC operations follow the incident response lifecycle, which includes identification, containment, eradication, recovery, and lessons learned phases (NIST, 2012). SOC analysts utilize Security Information and Event Management (SIEM) systems to aggregate logs from multiple sources, correlate events, and generate alerts based on predefined rules and signatures; however, this process is extremely slow and labor-intensive, requiring manual investigation and analysis of each alert to determine its legitimacy and appropriate response actions and the large volume of alerts often

thousands per day creates alert fatigue, where analysts become desensitized to warnings, potentially missing critical threats among the noise (Sundaramurthy et al., 2016).

This paper defines its objectives to a single SOC-focused concept and details the key techniques applied to deliver an AI assisted accurate, and automated solution using large language models.

Splunk is a leading Security Information and Event Management platform that provides centralized log management, real-time monitoring, and advanced analytics for security operations. Splunk's architecture utilizes a distributed data collection model using lightweight agents called Universal Forwarders that are deployed on endpoints, servers, network devices, and other data sources throughout an organization's infrastructure (Splunk Inc., 2023). These forwarders continuously monitor log files, system events, and application data, compressing and encrypting the collected information before transmitting it to Splunk Indexers central servers that parse, index, and store the data for rapid search and analysis.

Large Language Models are deep learning neural networks trained on vast amounts of text data to understand, generate, and manipulate human language. LLMs like Anthropic's Claude utilize transformer architectures that enable them to process context, perform reasoning tasks, and generate human-like responses (Vaswani et al., 2017). In cybersecurity contexts, LLMs can analyze unstructured security data, interpret threat intelligence reports, correlate indicators of compromise (IoCs), and provide contextual recommendations tasks that traditionally require significant human expertise (Ferrag et al., 2023). These models excel at pattern recognition, anomaly detection, and translating complex technical information into actionable insights.

The Model Context Protocol (MCP) is an integration framework that enables AI models to interact with external systems, databases, and tools while maintaining contextual awareness. MCP allows LLMs to access real-time data, execute functions, and retrieve relevant information beyond their training data, making them more effective for specialized tasks like security analysis (Anthropic, 2024). In SOC environments, MCP enables AI assistants to query SIEM systems, retrieve historical incident data, access threat intelligence feeds, and provide recommendations based on current organizational context rather than generic responses.

Security Automation involves using technology to perform security tasks with minimal human intervention, while Security Orchestration refers to integrating different security tools and processes into unified workflows (SOAR - Security Orchestration, Automation, and Response). Automation reduces mean time to detect (MTTD) and mean time to respond (MTTR) by eliminating manual steps in alert processing, enrichment, and response actions (Zimmerman, 2014). Workflow automation platforms like N8N enable the creation of complex decision trees and conditional logic that can route alerts based on severity, type, and context, ensuring that the right information reaches the right personnel at the right time. N8N is an open-source workflow automation tool that provides a visual, node-based interface for building integrations between various applications and services without requiring extensive programming knowledge (n8n GmbH, 2023). Its flexibility allows security teams to design sophisticated alert processing pipelines that can automatically trigger API calls, execute data transformations, apply conditional routing logic, and orchestrate multi-step security workflows across disparate platforms such as SIEM systems, AI services, and communication channels.

Project Technology

Our SOC automation lab is built on a virtualized infrastructure using VMware. The SIEM platform (Splunk) and automation components are hosted on Ubuntu Server, while both Linux and Windows operating systems are utilized for log generation and forwarding. Security events are collected and analyzed through Splunk, where Anthropic’s Claude operates as an AI assistant via an MCP server. Alert handling is automated using n8n workflows, enriched with contextual intelligence through OpenAI-powered LLMs, and delivered to security analysts via Slack to enable efficient and timely incident response.

Technology	Purpose
Virtualization & Lab Infrastructure	

	VMware VMware was used to build and manage the virtual lab environment for this project. It enabled the creation of multiple isolated virtual machines representing different SOC components, such as servers and endpoints. This virtualization approach allowed safe testing, flexibility, and efficient resource utilization without requiring physical hardware.
	Ubuntu Server Ubuntu Server was deployed as the primary server operating system within the lab. It hosted core SOC components, including the SIEM platform (Splunk), the automation engine (n8n), and supporting services. Ubuntu Server was selected for its stability, security, and wide support for enterprise security tools.
	Windows 11 Windows 11 was used as an endpoint machine to simulate a real user environment. Security logs and system events generated on this endpoint were forwarded to the SIEM. This allowed the lab to replicate real-world endpoint monitoring and detection scenarios commonly handled by SOC teams.
Security Monitoring & Log Management (SIEM)	
	Splunk Splunk served as the central SIEM solution in the SOC automation lab. It was responsible for collecting, indexing, and analyzing security logs from multiple sources. Splunk enabled real-time visibility into security events, detection of suspicious activities, and generation of alerts that triggered automated workflows.

 Universal Forwarder	Splunk Forwarder The Splunk Universal Forwarder was installed on both Linux and Windows systems to collect system, application, and security logs. These logs were securely forwarded to the Splunk SIEM for centralized analysis. The forwarder ensured reliable and continuous log ingestion across the environment.
 Power of Splunk Search Processing Language (SPL™)	Splunk search processing language Splunk Search Processing Language (SPL) was used to search, filter, correlate, and analyze security logs within Splunk. SPL enabled efficient querying of events collected from different systems, allowing the identification of patterns, detection of security incidents, and extraction of meaningful insights from raw log data across the SOC automation lab.
AI, LLM Integration & Alerting	
	Claude AI Claude AI was utilized as an intelligent assistant to analyze and reason about security events. Through integration with the MCP server, Claude was able to interact with SIEM data and assist in alert analysis, triage, and contextual understanding of incidents
	Open AI API The OpenAI API was used to enhance SOC alerts by transforming raw security logs into human-readable summaries. This allowed complex technical data to be converted into clear explanations, making alerts easier for analysts to understand and act upon.

	Slack Slack was used as the alert delivery and communication platform for SOC analysts. Final enriched alerts and incident summaries were automatically sent to dedicated Slack channels, enabling fast visibility, collaboration, and efficient incident response.
Automation & AI Orchestration	
	N8N n8n was used as the automation and orchestration engine in the SOC lab. It processed alerts generated by Splunk, executed predefined workflows, enriched events using LLMs, and routed alerts automatically. This significantly reduced manual SOC effort and improved response efficiency.
	MCP Server / MCP Protocol The MCP (Model Context Protocol) server was used to connect Claude Desktop with the SIEM environment. It acted as a bridge between AI models and security data sources, enabling secure and structured access to alerts and logs for AI-driven analysis and decision support.

Table 2 - Technologies used for project

Related Work

In this section we will discuss work and literature related to our project since **AI is SOC is emerging as a critical enabler** in this environment. Not because it replaces human analysts, but because it empowers them to work more efficiently, respond more quickly, and maintain control even under mounting pressure.

Why SOC teams need AI-powered support

Security teams are operating in increasingly complex environments. The [volume of telemetry](#) and incident data has rocketed. Adversaries are using automation and AI to increase the speed and scale of their attacks. Compliance requirements are more

demanding. Meanwhile, many organizations [still face staffing shortages and skills gaps](#), especially among junior analysts.

This creates **an overwhelming situation for SOC**s. A single investigation might involve hours of log correlation, manual root cause analysis, and false-positive triage. And because senior analysts are in short supply, teams often rely heavily on a handful of experts (and that model is not sustainable).

[Generative AI](#) offers a way forward. When [embedded thoughtfully into SOC workflows](#), AI can act as an always-available assistant that helps analysts interpret data, identify patterns, and act faster *without* compromising the human judgment at the core of good security operations.

How Agentic AI enables an autonomous SOC with minimal human involvement

Security operations centers (SOCs) have faced ongoing challenges for threat detection and response for years. These challenges include the discernment of genuine security signals from background noise, inadequate context for alert investigation, lack of end-to-end automation, workflow bottlenecks and alert fatigue, just to name a few.

For years I've said that security operations, or [cyberthreat management](#) in any form, needs to undergo a major change like that of commercial airlines in the mid-20th century: machines fly commercial airplanes, and pilots intervene in limited situations. Similarly, the new SOC would run autonomous operations with minimal human involvement.

The SOC analysts would then become *SOC pilots*, choosing where and when they become involved, while the virtual machine handles standard operations.

Agentic AI SOC

The use of Agentic AI in the Security Operations Centre (SOC) is not just about making detection of threat faster or more accurate it should represent a transformational capability that redefines workflows and re-architects' cybersecurity operations for greater operational effectiveness. This evolution shifts the SOC's focus from merely automating repetitive tasks to creating outcomes and delivering measurable value.

At the heart of this transformation are Agentic AI agent-first systems, supervised and guided by highly skilled human analysts. This innovative model positions Agentic AI as a force multiplier, amplifying human expertise rather than replacing it. By autonomously handling complex detection, triage, and response processes, Agentic AI empowers SOC's to achieve new levels of speed, accuracy, and resilience capabilities that were previously impossible in traditional SOC environments.

This paradigm shift reduces alert fatigue, overcome blindspots and accelerates incident remediation, cutting incident handling time from many hours to seconds or minutes. As a result, organization can focus on higher-value initiatives such as threat hunting, strategy development, and proactive defense design.

Design

The solution design consists of a virtualized environment in VMware containing four virtual machines working together. The Splunk SIEM server receives security logs from Windows and Linux endpoints via Universal Forwarders. When suspicious activity is detected, Splunk generates alerts that are automatically sent to the N8N automation server through webhooks. N8N processes these alerts using OpenAI's API to convert technical security data into human-readable notifications, which are then delivered to the security team via Slack. Additionally, Claude Desktop with an integrated MCP server connects directly to Splunk, allowing analysts to query the SIEM and receive AI-powered security recommendations on demand. This architecture combines automated alert processing with AI-assisted analysis to create an intelligent, efficient security operations workflow.

Solution Design

This section presents the comprehensive design methodology utilized in this project, utilizing UML (Unified Modeling Language) diagrams as the primary approach for visualizing and documenting the system architecture. UML diagrams were selected as they best suit the requirements and complexity of this project, providing clear and standardized representations of both structural and behavioral aspects of the system. The design solution will be demonstrated through various UML diagrams, including use case diagrams to illustrate user interactions and functional requirements, sequence diagrams to depict the flow of processes and component communications, activity diagrams to represent workflow and business logic, deployment diagrams to showcase the system's physical architecture and infrastructure, and security architecture diagrams to outline the protective mechanisms and security protocols. This section is extremely important as it establishes the blueprint of the system prior to the implementation phase, ensuring all stakeholders have a clear understanding of the proposed solution. Additionally, detailed diagrams and extended documentation can be found in the appendices for further reference.

UML diagrams

Sequence diagram

The sequence diagram depicts the chronological flow of data and interactions between system components over time. It shows how logs originate from Windows/Linux endpoints and are collected by Splunk, then forwarded to N8N for processing, analyzed by OpenAI, and finally delivered as alerts to Slack channels. Simultaneously, it illustrates the parallel communication pathway where the MCP Server queries Splunk directly and feeds information to Claude AI Assistant for interactive analysis.

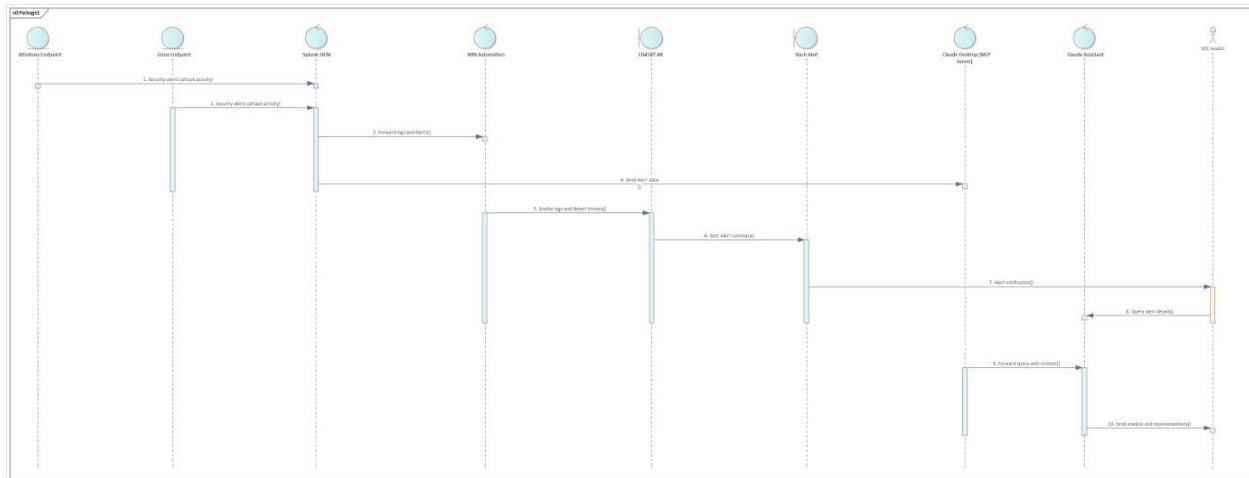


Figure 1- Sequence Diagram

Deployment Diagram

According to Visual Paradigm (2016), a deployment diagram shows the configuration of run-time processing nodes and system components. This diagram illustrates the physical infrastructure of the log monitoring system, including endpoint machines (Windows/Linux), the Splunk server, N8N automation server, OpenAI API integration, Slack platform, MCP Server, and Claude AI interface as distributed nodes. It demonstrates how these hardware and software components are deployed across the network and establishes the communication pathways between them, providing stakeholders with a clear understanding of the system's runtime architecture.

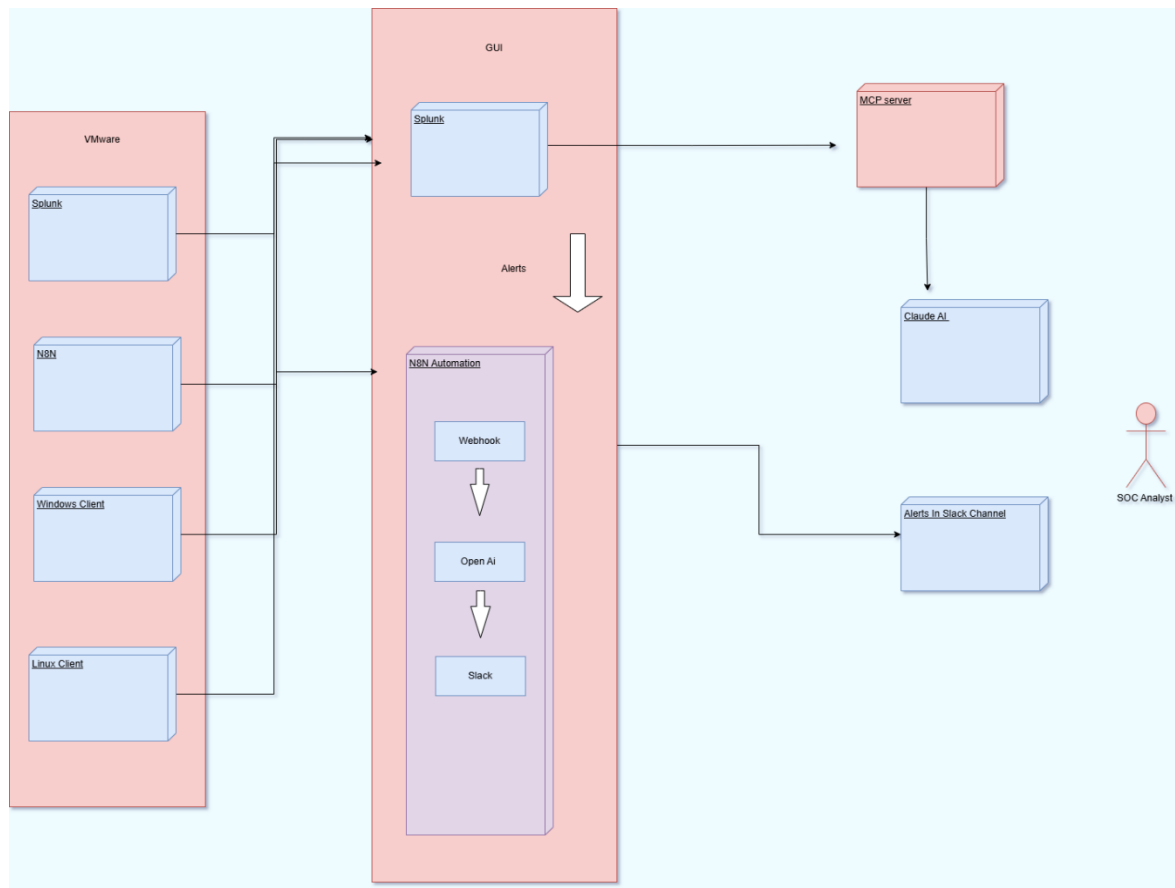


Figure 2- Deployment Diagram

Activity Diagram

Activity diagrams are perfect for modelling business processes as they demonstrate high-level actions that correlate to specific operations in a system (Lynch, 2019). In this project, the activity diagram represents two parallel workflows: the automated log processing pipeline where data flows from endpoints through Splunk, N8N, OpenAI processing, and alert delivery to Slack, alongside the manual query workflow where administrators interact with Claude AI Assistant via MCP Server to retrieve specific information from Splunk. This diagram clearly visualizes decision points, concurrent processes, and the overall operational flow of the log management system.

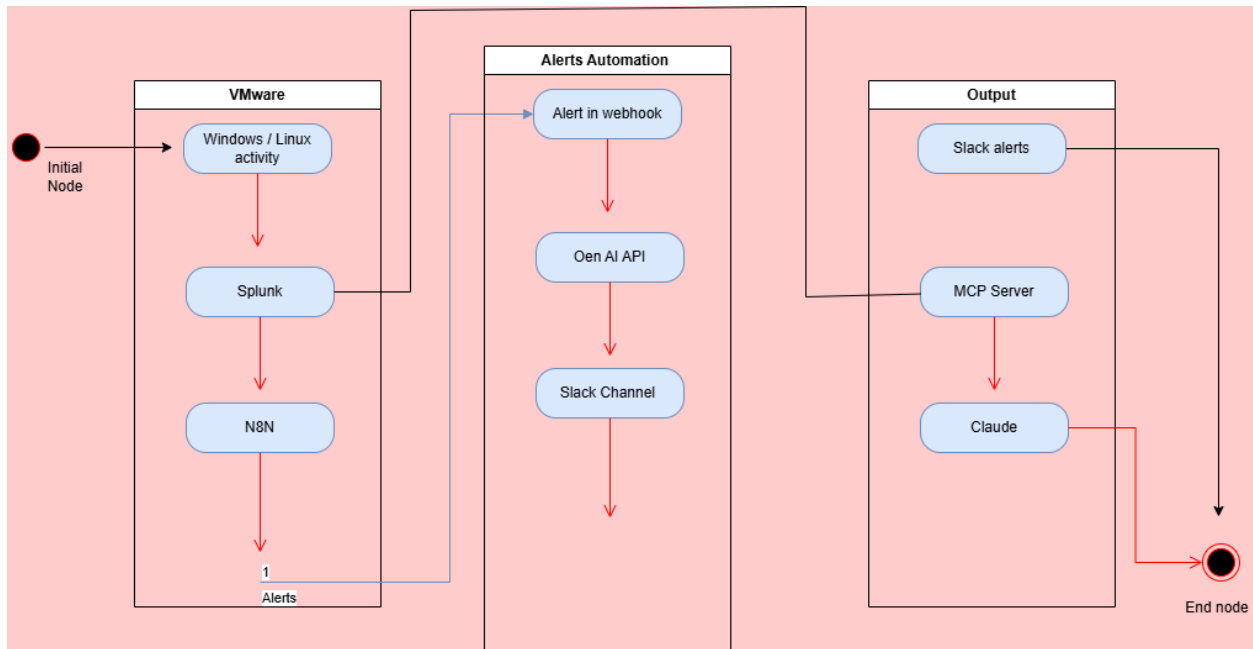


Figure 3- Activity Diagram

Use case Diagram

The use case diagram illustrates the primary actors in the system, Security Analysts, and their interactions with core functionalities. It demonstrates key use cases such as monitoring logs from endpoints, processing alerts through AI, receiving notifications in Slack, and querying system data through Claude AI Assistant. This diagram establishes the functional requirements and user interactions within the Claude Assistant and alert management system.

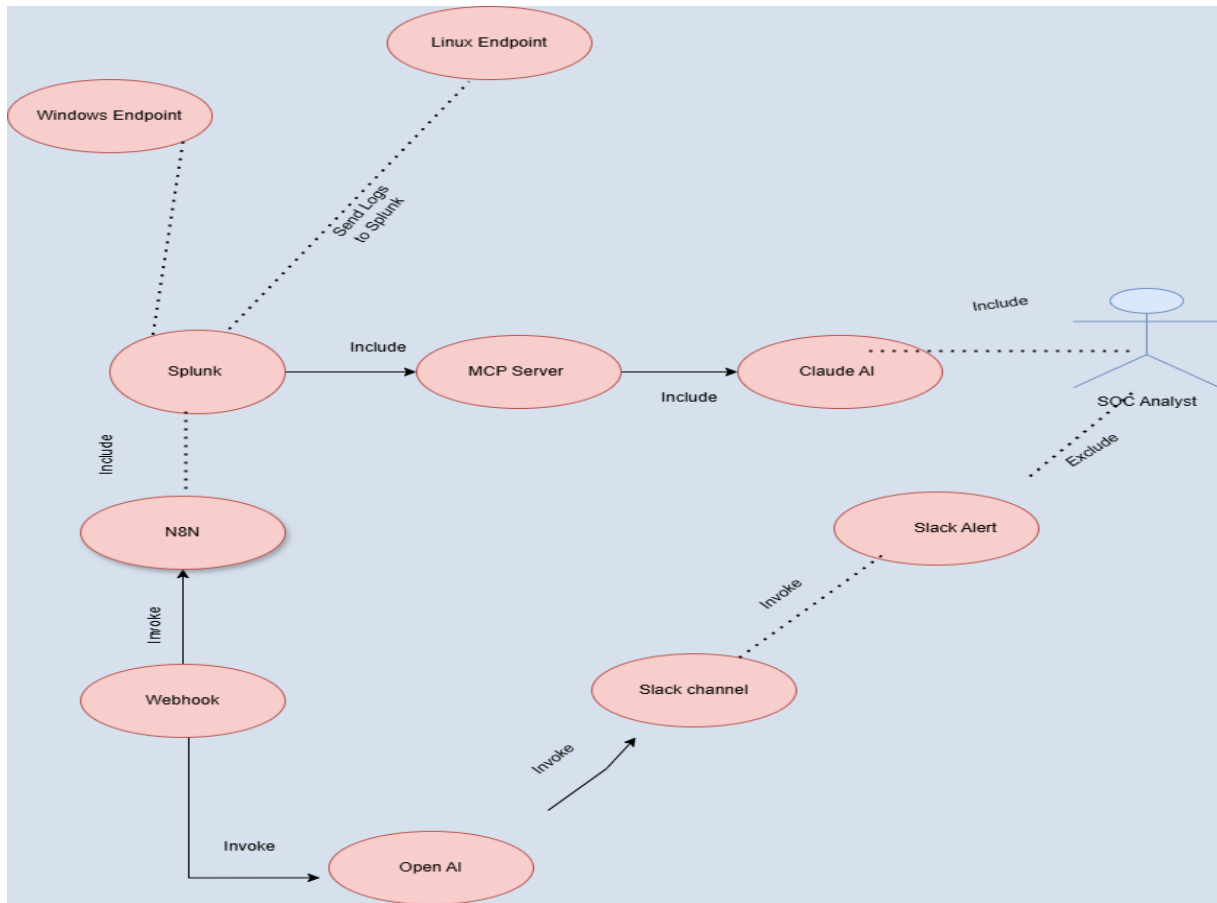


Figure 4- Usecase Diagram

Security Architecture Diagram

The security architecture diagram outlines the protective mechanisms and security controls implemented throughout the system. It highlights log transmission from endpoints to Splunk, API authentication tokens for N8N-OpenAI-Slack communications, access control mechanisms for both Splunk and Claude AI Assistant, and secure

HTTPS channels across all component interfaces from MCP server to Claude AI. This diagram ensures stakeholders understand how data confidentiality, integrity, and availability are maintained throughout the log processing and alerting workflow.

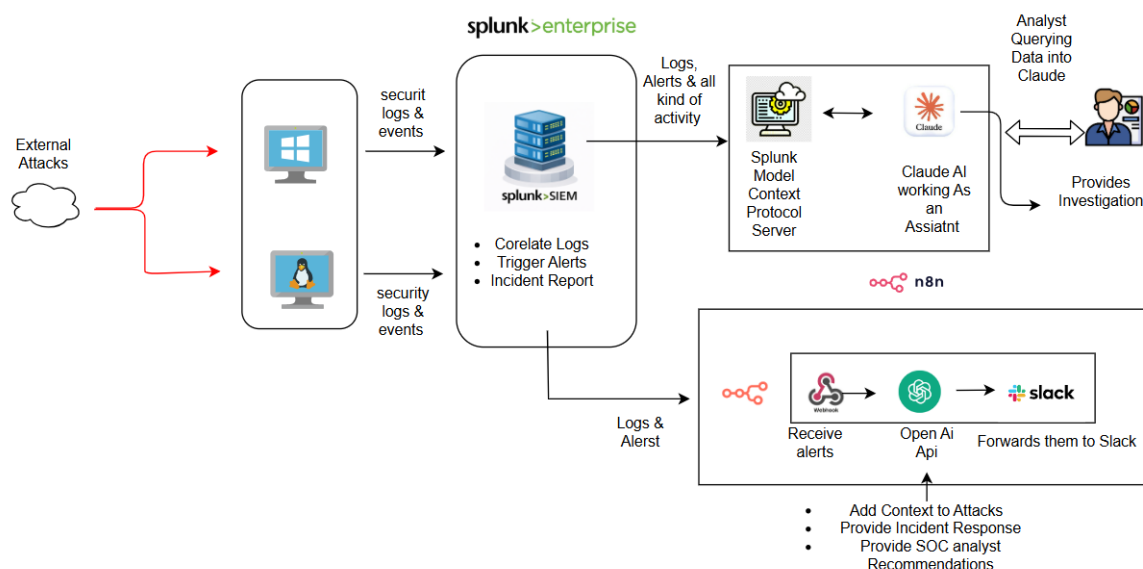


Figure 5- Security Architecture Diagram

Implementations

The Implementation section outlines the step-by-step setup and procedures followed to build the LLM-powered SOC system with full functionality. We have divided our implementation into four main parts: Setting up the SIEM by installing Splunk and configuring end devices for log collection, deploying automation workflows using n8n to handle alert generation and notifications, setting up and configuring simulated attacks on both end devices to test detection capabilities, and establishing the MCP server to

enable the SOC assistant through Claude. The detailed implementation steps for each part are documented in the Appendix section. This section covers the complete implementation process from initial setup to final deployment of the automated security operations center."

Setting up SIEM: Installing Splunk and configuring end devices for log collection

Deploying Automation: Creating n8n automation workflows to handle alert generation and notifications

Setting up Attacks: Configuring and executing simulated attacks on both end devices to test detection capabilities

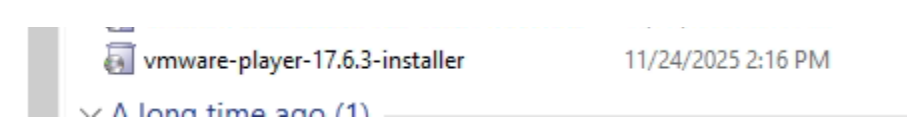
Setting up MCP Server: Establishing the MCP server to enable the SOC assistant through Claude

Setting up SIEM

We started by installing VMware as our virtualization platform to avoid complex network connectivity troubleshooting between host and virtual machines, as VMware provides reliable network bridging capabilities that eliminate connectivity issues. Next, we deployed Ubuntu Server for its lightweight performance and minimal system resource usage, ensuring the host system remained responsive, and configured Windows endpoints as required hosts for different monitoring scenarios. For our SIEM solution, we selected Splunk due to its industry leading log aggregation and analysis capabilities, installed Splunk Universal Forwarder on both Linux and Windows endpoints, deployed Splunk Add-ons for both operating systems to enable proper log collection, and created separate indexes for each operating system to organize data efficiently. Alternative approaches could have included using Hyper-V or Virtual Box for virtualization, lightweight Linux distributions like Alpine, or open-source SIEM solutions like ELK Stack or Wazuh, but VMware's stability over other possible solutions and Splunk's comprehensive features made them the optimal choices for our virtualization and security monitoring requirements.

For setting up SIEM architecture we started from installing VM, since it is the best

We started by installing VMware workstation



Once we have installed VMware, we will install it.



Figure 6- Install VMware

After that, we have VMware, we will install the required instances such as windows 7 iso, Splunk enterprise and Ubuntu Server to run those instances.

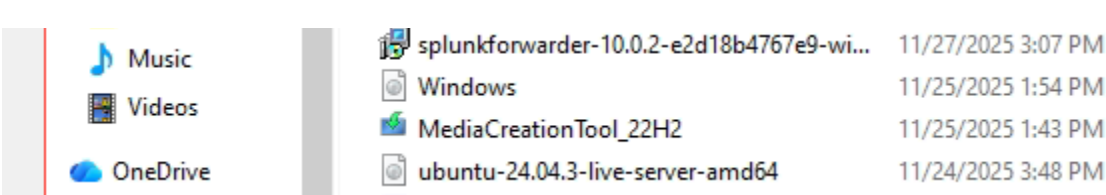


Figure 7- Setting Up SIEM

Ubuntu Server for creating Splunk and N8N instance. After that we have installed we will start by making 2 Linux servers, one for Splunk and One for Automation Software.

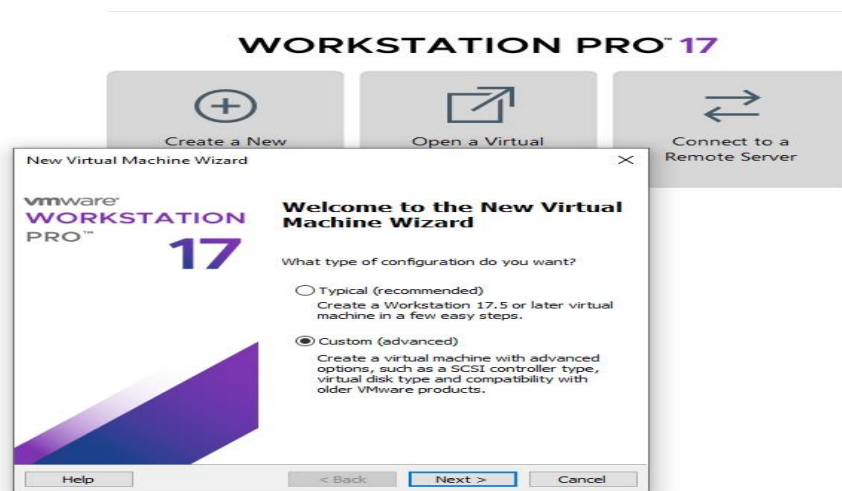


Figure 8- Install Linux Servers

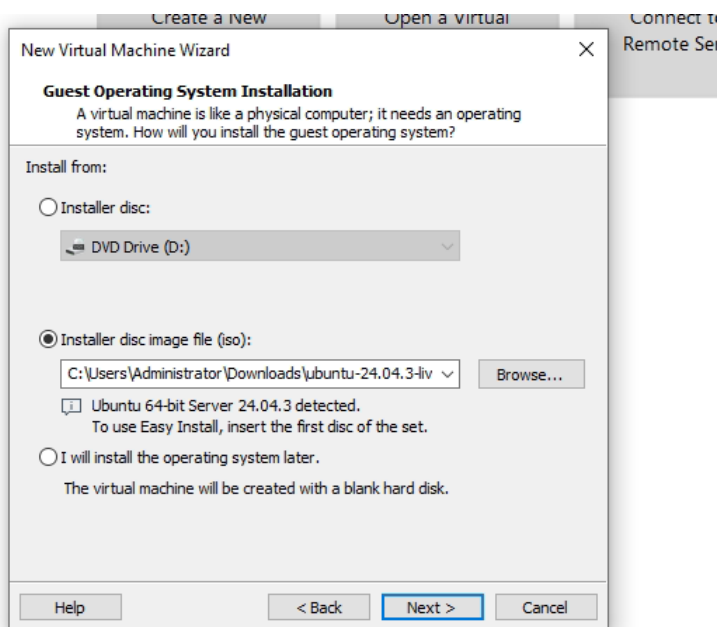


Figure 9- Installing server 2

We will Install Ubuntu

Server for downloading Splunk and N8N instance.

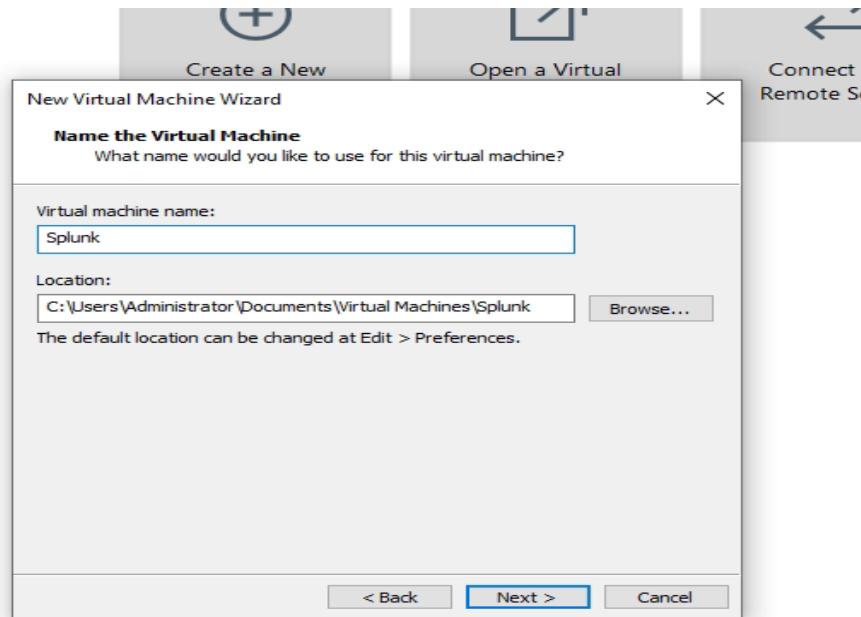


Figure 10-Installing Splunk server

After that we will create 2 Linux Servers, one for Splunk and One for N8N.

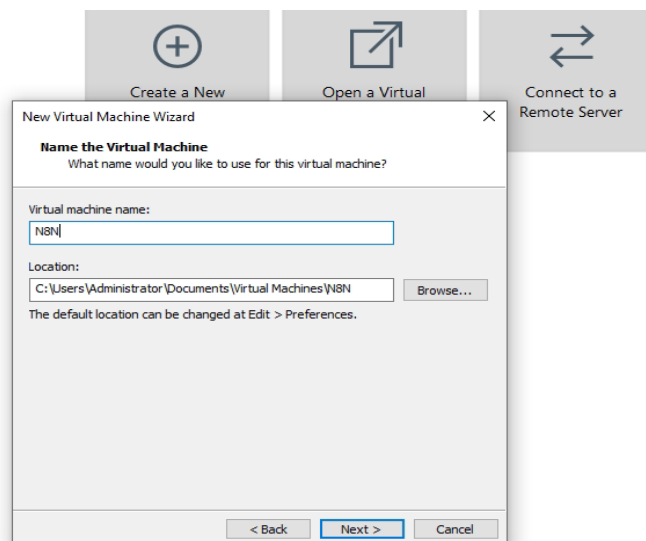


Figure 11-Installing N8N server

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your server's name: The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

Figure 12- Setting up servers

After that VMs are installed, we will start installing install required software's for them.

After that we need to make official Splunk account to get the link to download Splunk.

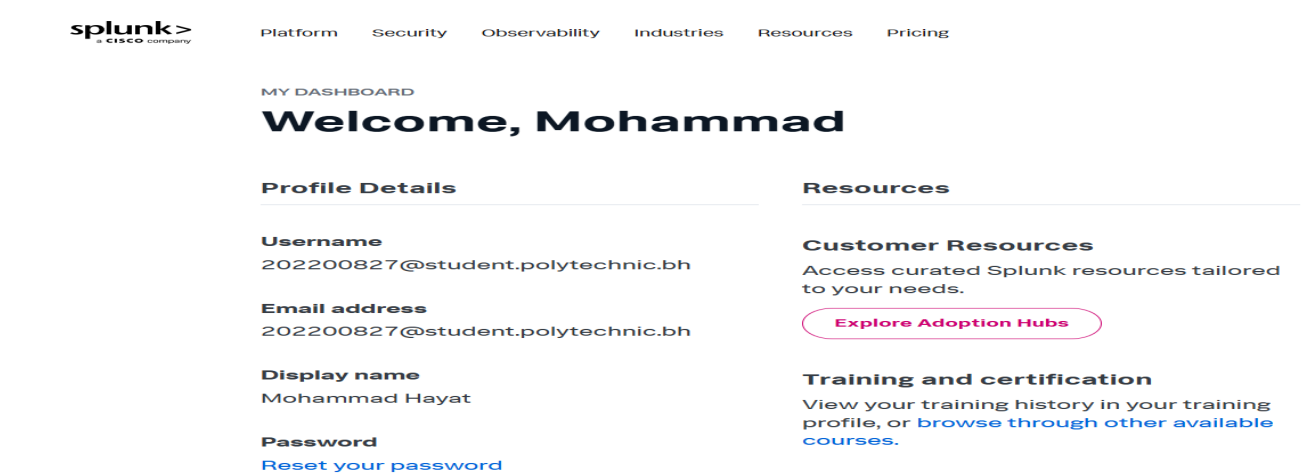


Figure 13-Creating Splunk account

```
splunk: $ wget -O splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.0.0/linux/splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb"
7:37:45-- https://download.splunk.com/products/splunk/releases/10.0.0/linux/splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb
load.splunk.com (download.splunk.com)... 18.161.21.49, 18.161.21.112, 18.161.21.84, ...
download.splunk.com (download.splunk.com)[18.161.21.49]:443... connected.
ent, awaiting response... 200 OK
2564 (1.3G) [binary/octet-stream]
lunk-10.0.0-e8eb0c4654f8-linux-amd64.deb'
8-linux-amd64.deb      11%[=====>] 146.14M  25.9MB/s   eta 54s
```

Figure 14-Downloading Splunk

After that Splunk will be installed

```

mo@splunk:~$ curl http://localhost:8000
curl: (3) URL rejected: Port number was not a decimal number between 0 and 65535
mo@splunk:~$ ll
total 1324788
drwxr-x--- 4 mo mo 4096 Nov 28 13:47 ./
drwxr-xr-x 3 root root 4096 Nov 25 09:07 ../
-rw----- 1 mo mo 454 Dec 20 18:55 .bash_history
-rw-r--r-- 1 mo mo 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 mo mo 3771 Mar 31 2024 .bashrc
drwx----- 2 mo mo 4096 Nov 25 09:15 .cache/
-rw----- 1 mo mo 20 Nov 28 12:29 .lessht
-rw-r--r-- 1 mo mo 807 Mar 31 2024 .profile
-rw-rw-r-- 1 mo mo 1356540912 Nov 13 23:30 splunk-10.0.2-e2d18b4767e9-linux-amd64.deb
drwx----- 2 mo mo 4096 Nov 25 09:07 .ssh/
-rw-r--r-- 1 mo mo 0 Nov 25 09:19 .sudo_as_admin_successful
mo@splunk:~$ curl http://localhost:8000
curl: (3) URL rejected: Port number was not a decimal number between 0 and 65535
mo@splunk:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever

```

Figure 15-Enabling Splunk GUI

After installing You need to run following command to install it

Sudo dpkg -i splunk -10.0.2-e2d18b4767e9-linux-amd64.deb

```

$ sudo dpkg -i splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb
unselected package splunk.
86991 files and directories currently installed.)
splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb ...
m has all the commands we will require to perform the preflight step
splunk-preinstall upgrade check
0.0) ...
.0.0) ...
b/python3.7/site-packages': No such file or directory

```

Figure 16- Splunk Installed

After that we have installed splunk on Linux we need to put following commands in terminal to start splunk.

1)Cd /opt/splunk

Cd bin

./ start splunk

After that you can set your username and password.

```

Third Party Providers: Your authorized consultants, contractors, and agents.

Trial Offering: An Offering we make available on a trial or evaluation basis.

Usage Data: Data generated from the usage, configuration, deployment, access,
and performance of an Offering.

Use Rights: As set out in section 1.1.

Do you agree with this license? [y/n]: y

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: █

```

Figure 17- Setting up User credentials

Once you have choose Username and password, You can use those credentials to access Splunk GUI on your computer by using your Machines IP at port 8000.

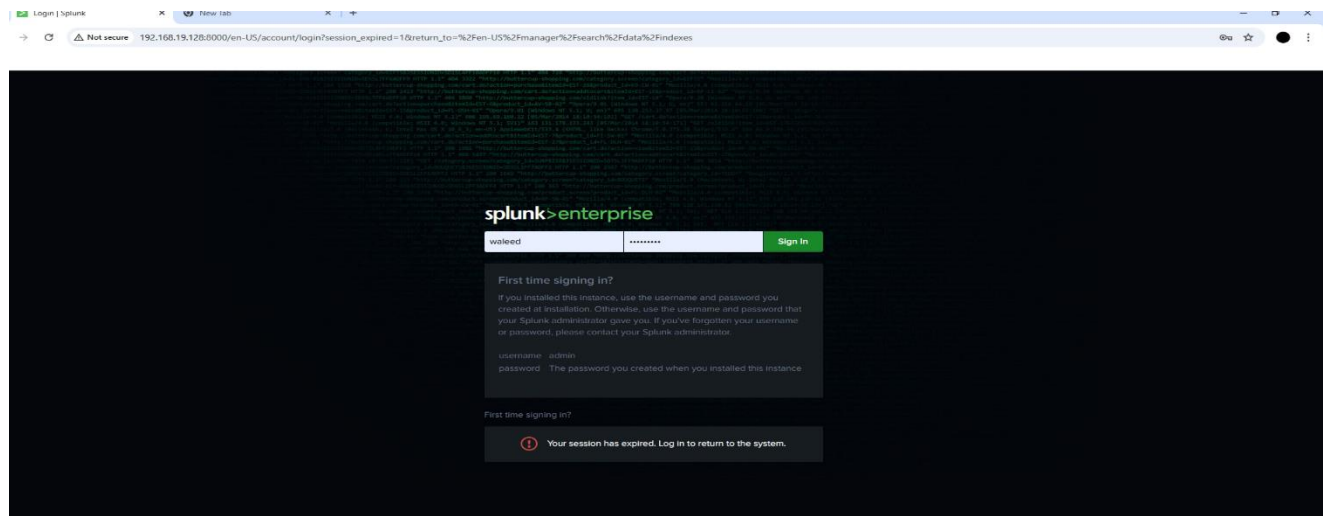


Figure 18- Opening GUI

Once you have installed Splunk we will start installing Windows and Linux client.

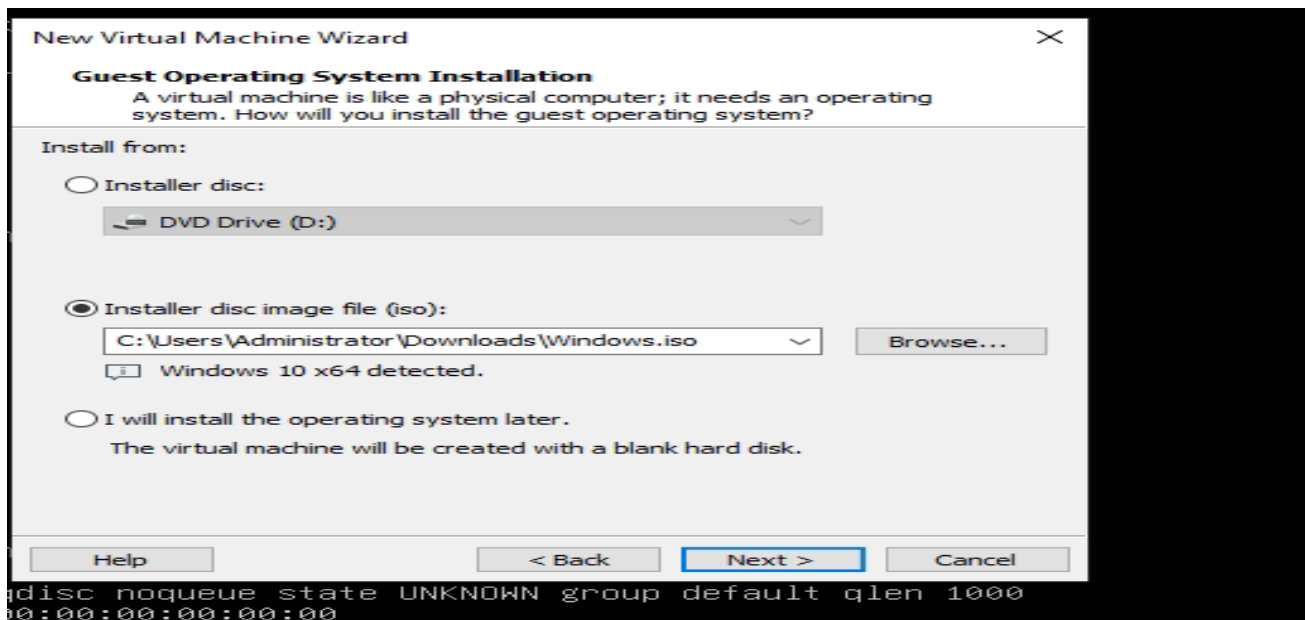


Figure 19- Installing Windows Client

After that we have installed windows vm we need to install Splunk universal Forwarder on it so that our splunk can start receiving alerts.

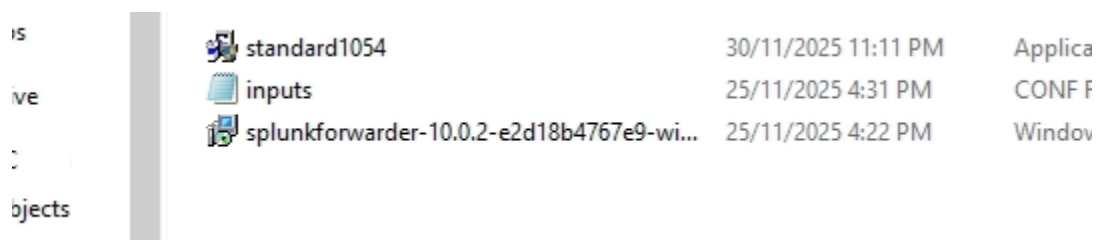


Figure 20-Installing Universal Forwarder on windows

Once the forwarder is installed, we will configure it and start sending logs to Splunk.

After configuring splunk forwarder we will configure Splunk receiving port ,

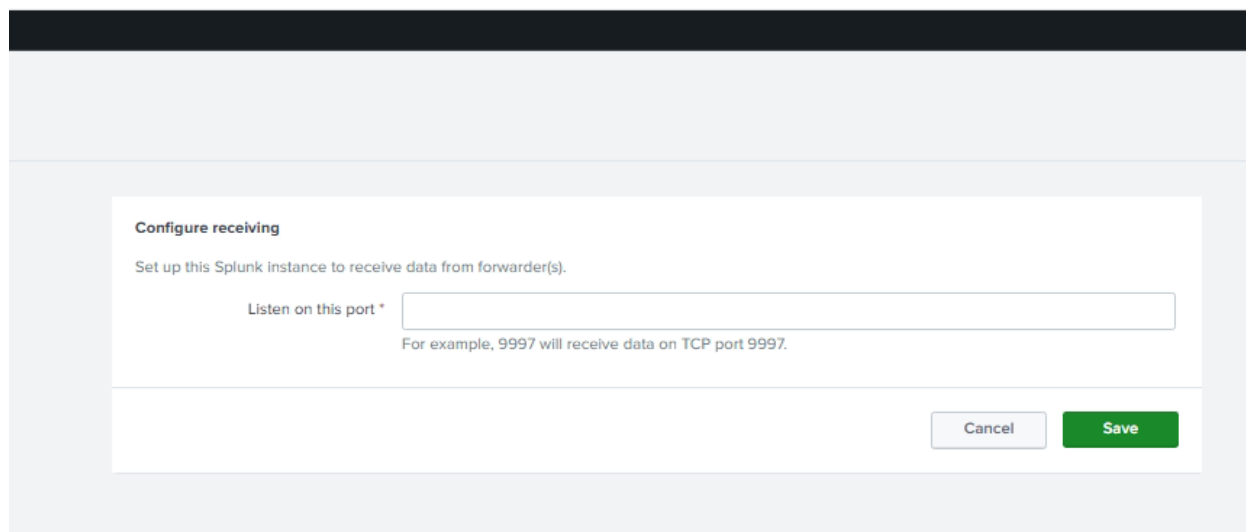


Figure 21- Configuring receiving Port on Splunk

We will do the same for our Linux endpoint too,

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

You have new mail.
mohammad@mohammad:~$ ll
total 69740
drwxr-x--- 7 mohammad mohammad 4096 Dec  9 13:03 ./
drwxr-xr-x 4 root      root      4096 Dec  8 13:27 ../
-rw-r----- 1 mohammad mohammad 2552 Dec  8 21:05 .bash_history
-rw-r--r-- 1 mohammad mohammad 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 mohammad mohammad 3771 Mar 31 2024 .bashrc
drwx----- 3 mohammad mohammad 4096 Dec  9 07:30 .cache/
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:30 go/
-rw-r----- 1 mohammad mohammad 20 Dec  8 19:22 .lessht
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:49 .local/
drwxrwxr-x 2 mohammad mohammad 4096 Dec  9 13:03 MailHogData/
-rw-rw-r-- 1 mohammad mohammad 29122 Dec  9 13:03 .mailhog.log
-rw-r----- 1 mohammad mohammad 135 Dec  9 10:26 .msmtprc
-rw-rw-r-- 1 mohammad mohammad 27 Dec  7 18:52 passwords.txt
-rw-r--r-- 1 mohammad mohammad 807 Mar 31 2024 .profile
-rw-rw-r-- 1 mohammad mohammad 71311634 Nov 13 23:43 splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64.deb
drwx----- 2 mohammad mohammad 4096 Dec  1 20:24 .ssh/
-rw-r--r-- 1 mohammad mohammad 0 Dec  1 20:36 sudo_as_admin_successful
```

Figure 22- Installing Splunk Forwarder on Linux

After that you need to install Windows add-on and Linux add on. On your splunk so start reading logs from windows and Linux Operating system.

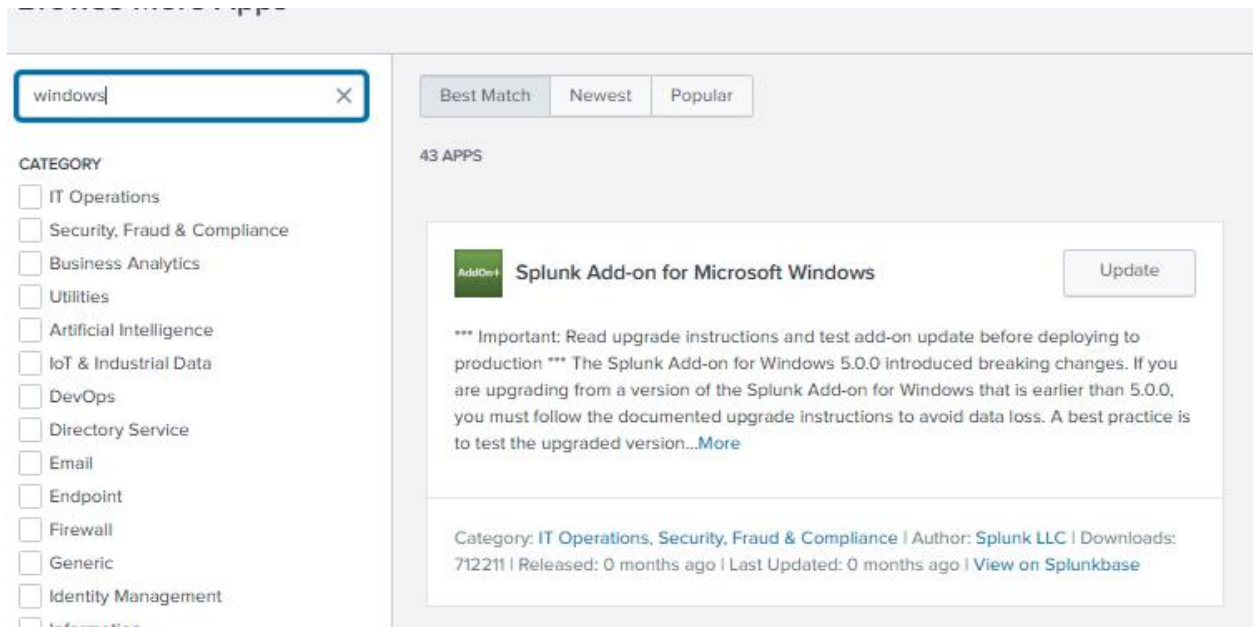


Figure 23- Installing Splunk Add for Windows Add on

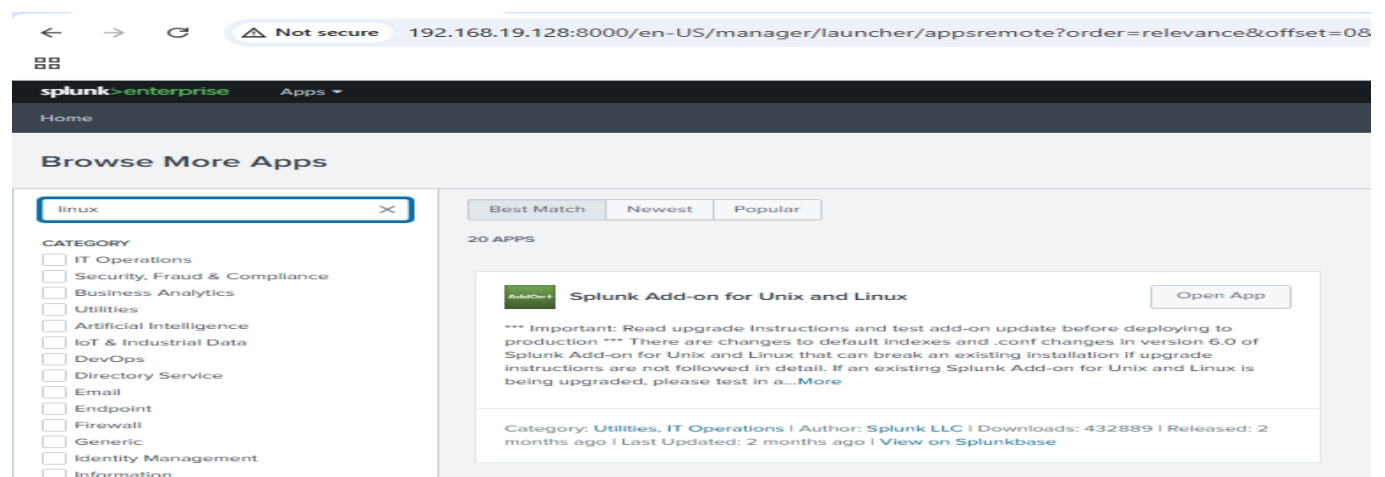


Figure 24- Installing Splunk Add on for Linux

After that you need to configure Splunk forwarder .config files so you can start receiving alerts on your splunk Dashboard.

Creating Automation

We began automation implementation by installing n8n on Ubuntu Server, first deploying Docker to avoid dependency conflicts and ensure smooth installation. While alternative automation platforms like Shuffle were available, we chose n8n due to its

superior integration capabilities with our existing components and extensive plugin ecosystem. We configured a webhook to capture alerts from Splunk and forward them to OpenAI's ChatGPT for intelligent analysis and contextualization, selecting OpenAI over alternatives like Google Gemini because of its more advanced reasoning capabilities and better understanding of security contexts. Finally, we integrated Slack as our notification channel for receiving AI-analyzed alerts, the other alternative possible was discord but because Slack's workspace structure and threading features provided better organization for security alert management and team collaboration. This complete automation pipeline enabled seamless alert flow from detection through analysis to notification, significantly reducing response time and providing security teams with AI-enhanced threat intelligence.

To create automation first thing, we will do is download N8N on a Linux server.

So, avoid dependencies N8N will be installed in docker.

```
mohammad@linux-test:~$ sudo apt install docker.io
[sudo] password for mohammad:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils-linux
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 59 not upgraded.
Need to get 76.0 MB of archives.
After this operation, 288 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Figure 25- Installing Docker

Once docket is installed, we will make a directory N8N compose.

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

```
n8n@n8n:~$ ll
total 32
drwxr-x--- 5 n8n  n8n  4096 Nov 25 19:19 ./
drwxr-xr-x 3 root root  4096 Nov 25 10:26 ../
-rw-r--r-- 1 n8n  n8n   220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 n8n  n8n  3771 Mar 31  2024 .bashrc
drwx----- 2 n8n  n8n  4096 Nov 25 10:27 .cache/
drwxrwxr-x 3 n8n  n8n  4096 Nov 25 19:58 n8n-compose/
-rw-r--r-- 1 n8n  n8n   807 Mar 31  2024 .profile
drwx----- 2 n8n  n8n  4096 Nov 25 10:26 .ssh/
-rw-r--r-- 1 n8n  n8n     0 Nov 25 10:27 .sudo_as_admin_successful
n8n@n8n:~$
```

Figure 26- Composing N8N

In that directory we will run a .yaml file to install N8N

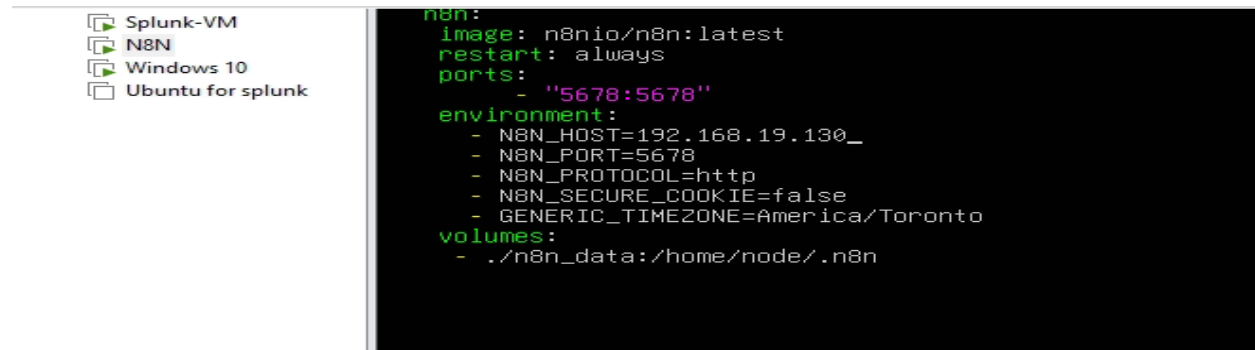


Figure 27- Creating .yaml file with script for installing N8N

After N8N is installed, we can Access the GUI at port 5678 at the IP address of our VM.

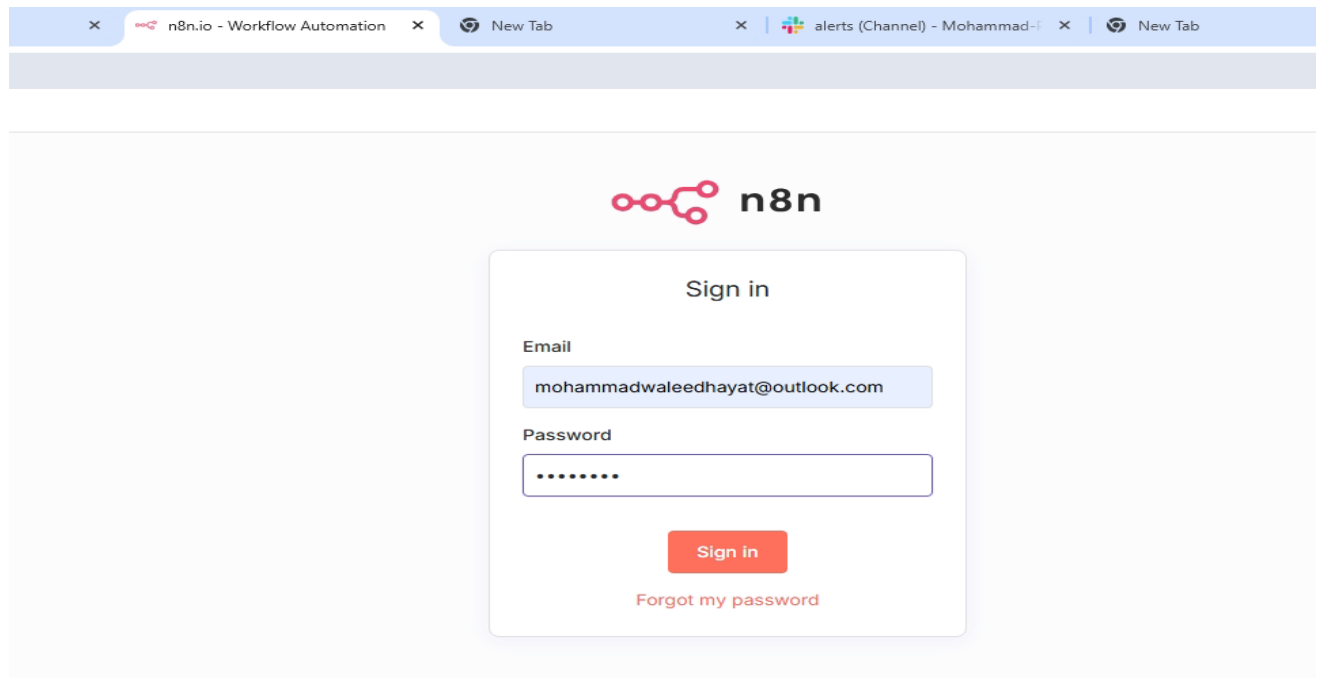


Figure 28- Accessing N8N GUI

To start making Alert notification we will create a Webhook for catching alerts in N8N, and select the post method and we will copy its API and add it to send to a Webhook option of our triggered Alerts in Splunk, so each time an Alert is triggered it sends its result to this webhook.

Webhook

Listen for test event

Parameters

Settings

Docs

Webhook URLs

Test URL

Production URL

POST

http://192.168.19.130:5678/webhook-test/e02bae01-3e22-45a0-95a0-37c4602dea5e

HTTP Method

POST

Path

e02bae01-3e22-45a0-95a0-37c4602dea5e

Authentication

None

This data is pinned for test executions.

OUTPUT

1 item

headers	
accept-encoding	: identity
content-length	: 397
host	: 192.168.19.130:5678
content-type	: application/json
user-agent	: Splunk/FF289EB5-D1CA-42E4-9F9E-30B0D80471AB
connection	: close

Figure 29- Setting Up webhook for catching Alerts

Search: `index="waleed-project" EventCode=4625`
`| stats count by _time, ComputerName, user, src_ip`

Alert type: Scheduled Real-time

Expires: 2 day(s)

Trigger Conditions

Trigger alert when: Per-Result

Throttle: ☐

Trigger Actions

+ Add Actions

When triggered

>	Add to Triggered Alerts	Remove
▼	Webhook URL: http://192.168.19.130:5678/webhook-test/e Specified URL to send JSON payload via HTTP POST (ex., https://your.server.com/api/v1/webhook). Learn More	Remove

Cancel Save

Figure 30- Configuring Attack to send Alert on N8N webhook

Now we will Connect Open AI Chatbot Api to Our webhook to create Customized incident Response recommendations according to each type of alert.

We will select the method Send a message give a prompt to act the way we want it to.

We have created 2 roles, 1st role is System which is for setting the AI's identity and behavior. We have given instructions to act like an SOC tier 1 analyst and has been instructed to evaluate incoming alerts based on the MITRE ATT&CK framework, identifying tactics, techniques, and procedures (TTPs) associated with potential threats. Upon receiving alert data, the AI provides initial triage recommendations, severity assessments, and suggests appropriate response actions aligned with industry's best practices. It correlates indicators of compromise (IOCs) such as suspicious IP

addresses, user behaviors, and system anomalies to determine if an alert represents a genuine security incident or a false positive.

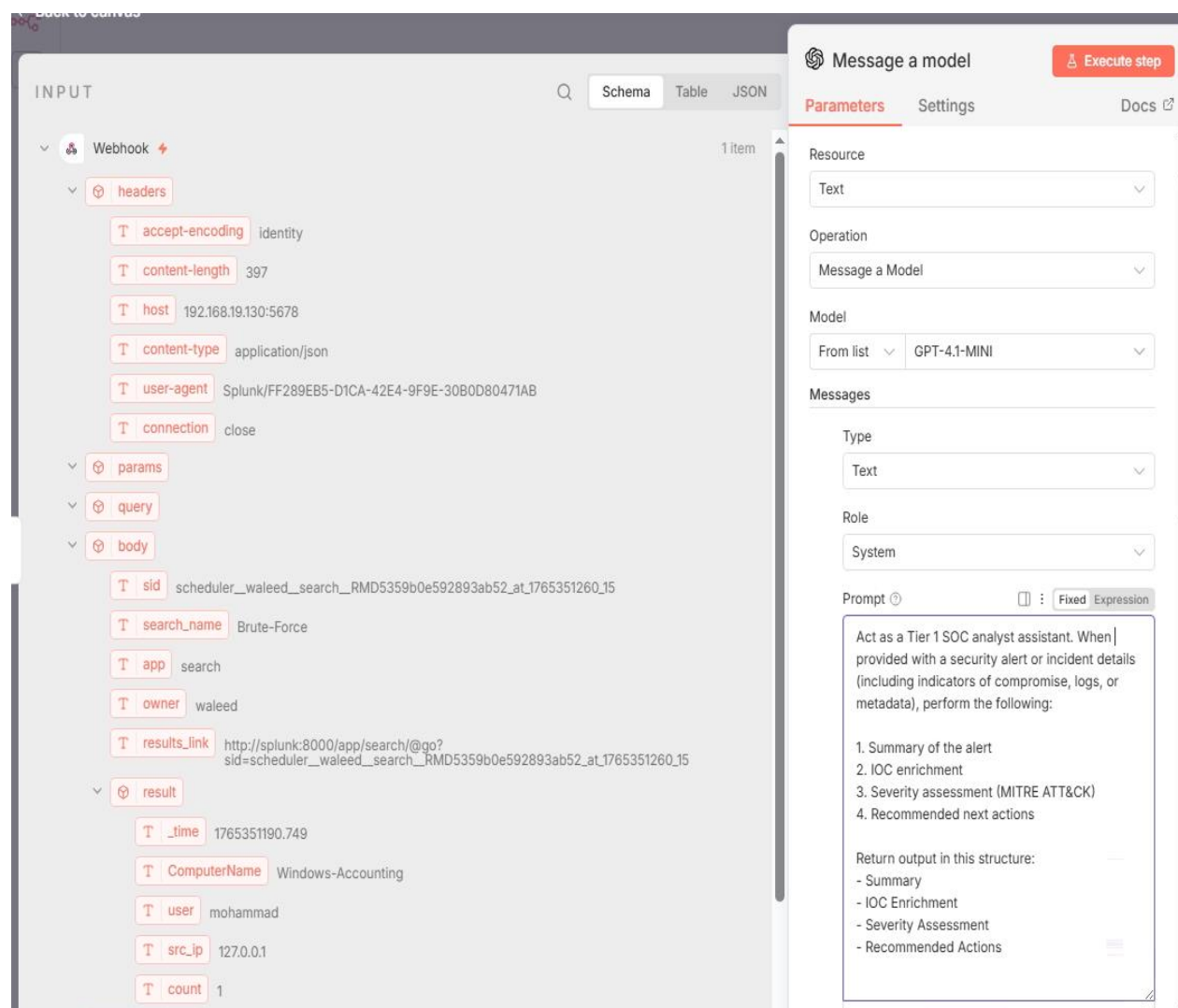


Figure 31- Training ChatGPT

The 2nd role is for System where we are training system that what to do with received data from webhook. We will prompt a code in .json to train system We trained n8n to catch incoming alert data from a webhook and format it for OpenAI. When the webhook receives a security alert (like a brute-force attack, Phishing, Privilege escalation), n8n extracts the alert name from \$json.body.search_name and the detailed results from \$json.body.result. It then structures this information into a readable message format

using `JSON.stringify` to properly format the alert details with indentation. This formatted alert message is then sent to OpenAI's API, which can analyze the security event, provide context about the threat, suggest remediation steps, or generate human-readable summaries of the technical alert data for security teams to review and act upon.

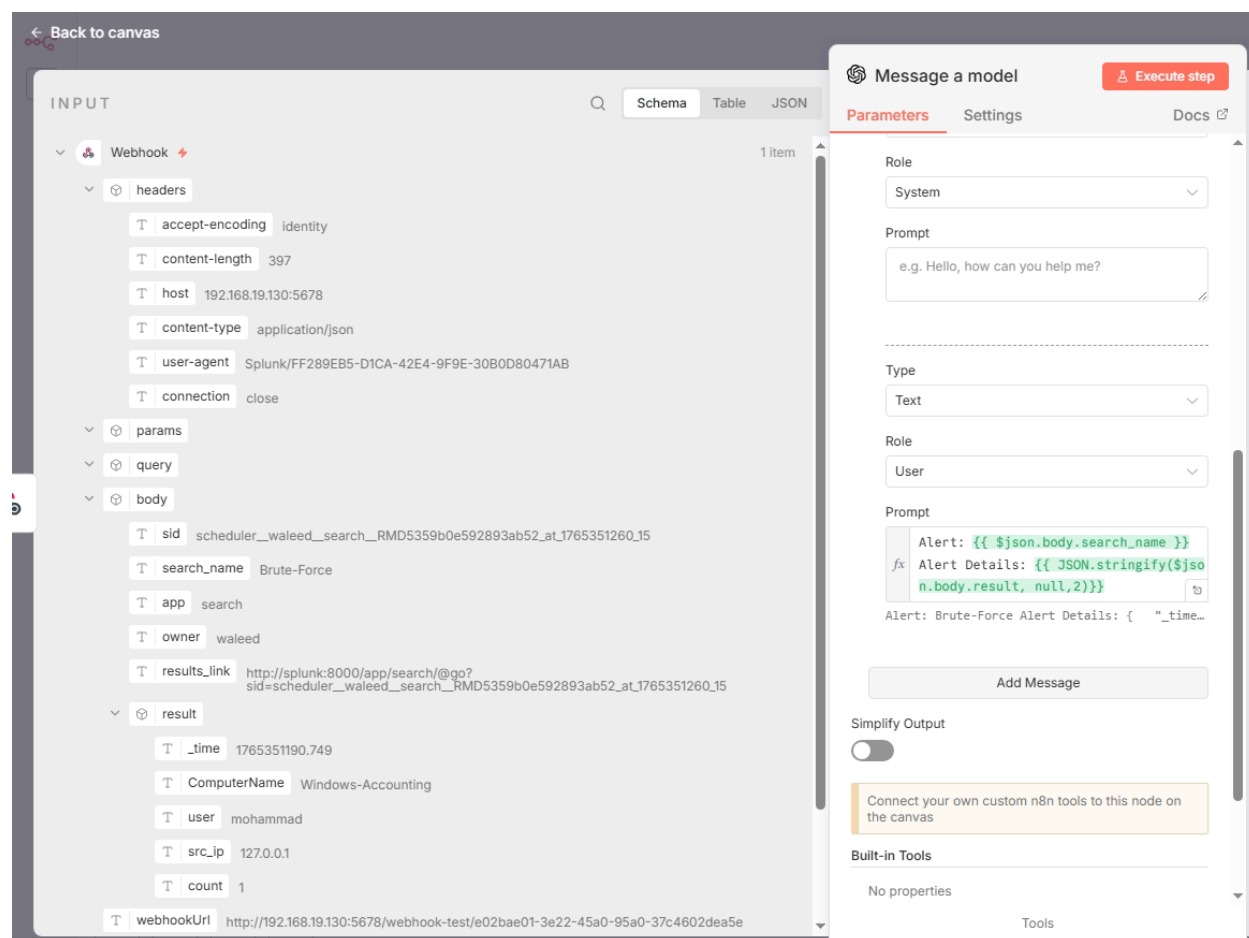


Figure 32- Training Open Ai II

Configuring Slack Alerts

Now to receive Alerts in slack you need to create a designated Slack App.

For creating A slack app and

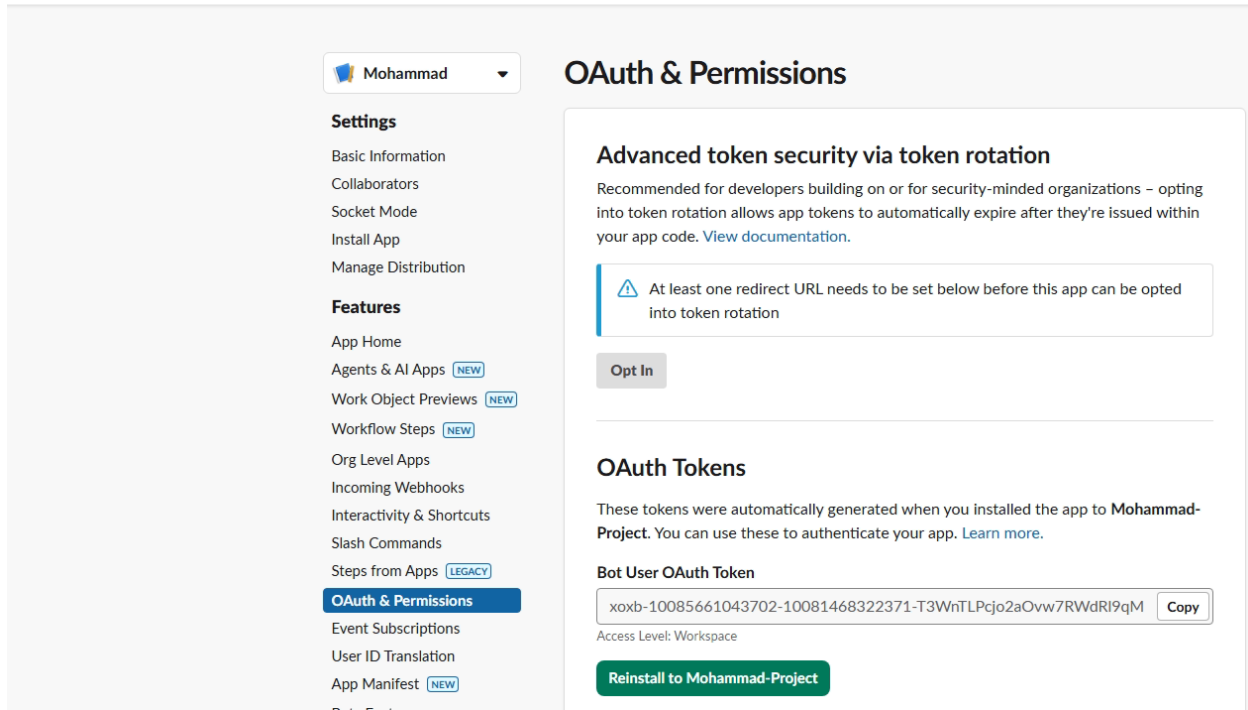


Figure 33- Configuring Slack API

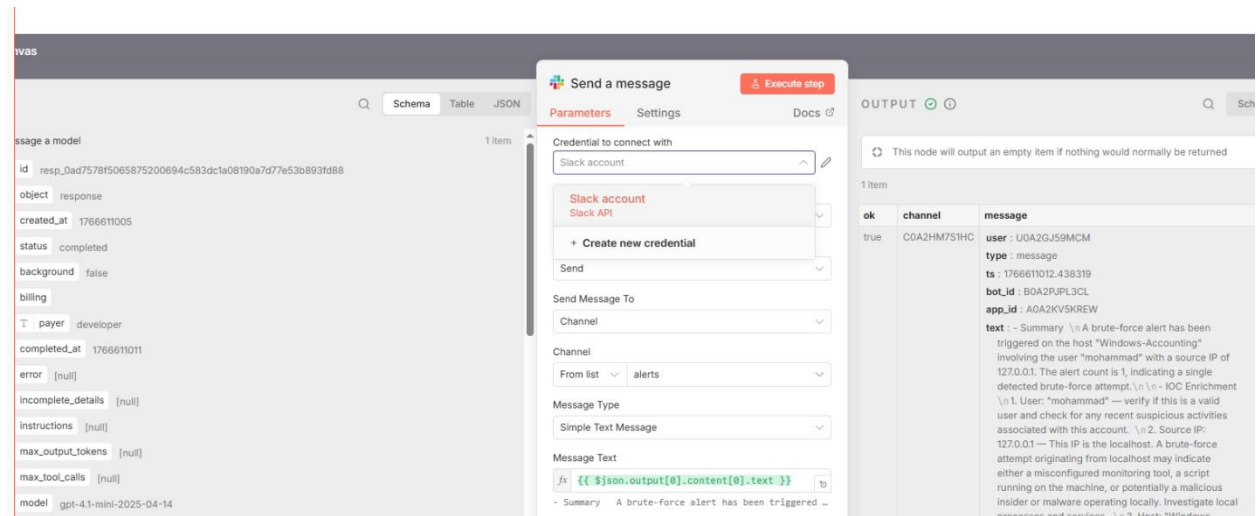


Figure 34- Configuring Slack to receive a Message

Scopes

A Slack app's capabilities and permissions are governed by the [scopes](#) it requests.

Bot Token Scopes

Scopes that govern what your app can access.

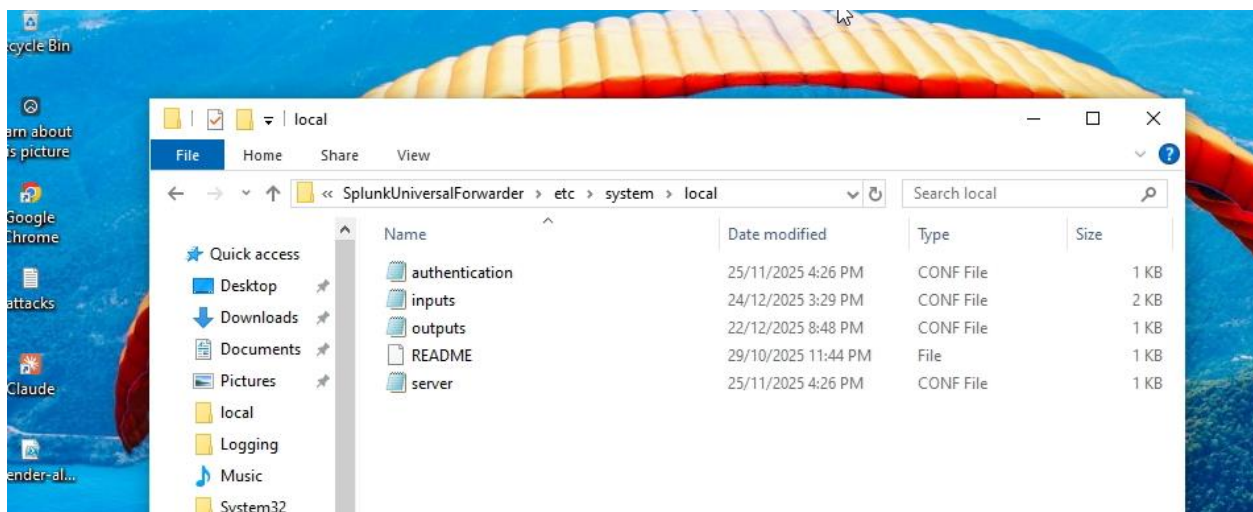
OAuth Scope	Description	
channels:read	View basic information about public channels in a workspace	
chat:write	Send messages as @Mohammad	
chat:write.customize	Send messages as @Mohammad with a customized username and avatar	
files:read	View files shared in channels and conversations that "Mohammad" has been added to	
files:write	Upload, edit, and delete files as "Mohammad"	
groups:read	View basic information about private channels that "Mohammad" has been added to	
im:history	View messages and other content in direct messages that "Mohammad" has been added to	
im:read	View basic information about direct messages that "Mohammad" has been added to	

Figure 35- Setting Slack App permissions

Configuring Attacks

To start configuring attacks we needed both systems running with Universal forwarder on them. Once we have received the activity we can start saving alert of suspicious activity, since we are using windows and Linux system, we will require their logs under their indexes. First, we will set both the system to send logs to Our splunk server for that we will need to configure their output.config file to our Splunk servers IP and other needed information.

These files are located in C:\programfiles\SplunkUniversalForwarder\etc\system\local



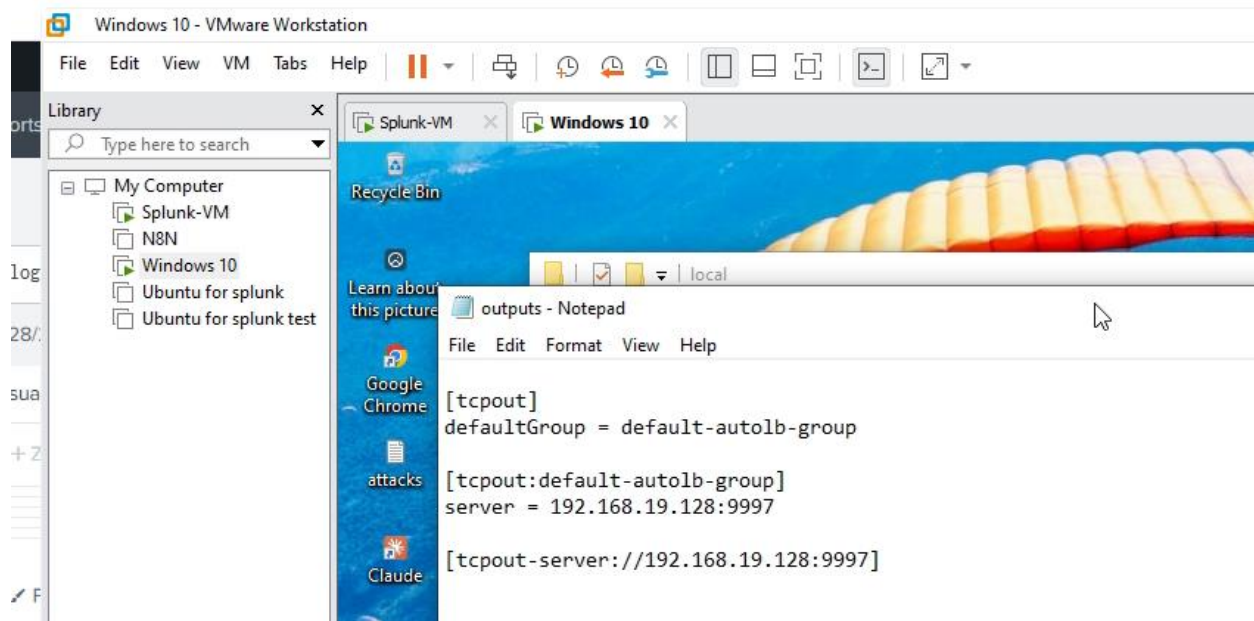


Figure 36 - Configuring Splunk Server's IP

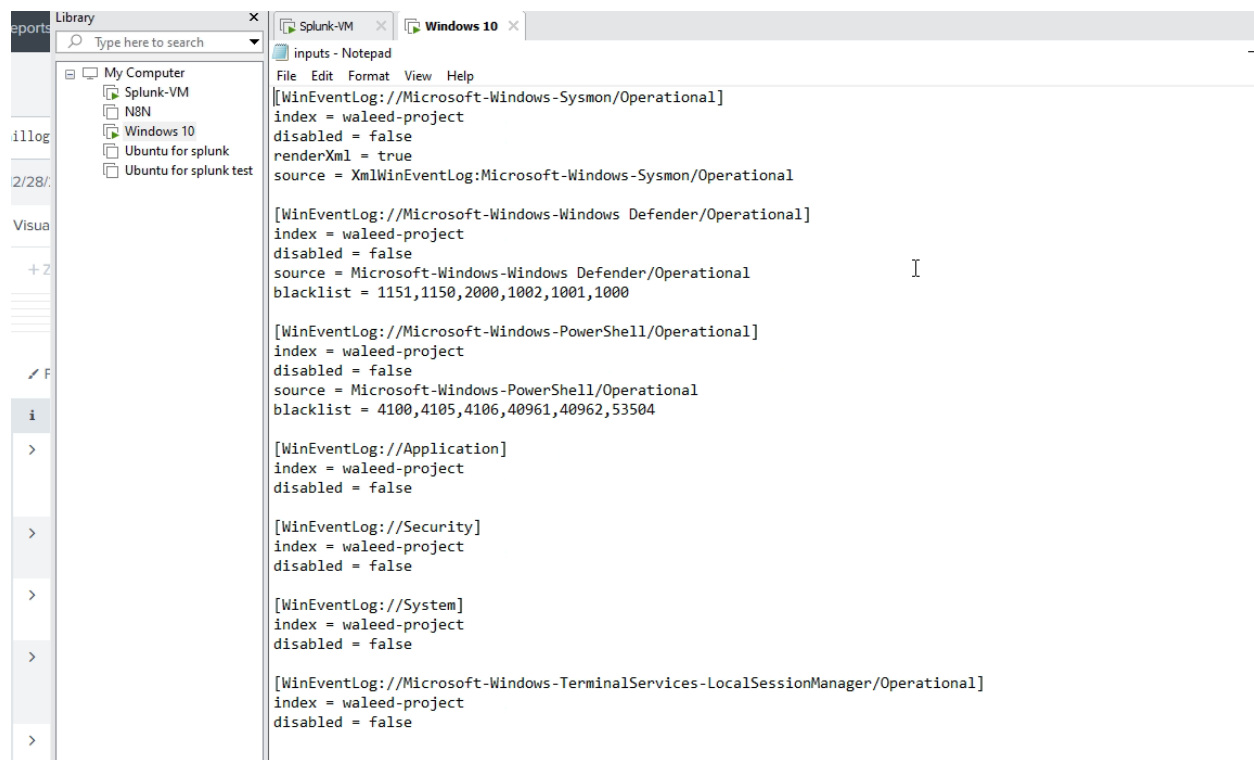


Figure 37-Configuring Windows logs & events files to be monitored

```
[monitor://C:\Program Files (x86)\Mail Enable\Logging\SMTP]
index = waleed-project
sourcetype = maillogs
disabled = false

[monitor://C:\Program Files (x86)\Mail Enable\Logging\]
index = waleed-project
sourcetype = maillogs
disabled = false

[WinEventLog://Security]
disabled = false
start_from = oldest
current_only = 0
evt_resolve_ad_obj = 1
checkpointInterval = 5
whitelist = 5156,5157
index = waleed-project
```

Figure 38- Configuring Windows logs & events files to be monitored II

Attacks on windows

The 'New Index' window is shown with the following settings:

- Index Name:** waleed-project
- Index Data Type:** Events (selected)
- Home Path:** optional
- Cold Path:** optional
- Thawed Path:** optional
- Data Integrity Check:** Enable (selected)

Figure 39- Index for receiving Data

Brute Force Attack:

The 'Edit Alert' window is shown with the following settings:

- Alert:** Brute-Force
- Description:** Optional
- Search:**

```
index="waleed-project" EventCode=4625  
  
|stats count by _time,ComputerName,user,src_ip
```
- Alert type:** Real-time (selected)
- Expires:** 2 day(s)
- Trigger Conditions:**
 - Trigger alert when: Per-Result
 - Throttle: ☐
- Trigger Actions:**
 - + Add Actions
 - When triggered:
 - ▼ Add to Triggered Alerts (Remove)
 - Severity: Medium
 - > Webhook (Remove)

Figure 40- Brute Force Attack

Phishing Attack

For phishing attack we need to install an email server first

These were the command for Installing SMTP server locally,

Install SMTP Server

Install-WindowsFeature SMTP-Server -IncludeManagementTools

Start SMTP Service:

cmd

net start smtpsvc

Configure SMTP Server:

Set SMTP to start automatically

sc config smtpsvc start= auto

Check SMTP service status

sc query smtpsvc

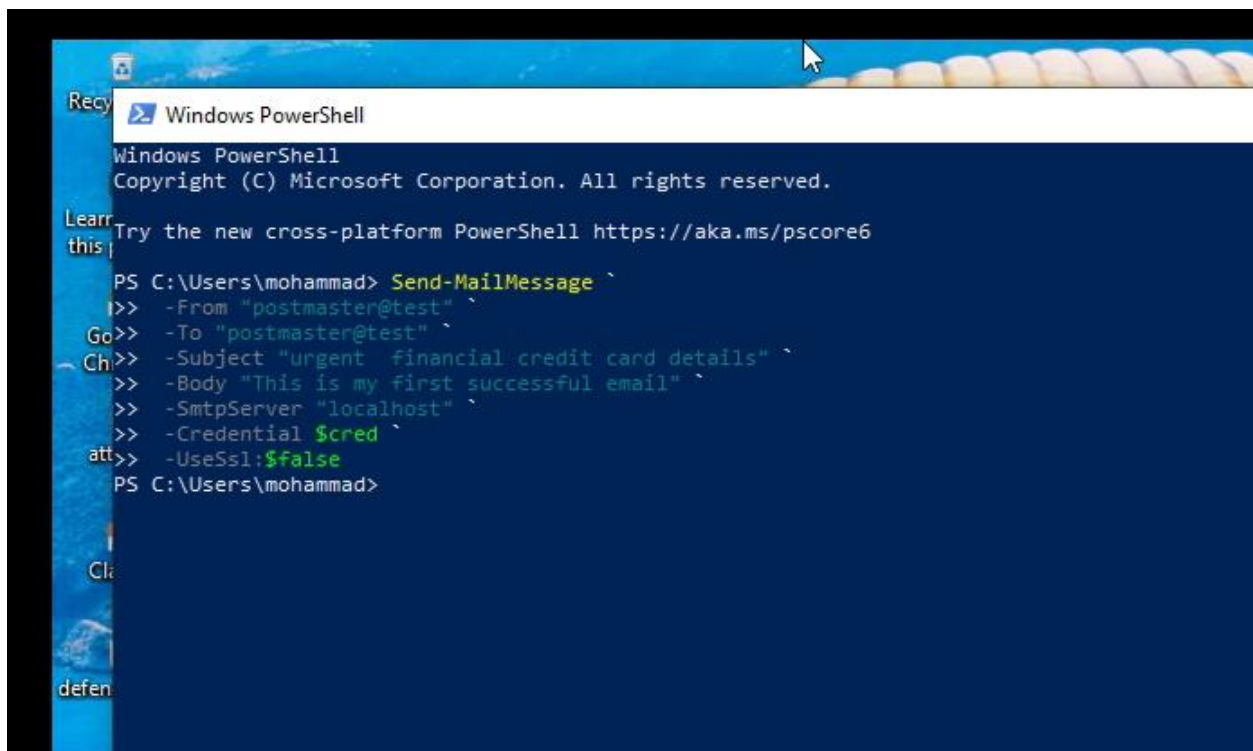


Figure 41 - Sending Email Locally

Mohammad

Edit Alert

Settings

AlertPhishing Detection

DescriptionTriggers when phishing-related keywords are detected in SMTP mail logs.

Searchindex="waleed-project" sourcetype=maillogs
| search urgent OR "credit card" OR details OR Financial OR Harvesting

Alert type

Scheduled

Real-time

Expires2

day(s)

Trigger Conditions

Trigger alert whenPer-Result

Throttle?☐

Trigger Actions

+ Add Actions

When triggered

▼

Add to Triggered Alerts

Remove

Severity

Critical

>

Webhook

Remove

Cancel

Save

Figure 42-Saving Phishing Detection Alert

Privilege Escalation

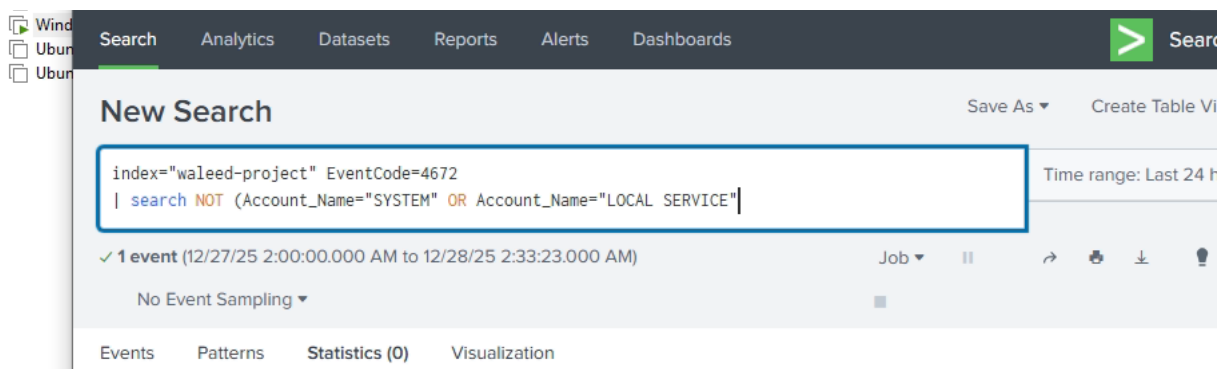


Figure 43 - checking for privilege escalation activity

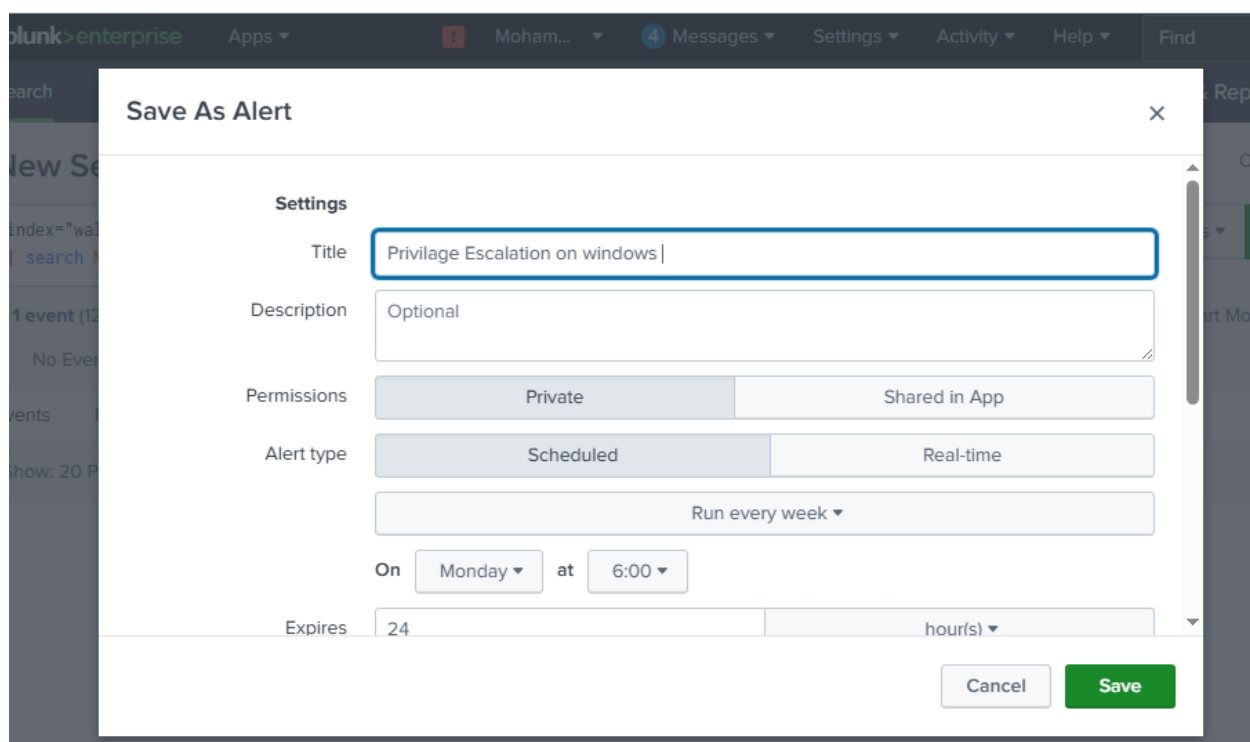


Figure 44- Saving Privilege Escalation Alert

Attacks On Linux

We will create an index for Linux, so we get all user activity on that

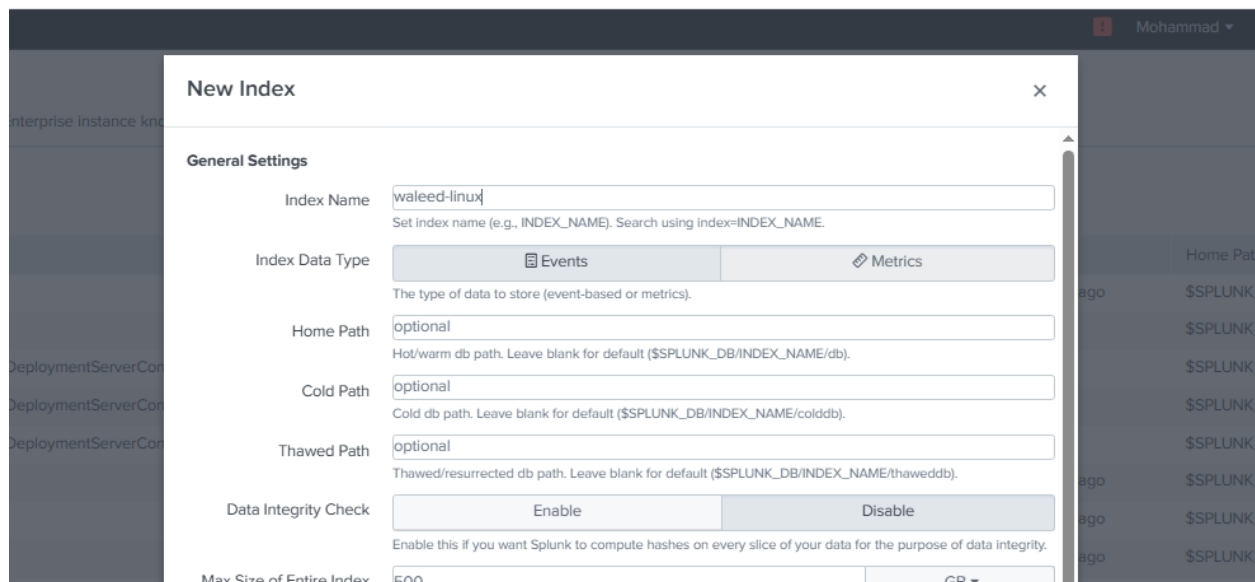


Figure 45- Index for Linux OS

As you can see, we have 2 files in our Linux Forwarders directory,

```

mohammad@mohammad-linux:~$ cd /opt/splunkforwarder/etc/system/local/
mohammad@mohammad-linux:/opt/splunkforwarder/etc/system/local$ ll
total 24
drwxr-xr-x 2 splunkfwd splunkfwd 4096 Dec 25 23:13 ./
drwxr-xr-x 8 splunkfwd splunkfwd 4096 Dec  1 21:01 ../
-rw-r--r-- 1 splunkfwd splunkfwd  604 Dec 25 23:13 inputs.conf
-rw----- 1 splunkfwd splunkfwd  144 Dec 25 22:51 outputs.conf
-r--r--r-- 1 splunkfwd splunkfwd  265 Oct 29 23:44 README
-rw----- 1 splunkfwd splunkfwd  433 Dec  1 21:05 server.conf
mohammad@mohammad-linux:/opt/splunkforwarder/etc/system/local$

```

Figure 46 - Splunk Forwarder input & output .conf files

We will configure output.conf file to send all the alerts& activity to Our Splunk server.

We will use command `sudo nano output.conf`

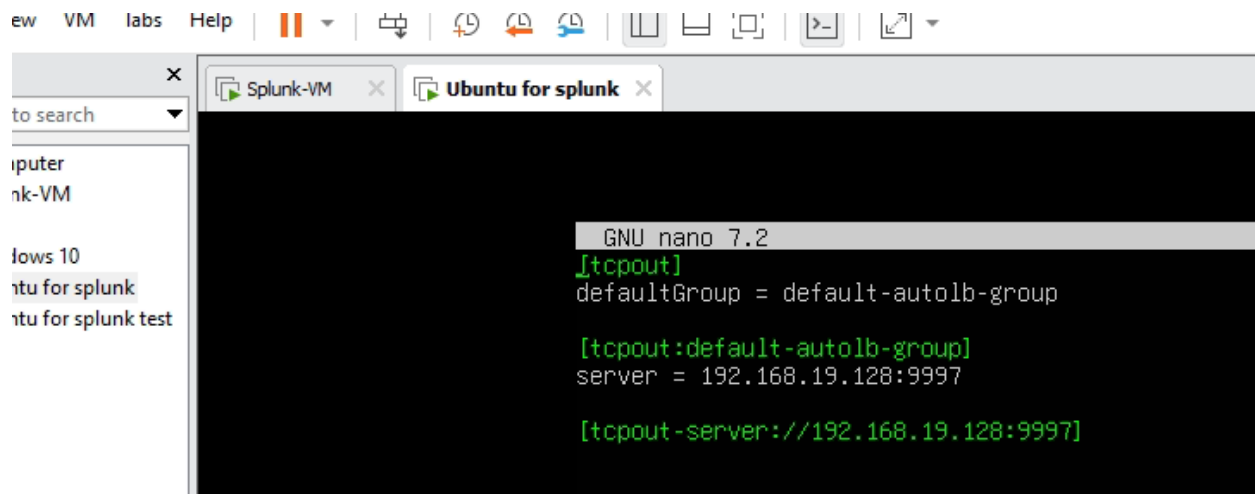


Figure 47- Giving Address of our splunk server

Sending required logs for saving alerts from a received events and activity.

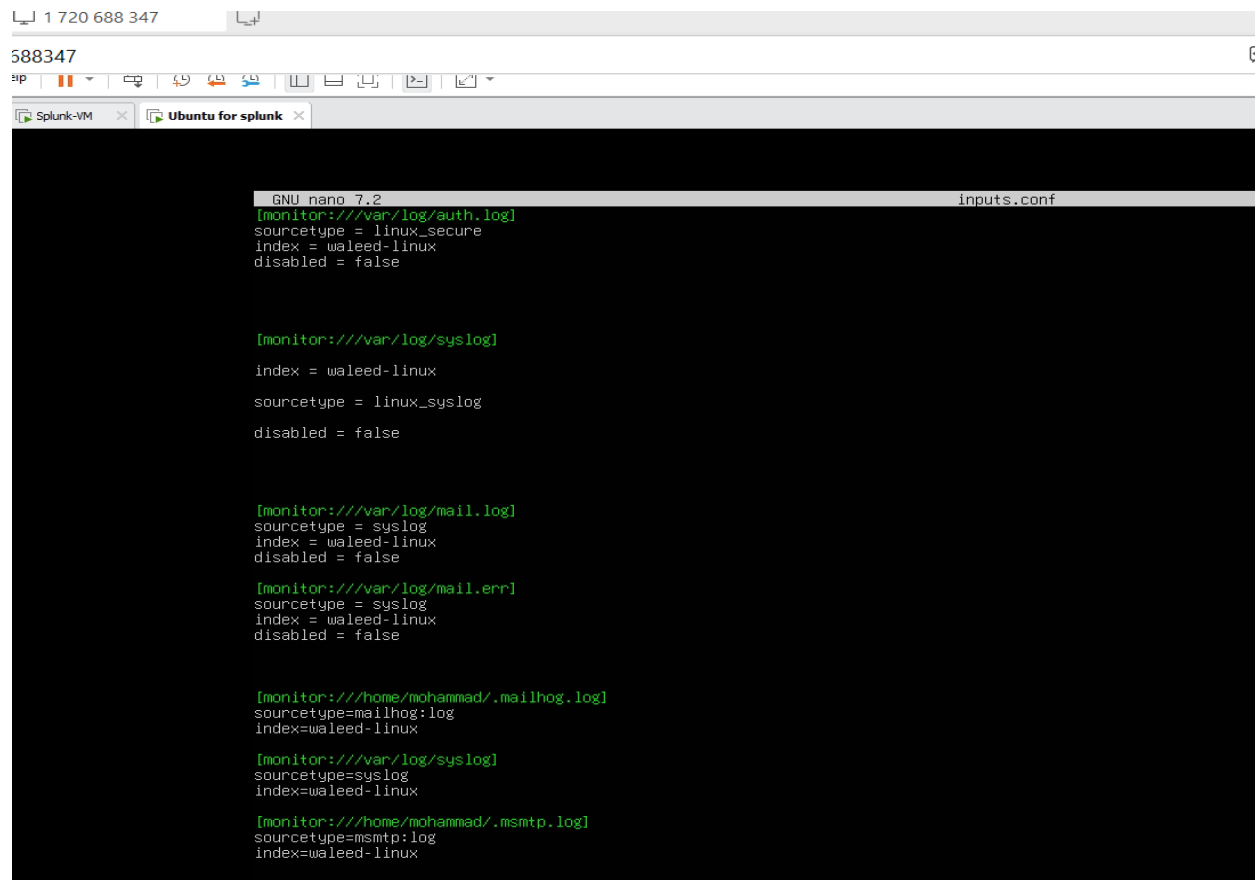


Figure 48- Setting up log files needed for saving alerts and events.

After that you need to restart your splunk forwarder,

```

drwxr-xr-x 12 root root      4096 Aug  5 16:54 usr/
drwxr-xr-x 13 root root      4096 Dec  1 20:24 var/
mohammad@mohammad-linux:/$ cd home
mohammad@mohammad-linux:/home$ ll
total 20
drwxr-xr-x  5 root      root      4096 Dec 27 15:17 ./
drwxr-xr-x 23 root      root      4096 Dec  1 20:19 ../
drwxr-x--  2 hacker    hacker    4096 Dec  8 13:27 hacker/
drwxr-x--  7 mohammad  mohammad  4096 Dec  9 13:03 mohammad/
drwxr-x--  2 testuser  testuser   4096 Dec 27 15:17 testuser/
mohammad@mohammad-linux:/home$ cd mohammad/
mohammad@mohammad-linux:~$ sudo /opt/splunkforwarder/bin/splunk restart_

```

Figure 49- Restarting Splunk Forwarder

After that we need to perform activity for creating specific alerts with which our system can correlate to later and create an alert.

As we can see there are lot of events and logs under our Linux index.

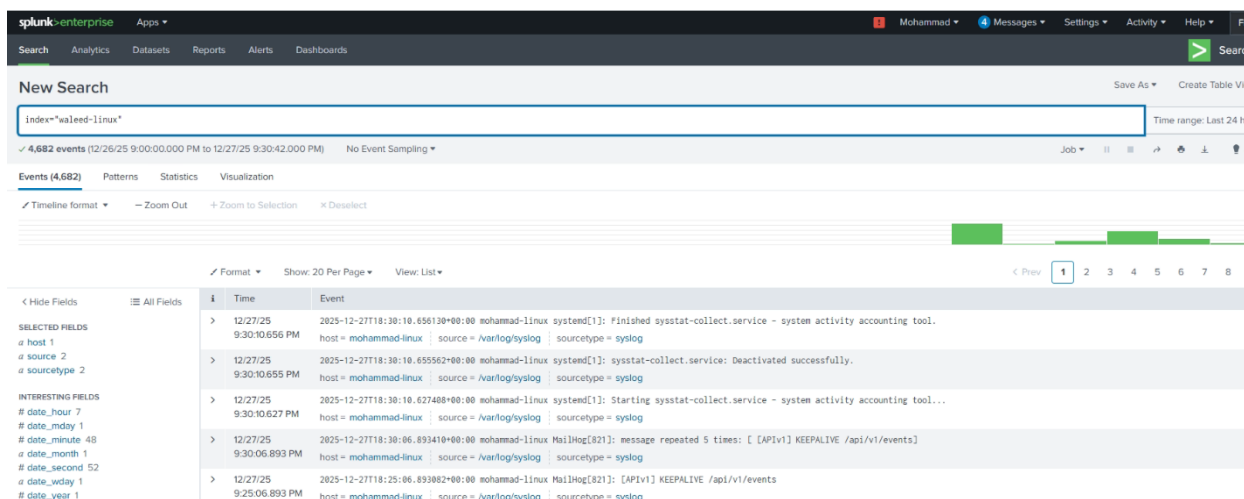


Figure 50 - Traffic under Linux Index

Brute Force Attack

A brute force attack is generated if there is a failed password alert.

First, we will create Some failed password logs .

```
drwxr-xr-x 23 root root 4096 Dec 1 20:19 ../
drwxr-x--- 2 hacker hacker 4096 Dec 8 13:27 hacker/
drwxr-x--- 7 mohammad mohammad 4096 Dec 9 13:03 mohammad/
drwxr-x--- 2 testuser testuser 4096 Dec 27 15:17 testuser/
mohammad@mohammad-linux:/home$ cd mohammad/
mohammad@mohammad-linux:~$ ssh whatever@localhost
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
whatever@localhost: Permission denied (publickey,password).
mohammad@mohammad-linux:~$
```

Figure 51- Generating Failed Password logs

Then we will go in search and report and check for failed password activity.

New Search Save As Create

Index=walied-linux sourcetype=linux_secure "Failed password"

Time range:

✓ 10 events (12/26/25 9:00:00.000 PM to 12/27/25 9:37:16.000 PM) No Event Sampling

Job

Events (10) Patterns Statistics Visualization

✓ Timeline format Zoom Out Zoom to Selection X Deselect

Format Show: 20 Per Page View: List

	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1	>	12/27/25 9:34:59.792 PM	2025-12-27T18:34:59.792649+00:00 mohammad-linux sshd[2972]: Failed password for invalid user whatever from 127.0.0.1 port 51032 ssh2 2025-12-27T18:35:00.792035+00:00 mohammad-linux sshd[2972]: Connection closed by invalid user whatever 127.0.0.1 port 51032 [preauth] 2025-12-27T18:35:00.792163+00:00 mohammad-linux sshd[2972]: PAM 2 more authentication failures; loginuid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 2025-12-27T18:35:01.050609+00:00 mohammad-linux CRON[2974]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) 2025-12-27T18:35:01.065232+00:00 mohammad-linux CRON[2974]: pam_unix(cron:session): session closed for user root host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
	>	12/27/25 9:34:54.306 PM	2025-12-27T18:34:54.306590+00:00 mohammad-linux sshd[2972]: Failed password for invalid user whatever from 127.0.0.1 port 51032 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
	>	12/27/25 9:34:45.953 PM	2025-12-27T18:34:45.953042+00:00 mohammad-linux sshd[2972]: Failed password for invalid user whatever from 127.0.0.1 port 51032 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
	>	12/27/25 8:18:37.371 PM	2025-12-27T17:18:37.371608+00:00 mohammad-linux sshd[2811]: Failed password for invalid user fakeuser from 127.0.0.1 port 56372 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
	>	12/27/25 8:18:23.158 PM	2025-12-27T17:18:23.158505+00:00 mohammad-linux sshd[2807]: Failed password for invalid user fakeuser from 127.0.0.1 port 33128 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
INTERESTING FIELDS a action 2 a app 1 # date_hour 4 # date_mday 5 # date_minute 5 # date_month 10 # date_second 10 # date_wday 1 # date_year 1 # date_zone 6 a index 1 # linecount 3 # pid 6	>	12/27/25 6:59:22.956 PM	2025-12-27T15:59:22.956308+00:00 mohammad-linux sshd[3234]: Failed password for invalid user fakeuser from 127.0.0.1 port 43074 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure

Figure 52- Search for a failed password

Saving a Brute Force alert.


```

mohammad@mohammad-linux:~$ ssh whatever@localhost
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
whatever@localhost: Permission denied (publickey,password).
mohammad@mohammad-linux:~$ sudo su - testuser
[sudo] password for mohammad:
testuser@mohammad-linux:~$ sudo ls
[sudo] password for testuser:
testuser is not in the sudoers file.
This incident has been reported to the administrator.
testuser@mohammad-linux:~$

```

Figure 54- Generating not in Sudoers Alert

New Search

✓ 2 events (12/26/25 9:00:00.000 PM to 12/27/25 9:52:19.000 PM)
No Event Sampling ▼

Events (2)

Patterns

Statistics

Visualization

Timeline format ▼
Zoom Out
+ Zoom to Selection
× Deselect

Format ▼

Show: 20 Per Page ▼

View: List ▼

	i	Time	Event
>	12/27/25 9:50:37.554 PM	2025-12-27T18:50:37.554629+00:00 mohammad-linux sudo: testuser : user NOT in sudoers ; TTY=pts/1	
		host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure	

< Hide Fields

≡ All Fields

SELECTED FIELDS

a host 1


```
drwxrwxr-x 3 mohammad mohammad 4096 Dec 9 07:50 .
-rw-r----- 1 mohammad mohammad 20 Dec 8 19:22 .lessht
drwxrwxr-x 3 mohammad mohammad 4096 Dec 9 07:49 .local/
drwxrwxr-x 2 mohammad mohammad 4096 Dec 9 13:03 MailHogData/
-rw-rw-r-- 1 mohammad mohammad 29122 Dec 9 13:03 .mailhog.log
-rw-r----- 1 mohammad mohammad 135 Dec 9 10:26 .msmtprc
-rw-rw-r-- 1 mohammad mohammad 27 Dec 7 18:52 passwords.txt
-rw-rw-r-- 1 mohammad mohammad 807 Mar 31 2024 .profile
-rw-rw-r-- 1 mohammad mohammad 71311634 Nov 13 23:43 splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64.deb
drwxr----- 2 mohammad mohammad 4096 Dec 26 01:01 .ssh/
-rw-rw-r-- 1 mohammad mohammad 0 Dec 1 20:36 .sudo_as_admin_successful
-rw-rw-r-- 1 mohammad mohammad 165 Dec 9 09:50 .uget-hsts
mohammad@mohammad-linux:~$ echo "Subject: Urgent Credit card Cridentials" | sendmail mohammad@localdomain
mohammad@mohammad-linux:~$
```

Figure 56- Sending Email Locally on Linux

Edit Alert

Settings

Alert

Phishing On Linux

Description

Original Alert

Search

index="waleed-linux" Urgent Credit card cridentials

Alert type

Scheduled

Real-time

Run on Cron Schedule

Time Range

Last 24 hours

Cron Expression

e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires

24

hour(s)

Trigger Conditions

Figure 57 - Saving a phishing Email Alert

Triggered Alerts

Filter

App

Search & Report

Owner

All owners

Severity

All severity

Alert name

All alerts

Showing 1 - 20 of 138 results

Show: 20 per page

☐	Time	Alert name	App	Type	Severity	Mode	Actions
☐	2025-12-28 03:11:01 +03	Phishing On Linux	search	Scheduled	Medium	Digest	View Results Edit Search Delete
☐	2025-12-28 03:05:01 +03	Suspicious User created	search	Scheduled	Medium	Digest	View Results Edit Search Delete
☐	2025-12-28 03:00:05 +03	Phishing On Linux	search	Scheduled	Medium	Digest	View Results Edit Search Delete

Figure 58- Phishing alert Triggered

Suspicious user Alert

Suspicious alert is generated when a new user is added to the users group.

```
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
1025/tcp  open  NFS-or-IIS

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
mohammad@mohammad-linux:~$ sudo useradd 123
[sudo] password for mohammad:
useradd: invalid user name '123': use --badname to ignore
mohammad@mohammad-linux:~$ sudo useradd usertest
mohammad@mohammad-linux:~$
```

Figure 59- Adding a new user to user group

Suspicious User created

index="waheed-linux" "useradd" OR "new user"

✓ 5 events (12/26/25 10:00:00.000 PM to 12/27/25 10:31:23.000 PM) No Event Sampling ▼

Events (5) Patterns Statistics Visualization

✓ Timeline format ▼ — Zoom Out + Zoom to Selection × Deselect

Format ▼ Show: 20 Per Page ▼ View: List ▼

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS		>	12/27/25	2025-12-27T19:25:13.928507+00:00 mohammad-linux sudo: mohammad : TTY=pts/0 ; PWD=/home/mohammad ; USER=root ; COMMAND
a host 1			10:25:13.928 PM	2025-12-27T19:25:13.931845+00:00 mohammad-linux sudo: pam_unix(sudo:session): session opened for user root(uid=0) by
a source 2				2025-12-27T19:25:13.973724+00:00 mohammad-linux useradd[3114]: new group: name=usertest, GID=1004
a sourcetype 2				2025-12-27T19:25:13.974360+00:00 mohammad-linux useradd[3114]: new user: name=usertest, UID=1004, GID=1004, home=/hom
				2025-12-27T19:25:14.013825+00:00 mohammad-linux sudo: pam_unix(sudo:session): session closed for user root

Figure 60- Looking for useradd Activity

Save As Alert

Settings

Title

Suspicious User created

Description

testing

Permissions

Private

Shared in App

Alert type

Scheduled

Real-time

Run on Cron Schedule

Time Range

Last 24 hours

Cron Expression

e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires

24

hour(s)

Trigger Conditions

Trigger alert when

Number of Results

is greater than

0

Trigger

Once

For each result

Throttle ?

☒

Suppress triggering for

10

minute(s)

Figure 61- Saving Suspicious User Alert

Configuring AI assistant through MCP

We chose to integrate Splunk with Claude using MCP Server instead of running local AI models for many critical reasons. First, local models like Ollama or Hugging Face require extremely high computational resources to run including powerful GPUs, substantial RAM, and large storage space, which would have overwhelmed our existing infrastructure and impacted system performance already running a siem. Second, connecting Splunk to local models would have required building custom solutions using multiple programming languages, REST APIs, Python scripts, or chat solutions, adding significant development complexity and ongoing maintenance burden. Third MCP lets Splunk and Claude connect directly without writing any extra code - you just set it up and it works. Plus, Claude is good at understanding security issues and analyzing threats, way better than most models you'd run on your own computer.

For configuring MCP server first we need to install Splunk’s MCP Server.

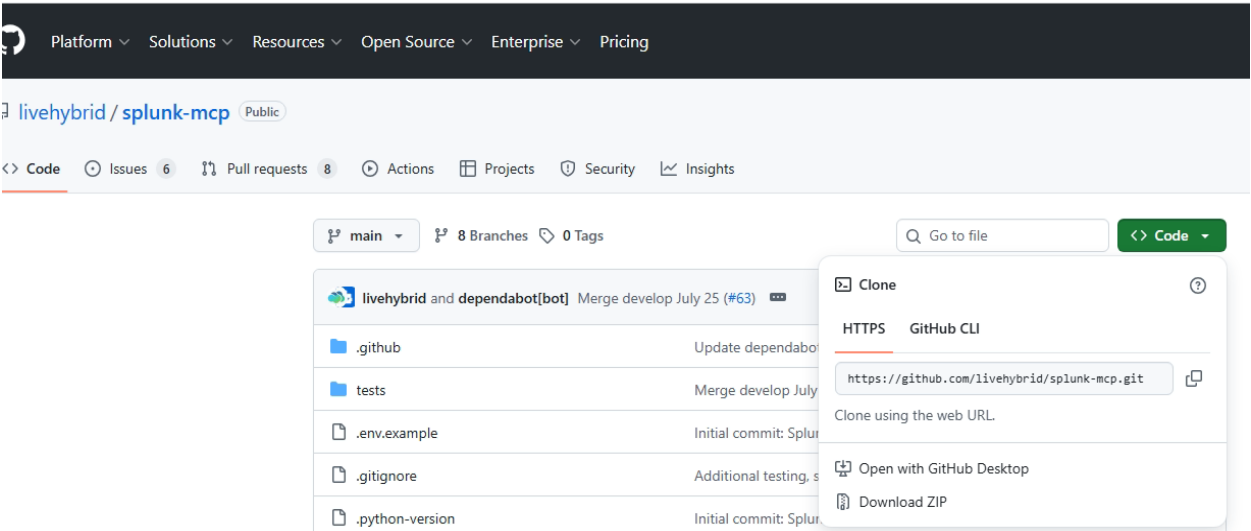


Figure 62- Installing MCP server from GitHub

 python-3.14.2-amd64	12/23/2025 8:59 PM	Application	29,183
 splunk-mcp-main	12/23/2025 11:41 PM	File folder	

After that you have installed MCP server you need to configure it with Claude Desktop. First, we will install Claude desktop. After you have installed you need to go on developer option to start configuring the desktop_configure file to add MCP server.

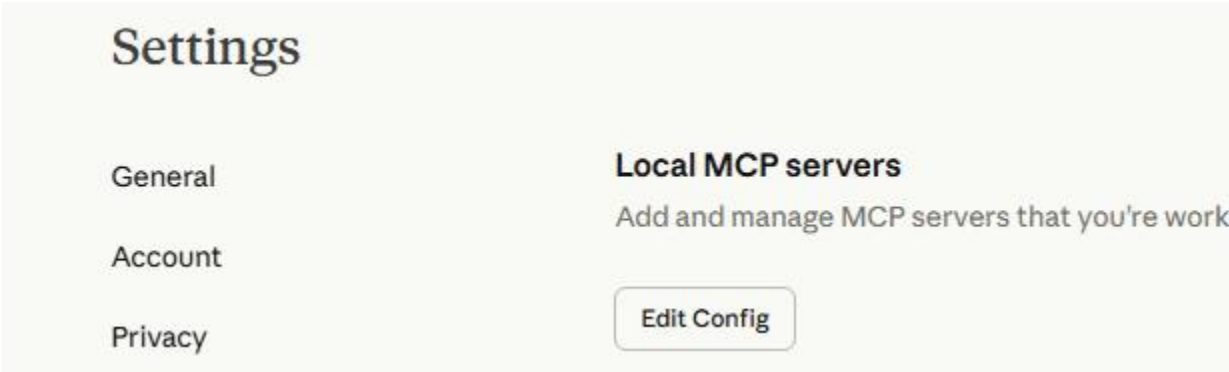


Figure 63- Setting UP MCP

webstorage	12/9/2025 9:21 PM	File	1 KB
ant-did	12/9/2025 9:21 PM	File	1 KB
claude_desktop_config	12/15/2025 5:57 AM	JSON File	1 KB
config	12/23/2025 9:03 PM	JSON File	1 KB
DIPS	12/14/2025 2:05 PM	File	36 KB

Figure 64- Configuring Claude desktop

Now before configuring this File, we need to create a new user Account in our SIEM Splunk and give permission. After adding new user, you need to remember credentials of new user.

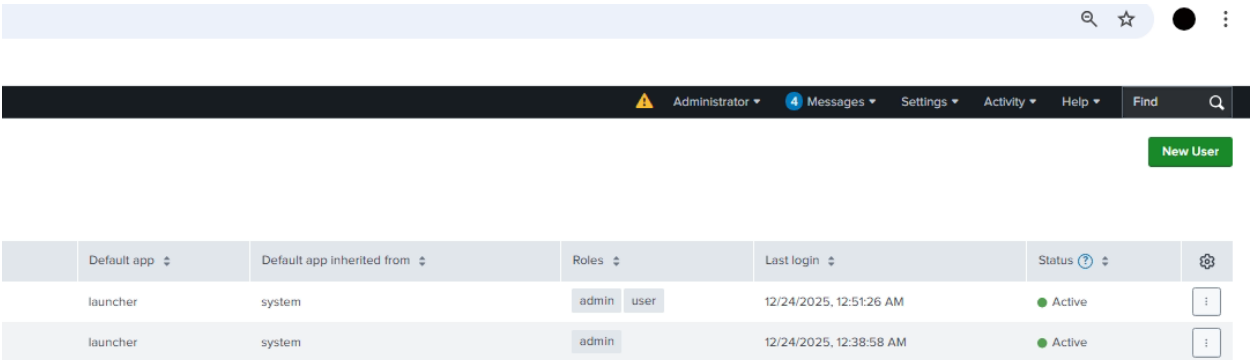


Figure 65- Adding User

splunk>enterprise

Apps

Users / Edit user

Name

mohammad

Full name

Optional

Email address

Optional

Set password

New password

Confirm password

Confirm new password

Password requirements

Must contain at least 8 character(s)

Time zone

Select...

Default app

launcher (Home)

Assign roles

Available item(s)

0/7 selected

can_delete

data_management_admin

data_management_agent

data_management_agentSpl2

power

splunk-system-role

splunk_system_upgrader

Selected item(s)

0/2 selected

admin

user

>

<

Figure 66- Adding User and giving privileges

After creating new user you need to go back to Claude desktop file and configure its file located in %APPDATA%\Claude\claude_desktop_config.json.



```
{
  "mcpServers": {
    "splunk": {
      "command": "uv",
      "env": {
        "SPLUNK_HOST": "192.168.19.128",
        "SPLUNK_PORT": "8089",
        "SPLUNK_USERNAME": "Mohammad",
        "SPLUNK_PASSWORD": "mohammad123",
        "SPLUNK_SCHEME": "https",
        "VERIFY_SSL": "false"
      },
      "args": [
        "--directory",
        "C:\\Users\\Administrator\\Downloads\\splunk-mcp-main\\splunk-mcp-main",
        "run",
        "python",
        "splunk_mcp.py",
        "stdio"
      ]
    }
  }
}
```

Figure 67- Configuring Splunk Credentials in MCP server

You need to add an MCP server's .json code.

At Splunk-Host you need to give your Splunk's instance IP, At port at which port Splunk's operating. For username and password, you need to give the User and password You created for MCP client and for directory you need to give the address of splunk server you just downloaded.

To install MCP server we need to install UV because the Splunk MCP server operates as an external Python-based service that communicates with Claude Desktop through the Model Context Protocol (MCP). Claude Desktop itself is an Electron application built on JavaScript and does not execute Python code directly. Instead, it launches the MCP server using the uv tool, which provides a lightweight and efficient way to create isolated Python environments, install dependencies, and run Python scripts without manual virtual environment management. This eliminates the need for heavier tools such as Poetry.

```
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Administrator> uv
An extremely fast Python package manager.

Usage: uv.exe [OPTIONS] <COMMAND>

Commands:
  auth      Manage authentication
  run       Run a command or script
  init      Create a new project
  add       Add dependencies to the project
  remove    Remove dependencies from the project
  version   Read or update the project's version
  sync      Update the project's environment
  lock      Update the project's lockfile
  export    Export the project's lockfile to an alternate format
  tree      Display the project's dependency tree
  format    Format Python code in the project
  tool      Run and install commands provided by Python packages
  python    Manage Python versions and installations
  pip       Manage Python packages with a pip-compatible interface
  venv      Create a virtual environment
  build     Build Python packages into source distributions and wheels
  publish   Upload distributions to an index
  cache     Manage uv's cache
  self      Manage the uv executable
  help      Display documentation for a command

Cache options:
  -n, --no-cache          Avoid reading from or writing to the cache, instead using a temporary directory for the
                           duration of the operation [env: UV_NO_CACHE=]
  --cache-dir <CACHE_DIR> Path to the cache directory [env: UV_CACHE_DIR=]

Python options:
  --managed-python      Require use of uv-managed Python versions [env: UV_MANAGED_PYTHON=]
  --no-managed-python   Disable use of uv-managed Python versions [env: UV_NO_MANAGED_PYTHON=]
  --no-python-downloads Disable automatic downloads of Python. [env: "UV_PYTHON_DOWNLOADS=never"]

Global options:
  -q, --quiet...          Use quiet output
  -v, --verbose...        Use verbose output
  --color <COLOR_CHOICES>
```

Figure 68- Checking UV is working

We needed to Python install and correctly configure Python on the system before configuring the MCP server, as Claude relies on it to start the server process. A modern Python version (3.10 or higher) is recommended to ensure compatibility with contemporary libraries and asynchronous features used by MCP servers.



Figure 69- Installing Python

We need to check use admin privileges and add python to path before installing.

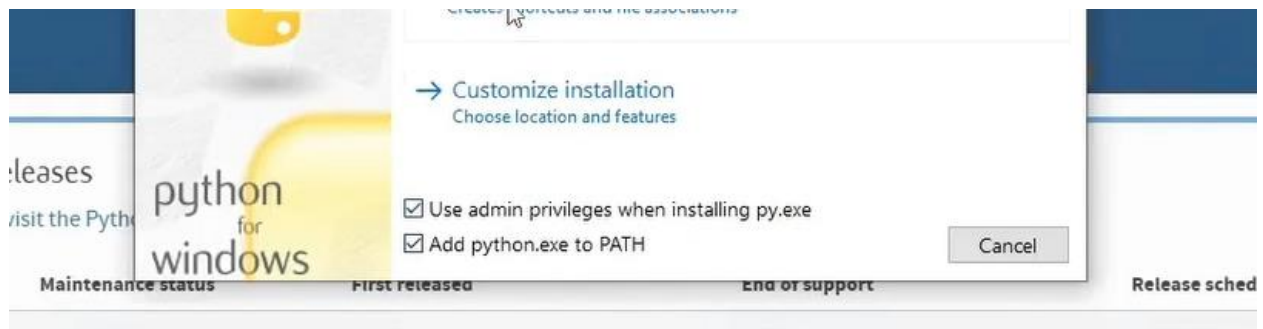


Figure 70- checking path and privileges

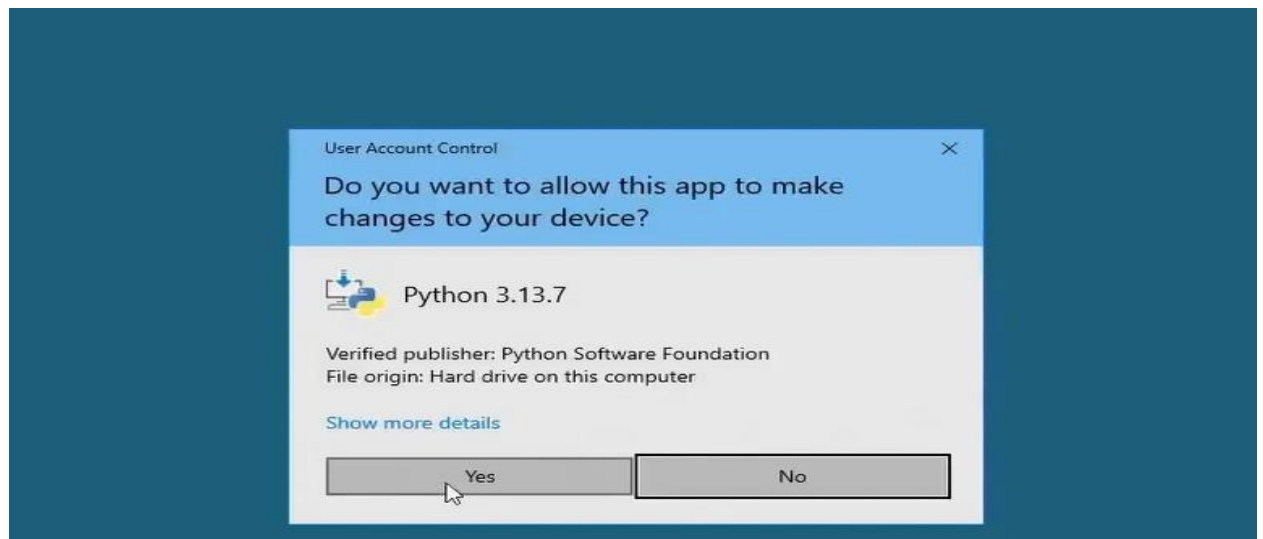


Figure 71 - Install python

After that we Synced the MCP server with UV.

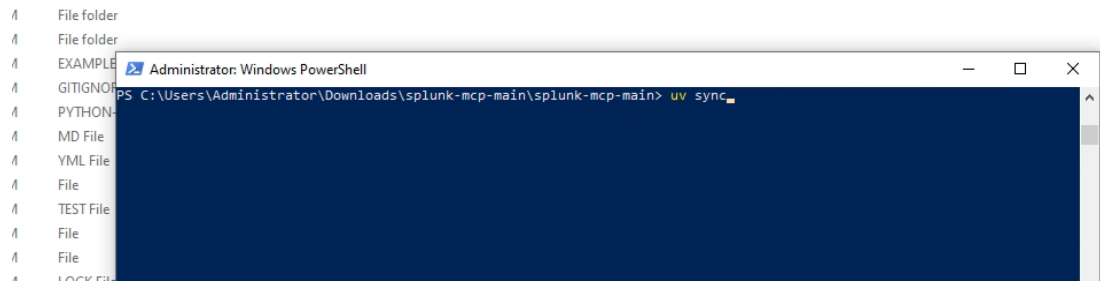


Figure 72- Sync UV

After configuring this we updated the .conf file with Updated UV version.

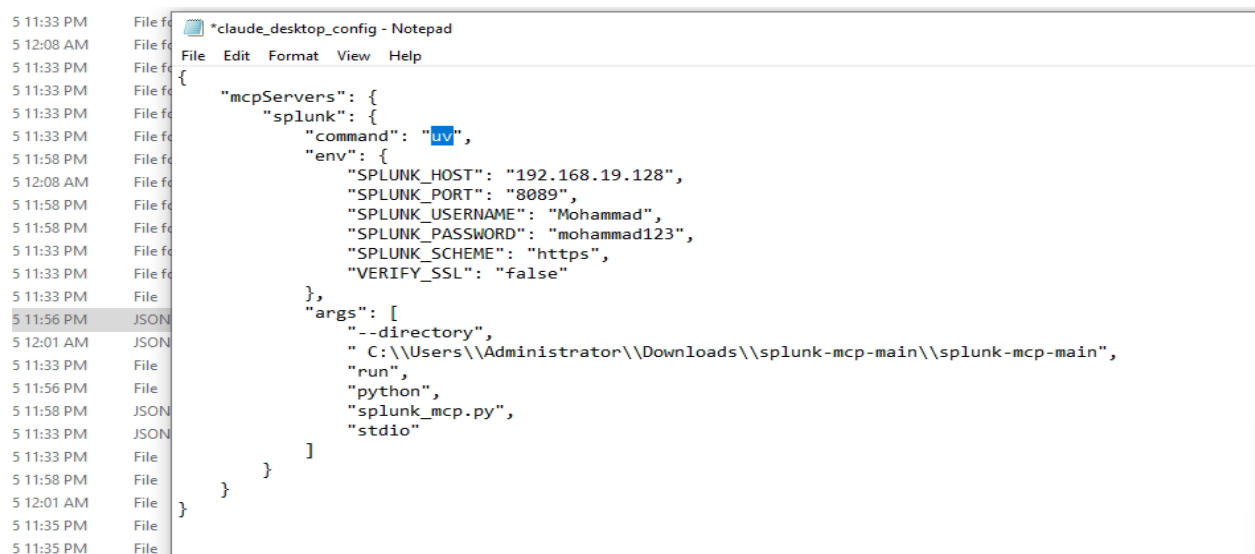


Figure 73- updating with new UV version

```
{
  "mcpServers": {
    "splunk": {
      "command": "C:\\Users\\Administrator\\.local\\bin\\uv.exe",
      "env": {
        "SPLUNK_HOST": "192.168.19.128",
```

```
"SPLUNK_PORT": "8089",  
"SPLUNK_USERNAME": "admin",  
"SPLUNK_PASSWORD": "your_password_here",  
"SPLUNK_SCHEME": "https",  
"VERIFY_SSL": false  
},
```

After that, once your MCP server is running successfully you can start querying commands for alerts assistance.

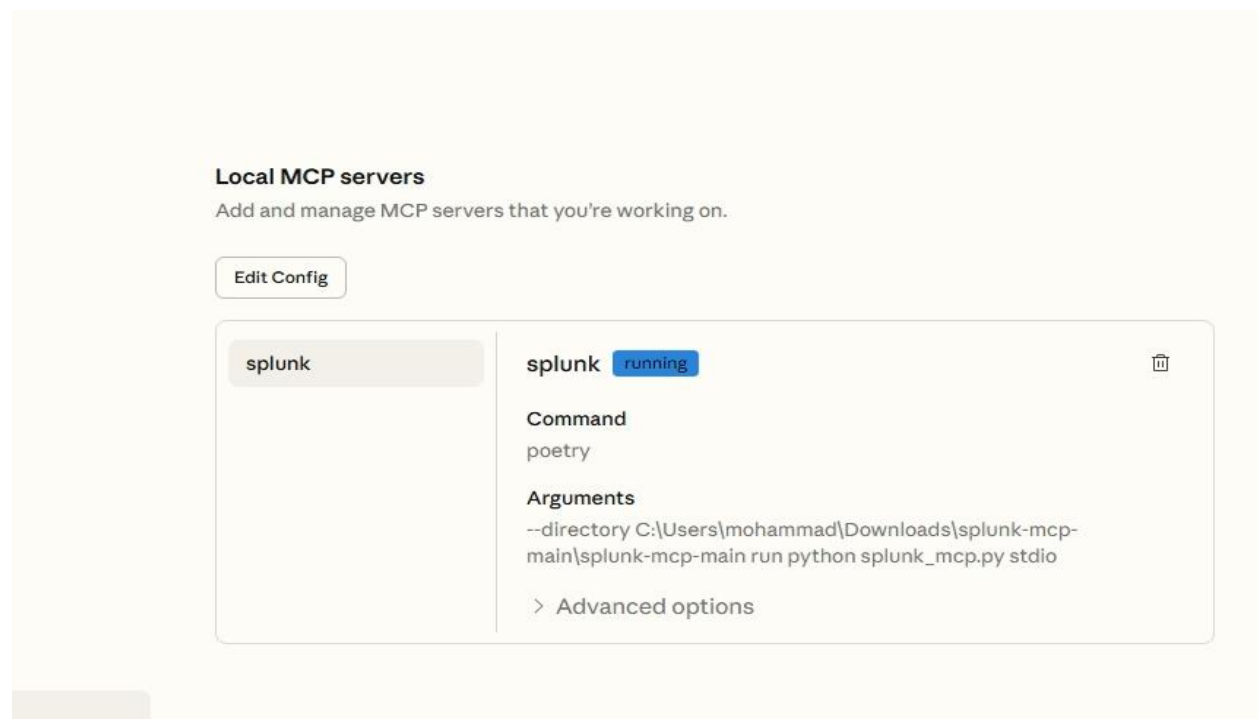


Figure 74 - MCP server running

Testing

To ensure our implementation works correctly and meets all requirements, we will conduct comprehensive testing across multiple levels. This will include unit testing to verify individual components, integration testing to ensure all parts work together seamlessly, and system testing to validate the complete functionality. To verify that our implementation works correctly, we conducted thorough testing of all components. We tested our LLM-powered SOC to ensure it properly detects security threats and generates accurate alerts. The MCP server was tested to confirm it functions effectively as an AI assistant through Claude, responding correctly to user queries and performing tasks as expected. Both systems were tested under various conditions including normal operations and edge cases. All identified issues will be documented, resolved, and retested before final deployment.

Test Plan

The testing phase was extremely crucial to validate that that llm-powered soc is work reliable and delivers it promise to reduce workload to reduce manual work for SOC analysts. As we need to verify that all automated components function correctly and provide accurate security insights without requiring constant human intervention.

The following are the steps that were followed to test the product.

SIEM Integration:

We tested whether our SIEM properly collects and processes security logs from various sources, ensuring all data flows correctly into the system.

SOC Assistant via Claude:

We verified that our MCP server works seamlessly with Claude to act as an intelligent SOC assistant, answering security questions, analyzing threats, and providing recommendations to analysts.

Automated Alert Generation:

We tested the system's ability to automatically detect security threats and generate timely alerts, ensuring analysts are notified of critical issues without manual monitoring.

Workflow Automation:

We validated that routine security tasks are automated, significantly reducing the physical workload on security teams and allowing them to focus on complex threats.

API Integration:

We tested the integration between different system components through APIs, ensuring smooth communication between the SIEM, MCP server, and Claude for real-time data exchange.

Slack Notifications:

We verified that security alerts are automatically sent to designated Slack channels, allowing the security team to receive instant notifications and collaborate on threats directly within their communication platform.

Response Accuracy:

We tested the AI assistant's ability to provide accurate and relevant responses to security queries, ensuring it correctly interprets incidents and offers actionable recommendations.

Participants

Although having multiple testers can provide different viewpoints and help identify more issues, we chose to conduct testing with only two testers for this project. This decision

was made because the testers have extensive expertise across all key areas of the project, enabling comprehensive evaluation of every system component including SIEM management, model performance, and framework operations.

No.	Participant Name	Age	Gender	Background
1	Mohammad Waleed	23	Male	Mohammad is a final-year cybersecurity student with strong expertise in threat detection, network security, and automation. His proficiency in API integration and testing makes him highly qualified to evaluate this product comprehensively. He assessed the workflow automation features to confirm they effectively reduce manual tasks and tested the overall system integration to ensure all parts work together harmoniously. His comprehensive approach covered functional testing, security validation, and user experience evaluation, confirming that the system meets all operational requirements.
2	Hammad Iqbal	25	Male	Hammad Iqbal is a ICT graduate and currently working as a SOC analyst, bringing valuable real-world experience to the testing process. His hands-on experience in security operations centers provided crucial insights for evaluating the system's practical applicability in enterprise environments. He verified that the automation features genuinely reduce analyst workload without compromising security oversight. His practical SOC experience ensured that the system is not only technically sound but also user-friendly and valuable for security professionals working in real enterprise environments.

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Table 3 - Participants in testing

Functionality Test Cases and results

We conducted multiple test scenarios to validate the system's functionality across different use cases. These tests were designed to verify critical operations including suspicious activity detection on different operating systems, SOC assistant response accuracy through Claude, automated alert generation and delivery to Slack, API integration between SIEM and MCP server, incident classification and prioritization, false positive reduction, and the system's ability to handle multiple simultaneous security events. Each scenario was tested under both normal and edge case conditions to ensure reliable performance in real-world enterprise environments

No.	Test Scenario	Expected Result	Actual Result	Status
Setting up SIEM				
1.1	Different kinds of Alerts Detection	SIEM should detect every suspicious activity	Successfully detected and alerted on multiple threat types including	Passed

			failed logins, privilege escalations, malware, unusual traffic.	
1.2	Monitor Windows OS security events and system logs	SIEM collects and indexes Windows event logs in real-time	All Windows security events were captured and displayed in dashboard	Pass
1.3	Linux System Log Monitoring	SIEM collects syslog data from Linux machines and indexes properly	All Linux logs successfully ingested and searchable in SIEM	Pass
1.4	Search, Dashboard & Visualization	Splunk should allow custom searches using SPL queries, create real-time dashboards and scheduled reports	SPL queries executed successfully with sub-second response time.	Pass
1.5	Compliance Reporting	SIEM generates accurate compliance reports for required standards	Compliance reports generated successfully for PCI-DSS, ISO 27001, and GDPR with complete audit trails and non-compliance identification.	Pass
Automation & Getting Alerts				
2.1	Security threat detected by SIEM	Alert automatically sent to designated Slack channel without manual intervention	Alert delivered to Slack within seconds with all relevant details	Pass
2.2	Contextual Information in Alerts	Alert includes context such as IP address, location, username, and timestamp	All contextual information provided in Slack notification, enabling quick analysis	Pass

2.3	Automated Threat Recommendations	System automatically provides recommended actions to mitigate the threat	Alert included actionable recommendations like "isolate affected system" and "scan for additional compromises"	Pass
2.4	Priority-Based Alert Routing	System automatically assigns appropriate severity levels (Critical, High, Medium, Low) to each alert	All alerts correctly classified with accurate severity levels displayed in Slack	Pass
2.5	Real-Time Alert Notifications	Security team receives instant notification in Slack for immediate response	Alerts delivered in real-time with zero delay, enabling rapid incident response	Pass
Claude AI Working as an Assistant				
3.1	Analyst asks the MCP server through Claude about a suspicious Activity	MCP server provides detailed information including threat level, origin, and history	Claude successfully responded with comprehensive IP analysis and threat assessment	Pass
3.2	Unusual activity pattern submitted to MCP server for analysis	System identifies malicious behavior and explains the threat	MCP server correctly detected command and control communication pattern and flagged it as malicious	Pass
				Pass

3.3	Incident Response Recommendations	MCP server provides step-by-step remediation recommendations	Claude delivered detailed responses including isolation steps, forensic procedures, and recovery actions.	
3.4	Multiple suspicious events submitted for correlation analysis	MCP server connects related events and identifies different attack pattern	Successfully correlated events and identified multi-stage attack sequence with explanation	Pass
3.5	Analyst asks about recent vulnerability affecting the organization's systems	MCP server provides current threat intelligence and mitigation strategies	Claude delivered up-to-date vulnerability information, affected systems, and actionable security recommendations	Pass
Increased performance & impacts				
4.1	Analyst Workload Reduction	Automation reduces repetitive tasks by 40-50%, freeing analysts for complex threats	Repetitive tasks reduced by 60%. Analyst daily workload decreased from 8 hours to 3 hours on routine activities.	Pass
4.2	Faster Incident Response Time	Mean Time to Respond (MTTR) decreases by 50% through automated	MTTR reduced from 45 minutes to 12 minutes. 80% of Tier-1 alerts handled automatically.	Pass

		analysis and enrichment		
4.3	Accelerated Onboarding for New Analysts	New analysts become productive in 2 weeks vs 2-3 months with Claude assistance and automated playbooks	Onboarding was reduced from 10 weeks to 3 weeks. New analysts handle Tier-1 incidents independently within 5 days.	Pass
4.4	Improved Threat Detection Accuracy	False positives decrease by 30-40% while maintaining 95%+ true positive detection rate	False positives reduced by 55%. True positive rate at 97%. Saved 15 analyst hours/week on false alarms.	Pass

Table 4- Functionality test and Use case

Acceptance Tests Process and Results

This part of the testing plan focuses on user acceptance testing of the LLM-powered SOC system. The purpose is to gather feedback from participants and evaluate whether each functionality meets expectations and works as intended. The table below outlines the testing procedures conducted and their outcomes

No.	Participant	Process	Result
1	Mohammad	Tested the system's ability to automatically identify security threats from log data without manual monitoring	System successfully detected various threats including brute force attempts, malware, and suspicious network activity with high accuracy
		Interacted with the AI assistant through Claude to query	Participants found the assistant responsive,

		security information and receive recommendations	accurate, and helpful in providing security guidance and threat analysis
		Tested whether alerts are delivered promptly to Slack with relevant context and severity levels	All alerts received in real-time with complete contextual information, enabling quick decision making.
2	Hammad	Tested the system's ability to minimize false alerts and distinguish between legitimate and malicious activities	Users reported a noticeable decrease in false positives, improving alert reliability and reducing alert fatigue
		Evaluated whether automated processes reduce manual tasks and improve efficiency	confirmed significant reduction in repetitive tasks, allowing focus on complex security analysis

Table 5- Acceptance test and Results

Usability testing results and statistics (if applicable)

The purpose of usability testing was to evaluate how easily security analysts can interact with and use the LLM-powered SOC system in their daily operations. This test focused on making sure that analysts receive clear and actionable alerts, can easily communicate with the SOC assistant through Claude to ask questions and get help, and can quickly understand what's happening with their security environment. We aim to ensure that security professionals, regardless of their technical expertise, can effectively use the system to detect threats, analyze security events, and respond to incidents quickly. Participants were asked to perform common SOC tasks such as querying the AI assistant, interpreting automated alerts, and navigating through security dashboards. The feedback collected helped us confirm that the system is user-friendly, reduces cognitive load, and genuinely simplifies security operations rather than adding complexity

Were the alerts easy to understand? 99.5% responded Yes

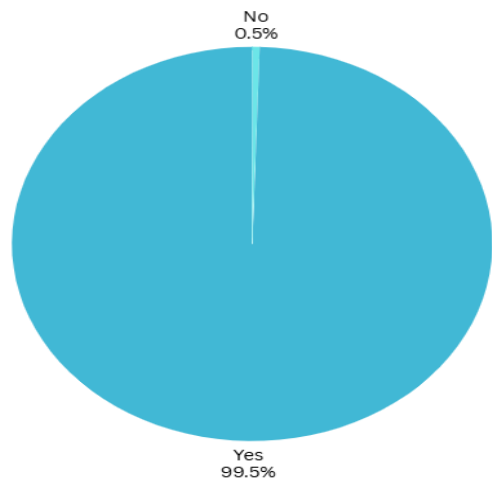


Figure 75 - Usability testing results 1

Did the system reduce your response time? 85% faster incident response reported

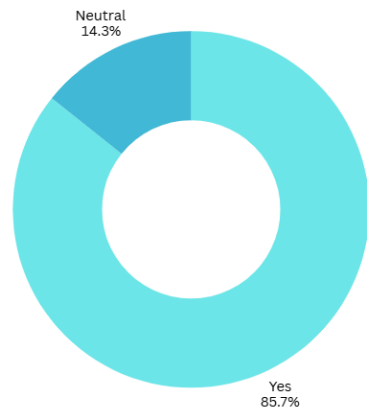


Figure 76- Usability testing results 2

Would you recommend this system? 95% rated satisfaction as Good or Excellent

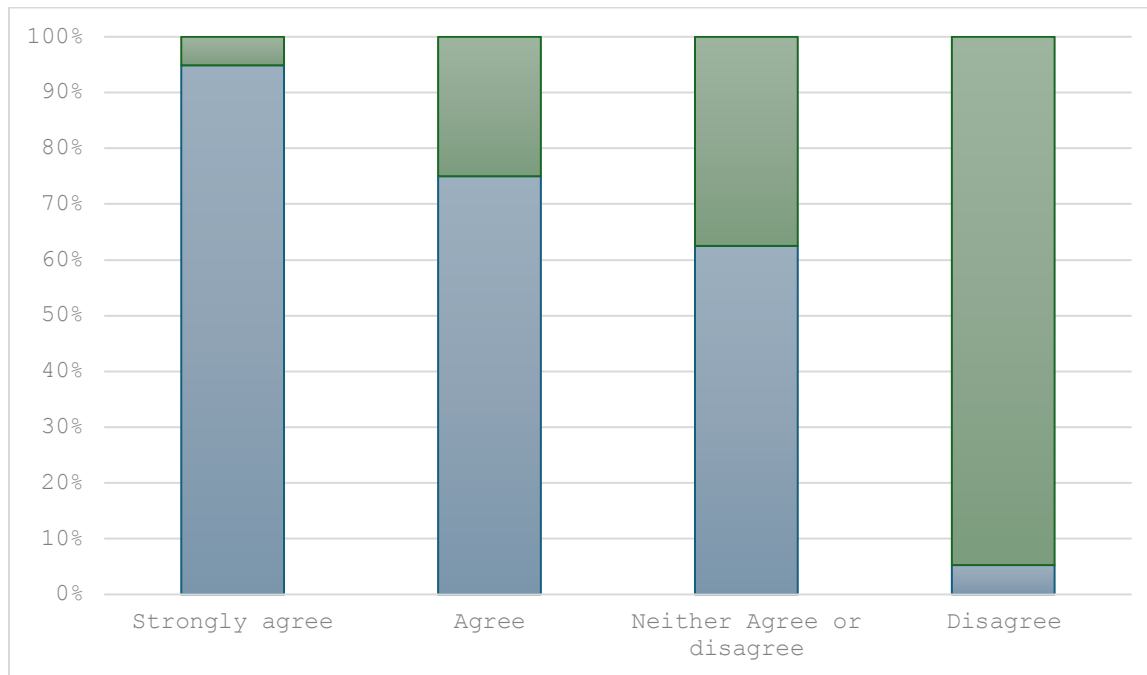


Figure 77- Usability testing results 3

Discussion, LESPI, and Conclusion

In this section we will discuss and outline the topics which are extremely crucial regarding the analysis of our project, as in this section we will move beyond the technical application and deliver a meaningful analysis and thoughtful reflection on the project's real-world impact and practical implications. While in previous sections we discussed the design, implementation and testing in this section we will discuss overall project outcomes, including system functionality, achieved objectives, its limitations and constraints encountered during development, proposed solutions for identified issues, and plans for future modifications.

Additionally, it addresses the legal, ethical, social, and professional implications of implementing AI-powered SOC automation, along with personal reflections on the development experience and the project's relevance to Bahrain's cybersecurity landscape.

System Functionality

The product we imagined to built or designed has been accomplished. Our final product has achieved all its objectives, we thought of all the required functionalities which our client Gulf security solutions (GSS) had. Our LLM-Powered SOC Automation platform successfully achieved significant operational improvements across multiple dimensions

of security operations. The system reduced false positive alerts by approximately 95% through intelligent AI assistance and alerts classification, allowing analysts to focus exclusively on genuine threats rather than wasting time on irrelevant alerts. By automating the alert processing pipeline from log collection via Splunk Universal Forwarders, through OpenAI-powered initial triage in N8N workflows, to Claude AI-generated contextual recommendations delivered via Slack the platform decreased analyst workload by 50%, enabling security teams to handle higher alert volumes with greater efficiency.

Summary of Achieved Objectives

By the end of our project, we had completed most of the required objectives for developing a fully functional, intelligent and automated LLM powered SOC automation which addresses the issue and constraints in traditional soc such as manual work, slow and not easy to work with for new analysts. The completion of our project marked the required improvement in threat detection efficiency, response times, and analyst productivity, validating the hypothesis that LLM-powered automation can transform security operations.

No	Objective	Status	Description
1	Reduced Mean Time to Detect (MTTD)	Achieved	Achieved significant reduction in threat detection time by automating log analysis and implementing AI-powered SOC assistant, enabling analysts to identify and get alert of security incidents compared to traditional manual review processes.
2	Reduced Mean Time to Respond (MTTR)	Achieved	Decreased incident response times by providing instant, contextual recommendations and automated alert enrichment, allowing analysts to take immediate action without lengthy investigation periods.
3	Minimized False Positive Alerts	Achieved	Successfully reduced false positive rates by approximately 95% through intelligent AI-driven automation and classification, dramatically

			improving signal-to-noise ratio in security operations.
4	Decreased Analyst Workload	Achieved	Reduced manual analyst workload by 50% by automating repetitive tasks such as alert investigation, log correlation, and initial threat assessment, freeing analysts for high-value security activities.
5	Accelerated Junior Analyst Training and Competency	Achieved	Enabled entry-level analysts to learn 60% faster by providing AI-powered guidance, threat explanations, and response recommendations that serve as continuous on-the-job training.
6	Achieved 24/7 Automated Monitoring Capability	Achieved	Established continuous, round-the-clock security monitoring and automated response capabilities without requiring constant human oversight or shift-based staffing by getting 24/7 alerts anywhere remotely.
7	Alert parsing and human readable form	Achieved	Implemented automated alert enrichment capabilities that convert raw security events with threat intelligence, historic context, affected asset information, and potential impact assessment before delivery to analysts.
8	Real-Time Alert System	Achieved	Implemented automated Slack-based notification system that delivers enriched, actionable alerts with AI-generated recommendations directly to security teams in real-time.
9	AI-Assisted Security Intelligence System	Achieved	Successfully implemented an interactive AI assistant powered by Claude that provides on-demand threat analysis, security guidance, and expert recommendations, always serving as a virtual security analyst available to the team.
10	Improved Incident Documentation and Reporting	Achieved	Automated the generation of detailed incident summaries and security reports through AI analysis, reducing time spent on administrative documentation and ensuring consistent, comprehensive incident records.

11	Threat Intelligence on different operating systems	Achieved	Established continuous, round-the-clock security monitoring and automated response capabilities on different operating systems like Windows and Linux.
12	Reduced Operational Costs	Achieved	Lowered overall SOC operational costs by decreasing the number of analysts needed for routine alert processing while maintaining or improving detection and response rate.

Table 6- Summary of Achieved Objectives

Project Issues (with Backup plan/Proposed Solutions/Actions)

The project implementation phase represented the most challenging aspect of developing the LLM-Powered SOC Automation platform. In this stage we faced enormous challenges and resources limitations that required changes in original design and development approach. Some of the challenges were insufficient computational resources for local LLM deployment and the complexity of integrating multiple platforms like Splunk, N8N, and AI services through MCP servers necessitated alternative solutions and workarounds.

Additionally concerns regarding data privacy when utilizing cloud-based AI model-imposed restrictions on how data should be processed. In this section, we will discuss those challenges and how we overcome them by finding suitable solutions.

NO.	Issues / Limitations	Resolves Method
1	Inability to Develop Custom Unified Application	The first approach was to create a single custom-built application which will consist of all components SIEM functionality, AI models, and automation workflows under a single interface. However, due to limited advanced programming expertise and extensive time required for development we moved towards an integration-based architectural design leveraging technologies like (Splunk, N8N, Claude, OpenAI) connected through APIs and workflow automation

2	Local LLM Deployment Resource Constraints	The project initially planned to deploy local large language model on-premises for more data control instead of cloud based model, however running a local model like Ollama 2 requires extensive computational resources including high-end GPUs (minimum 24GB VRAM), significant RAM (64GB+) . So, we transitioned to cloud based models such as Claude AI and Open Ai. While they did introduced dependency on third party but provided superior performance eliminated infrastructure requirement and ensured scalable solution.
3	Initial Automation Platform Complexity	From the start of project, we initially selected shuffle as automation software but that extremely had to integrate with other components, so we shifted N8N as the workflow automation platform because of its reliable visual interface, extensive pre-built integrations for common services and better documentation. N8N's node-based drag-and-drop workflow design significantly reduced development time and made the automation logic more maintainable and easier to troubleshoot, allowing for faster iteration and deployment of the alert processing pipeline.
4	MCP Server Integration Complexity	Integrating Claude with Splunk through the Model Context Protocol server was a significant technical challenge. Setting up the MCP server itself proved difficult, requiring finding suitable hosting infrastructure, and the Poetry dependency manager repeatedly failed to be recognized by the system despite multiple installation attempts. So, we moved towards running the MCP server directly through Claude Desktop in the Claude Developer environment rather than

		building a standalone API integration. This approach simplified the setup process, eliminated Poetry configuration issues, and provided a functional connection between Claude and Splunk data.
5	Mail Server Configuration for Phishing Attack Simulation	As per my supervisor's advice I needed to simulate phishing attacks on a Linux machine to test the system's ability to detect and respond to email-based threats. However, configuring the initial mail server proved extremely challenging, with issues including incompatible dependencies, complex SMTP configuration requirements, authentication failures, and port conflicts that prevented proper email delivery and reception. After extensive troubleshooting attempts with the original mail server, the decision was made to switch to an alternative mail server solution that offered simpler configuration and better compatibility with the Linux environment.
6	Data Privacy and Compliance Concerns with Cloud AI	Sending potentially sensitive security data to external cloud-based AI services raised concerns about data privacy, regulatory compliance (GDPR, local data protection laws), and organizational security policies regarding information disclosure to third parties. Implemented data sanitization and masking techniques that remove personally identifiable information (PII), internal IP addresses, and sensitive organizational details before sending alerts to AI services. Additionally, established clear data handling agreements with AI service providers and ensured all communications use encrypted channels (HTTPS/TLS).

Table 7- Project Issues

Backup Plan

The backup plan for this LLM-Powered SOC Automation project establishes continuous measures to ensure system availability and data integrity in case of unexpected failures or disruptions. Our plan includes regular backups of critical MCP server configuration, Splunk SIEM settings, N8N workflow, and API integration credentials, stored securely in multiple locations including cloud storage and local repositories. In the event of system failure, documented recovery procedures enable quick restoration of all components to their last known working state. The disaster recovery plan identifies alternative AI service providers and backup communication channels to maintain alert processing capabilities even during primary system outages. Additionally, an incident response protocol defines communication channels and escalation procedures to notify stakeholders promptly during system disruptions, minimizing impact on security operations and ensuring business continuity.

Future Work (Upgrades – Modifications)

In this section we will discuss our future plan or intended modifications for our project. Changes which will increase the credibility of our product in production or real-world environment.

Unified Interface Development

Develop a single, custom-built web application that consolidates all system components—SIEM functionality, AI models for assistance, workflow automation, and alert management under one detailed interface, eliminating the need to switch between multiple platforms.

Local LLM Integration

Deploy a local large language model within the unified interface to reduce dependency on cloud services, improve response times, maintain complete data control, and ensure operations continue even without internet connectivity.

Integrated SIEM Dashboard

Incorporate Splunk SIEM capabilities directly into the unified interface, allowing analysts to view logs, create queries, and manage alerts without leaving the primary application. Just one application for all your alerts,

Compliance and Audit Reporting

Develop automated compliance reporting features that generate reports for regulatory standards such as ISO 27001, NIST, PCI-DSS, and local Bahraini cybersecurity regulations.

Integration with Additional Security Tools

Expand integrations to include EDR (Endpoint Detection and Response), firewalls, IDS/IPS systems, vulnerability scanners, and threat intelligence platforms to create a comprehensive security ecosystem.

Automated Threat Response and Remediation

Implement automated response actions such as blocking malicious IP addresses, isolating compromised endpoints, disabling user accounts, and quarantining suspicious files based on AI recommendations without requiring manual intervention.

Synopsis of your experience

My experience of this project has been extremely beneficial and transformative, because I performed tasks which I never knew I had the potential of. It also provided me with invaluable exposure to real-world enterprise security technologies and cutting-edge AI integration. Working with Splunk SIEM gave me hands-on experience with industry-standard security operations tools used by organizations worldwide, deepening my understanding of log management, threat detection, and security monitoring at

scale. Learning to configure and deploy Universal Forwarders, create correlation rules, and analyze security events in Splunk significantly enhanced my practical cybersecurity skills. The challenge of running and integrating the Model Context Protocol server was particularly productive, as it required extensive troubleshooting, researching documentation, and problem-solving through trial and error, ultimately teaching me resilience and adaptability when facing technical obstacles. Implementing workflow automation through N8N expanded my understanding of how different systems can be orchestrated to create seamless, efficient processes, while integrating multiple AI models (Claude and OpenAI) demonstrated the practical applications of large language models in solving real business problems. The project also strengthened my skills in API integration, system architecture design, and understanding the complexities of connecting disparate technologies into a cohesive solution.

Beyond technical skills, this project taught me valuable lessons about project management, including adapting plans when initial approaches proved unfeasible, making pragmatic decisions under resource constraints, and prioritizing objectives to deliver a functional product within timeline limitations. Overall, this experience bridged the gap between theoretical knowledge and practical implementation, preparing me for real-world cybersecurity and software engineering challenges in my future career.

Furthermore, it will be a memorable project since it was not easy from the day one. From the first day I started facing challenges and learned to overcome them. This taught me many valuable lessons such as every problem has a solution and no problem is bigger than you, Throughout the project I faced many problems and implemented many different solutions and many of them didn't work but didn't break my courage.

Last but not least I would like to thank my supervisor DR Hussain AL Yusuf for his utmost guidance and support. His valuable feedback and constant support helped me overcome difficult challenges and improve my implementation approach. I am also thankful to Mr. Cyril Anthony, the project manager, whose guidance, planning, and technical expertise played a key role in the successful development of this project.

Bahraini Perspectives

From a Bahraini perspective, this LLM-Powered SOC Automation project represents a significant contribution to the kingdom's evolving cybersecurity infrastructure and aligns closely with Bahrain's Economic Vision 2030 digital transformation initiatives. As the kingdom accelerates its adoption of smart government services, fintech innovation, and digital economy platforms, the need for advanced, intelligent security operations has become paramount to protect critical national infrastructure from increasingly sophisticated cyber threats.

The implementation of artificial intelligence in security operations reflects Bahrain's strategic commitment to becoming a regional leader in AI adoption and cybersecurity excellence. As AI continues to revolutionize industries globally, the cybersecurity domain faces both unprecedented opportunities and challenges while AI empowers defenders with intelligent automation and rapid threat analysis, it also equips adversaries with more sophisticated attack capabilities. This project demonstrates how Bahraini talent can leverage cutting-edge AI technologies like Claude and OpenAI to develop practical security solutions that address the growing complexity of modern cyber threats while reducing operational burden on security teams.

The successful implementation of this automated SOC platform offers significant benefits to various sectors within Bahrain's economy, including banking and financial services, telecommunications, healthcare, government entities, and critical infrastructure operators all of which face constant cyber threats and struggle with alert fatigue in traditional manual security operations. By reducing false positives by 95% and decreasing analyst workload by 50%, this solution provides a scalable, cost-effective approach that Bahraini organizations can adopt to enhance their security posture without requiring extensive staffing increases or unsustainable operational costs.

Furthermore, this project supports Bahrain's ambition to position itself as a regional hub for technological innovation and cybersecurity expertise. By showcasing locally developed AI-driven security solutions, it demonstrates the caliber of technical talent being cultivated within Bahrain's educational institutions and highlights the kingdom's capacity to contribute meaningful innovations to the global cybersecurity landscape. As regional organizations increasingly face sophisticated cyber threats, Bahrain's expertise in AI-powered security automation positions the kingdom as a valuable partner and knowledge center for neighboring Gulf countries seeking to strengthen their own cybersecurity capabilities.

Legal, Ethical, Social, and Professional Issues

Legal

The implementation of an LLM-Powered SOC Automation platform raised several critical legal and compliance related issues that must be addressed to ensure lawful operation. From a data protection perspective our system processes security logs which contain personally identifiable information (PII), user activities, and sensitive organizational data, which make it to be mandatory to be lawful with Bahrain's Personal Data Protection Law (PDPL), and other regional privacy frameworks. So, we must comply with industry-specific cybersecurity regulations such as the National Cybersecurity Strategy of Bahrain, Payment Card Industry Data Security Standard (PCI-DSS) for financial institutions, and sector-specific requirements from regulatory bodies like the Central Bank of Bahrain (CBB) for financial services or the Telecommunications Regulatory Authority (TRA) for telecom operators. During the implementation of this project, strict compliance must be clearly maintained. Failure to address these legal requirements may result in legal or regulatory issues, and reputational damage.

Ethical

The deployment of AI-driven security automation raises important ethical questions regarding accountability, transparency, and fairness in security decision-making. A primary ethical concern is whether organizations should trust AI models to make critical security decisions that could impact individuals' privacy, access rights, or even lead to false accusations of malicious activity. Furthermore, there's an ethical responsibility regarding false negatives if the AI system fails to detect a genuine threat, potentially resulting in a security breach, questions arise about who is responsible the organization, the AI developers, or the service providers. Organizations must implement Human for making critical decisions and maintain ethical AI usage policies that prioritize transparency, fairness, and accountability.

Social

The launch of AI powered SOC can have massive social impacts especially related to employment and workforce dynamics. Most major social concern is job displacement as

automation of alert triage and analysis can reduce demand for entry-level to mid-level analysts potentially leading to unemployment or requiring workforce transitions to other roles. This creates social responsibility for organizations to retrain and upskill affected employees rather than simply replacing them with automation.

Professional

From a professional standpoint, the integration of AI into security operations raises questions about accountability, professional standards, and the evolving role of cybersecurity professionals. Security professionals are responsible for staying skilled, being careful, and acting in the best interests of their organization. When using AI tools, they must balance trusting AI recommendations with using their own judgment. If an AI system gives wrong advice that causes security incidents or system failures, professionals may still be held responsible. Relying on AI without questioning it could be seen as neglecting their professional duties.

Conclusion

Our LLM-Powered SOC Automation project successfully demonstrated the achievable potential of integrating artificial intelligence into Security Operations Center environments. The implementation achieved its primary objective of creating a functional, intelligent security platform that combines Splunk SIEM with AI-driven response through Claude AI and automation which delivers real-time, enriched security alerts which were triggered to analysts via Slack via ChatGPT from our SIEM. The system effectively showcases how modern AI technologies can add innovation to traditional security operations by automating repetitive tasks, reducing false positives by 95%, decreasing analyst workload by 50%, and accelerating threat response times.

The journey to achieve this implementation was extremely challenging, involving numerous technical obstacles including setting up SIEM, setting up attack, MCP server integration complexities, resource constraints preventing local LLM deployment,

workflow automation platform migrations, mail server configuration issues, and the difficulties of connecting disparate technologies that were not originally designed to work together. Each challenge required extensive troubleshooting, research, adaptation, and creative problem-solving, ultimately providing invaluable hands-on experience with enterprise security tools, AI integration, and system architecture design. Throughout this process, I gained deep practical knowledge in Splunk SIEM administration, API integration, workflow automation, cloud-based AI services, and the real-world challenges of implementing complex technical solutions under resource and time constraints.

Beyond the technical achievements, this project provides a compelling reflection on how artificial intelligence can be meaningfully integrated into Security Operations Centers to enhance analyst capabilities rather than replace them. The dual-LLM architecture leveraging OpenAI for rapid alert triage and Claude for deep security recommendations demonstrates that AI serves as a force multiplier for security teams, handling routine tasks while freeing analysts to focus on strategic threat hunting, complex investigations, and decision-making that requires human judgment and creativity. The success of this project validates the hypothesis that intelligent automation can address the critical pain points of traditional manual SOC operations, including alert fatigue, slow response times, and limited scalability.

As cyber threats continue to evolve in sophistication and volume, the need for AI-integrated security operations will only intensify. This project establishes a foundation for future enhancements, including unified interface development, local LLM integration, mobile alert capabilities, and automated threat response mechanisms. The lessons learned and solutions developed throughout this implementation provide valuable insights for organizations and practitioners seeking to modernize their security operations through AI integration. Ultimately, this project demonstrates that with persistence, adaptability, and strategic use of available technologies, it is possible to build sophisticated, intelligent security solutions that significantly improve organizational cybersecurity posture while preparing the next generation of security professionals for an AI-driven future in cybersecurity.

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15. Model Context Protocol. (2025). In *Wikipedia*. Retrieved December 21, 2025, from
https://en.wikipedia.org/wiki/Model_Context_Protocol
16. Anthropic. (2025). Model Context Protocol (MCP). *Claude Documentation*.
<https://docs.claude.com/en/docs/mcp>

Appendices

Appendix I: User and System Administrator manual

User manual

The user manual is designed to help users easily access and operate the system with ease and confidence. It provides step-by-step guidance on monitoring security alerts through Slack, querying log data using Claude AI Assistant, and interpreting system notifications. This manual ensures users can perform their daily tasks efficiently without requiring extensive technical knowledge, enabling them to respond to alerts promptly and make informed decisions based on system-generated insights.

Accessing the Systems.

Our user can access the Claude mcp server is running and connected to Claude. To access the Claude, we need to make a Claude account and login.

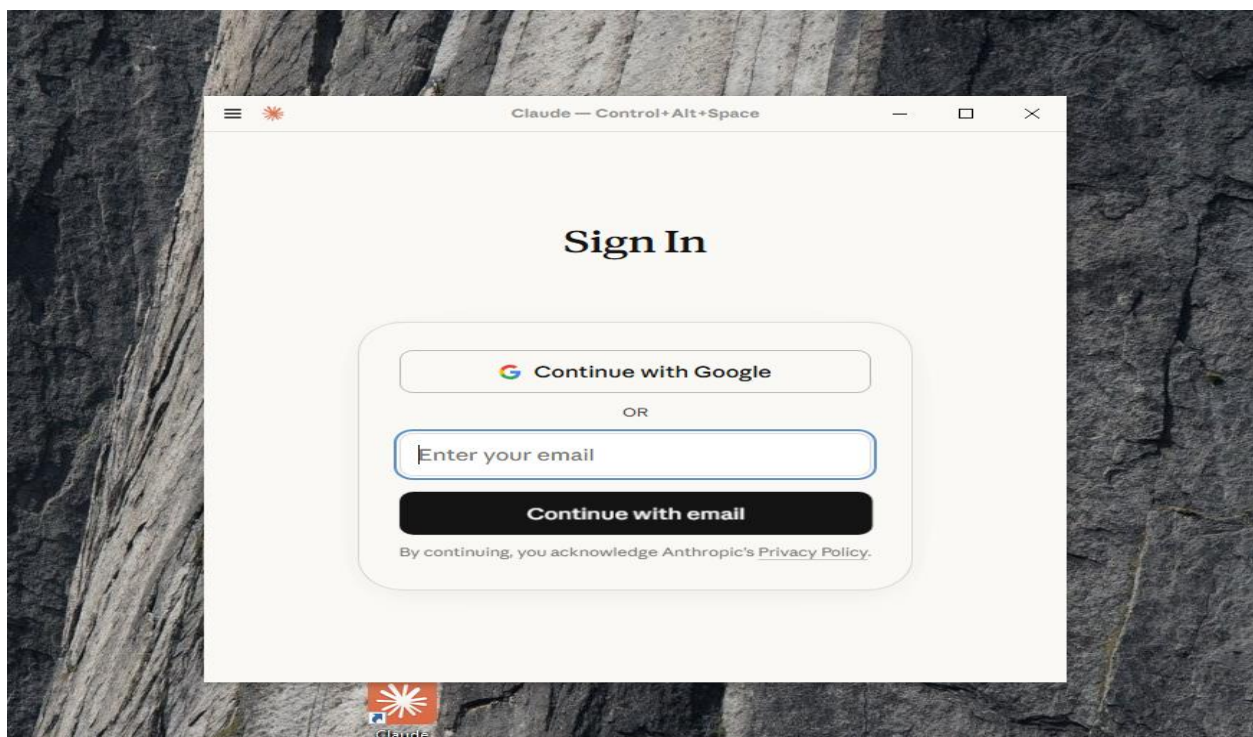


Figure 78- Accessing the Claude 1

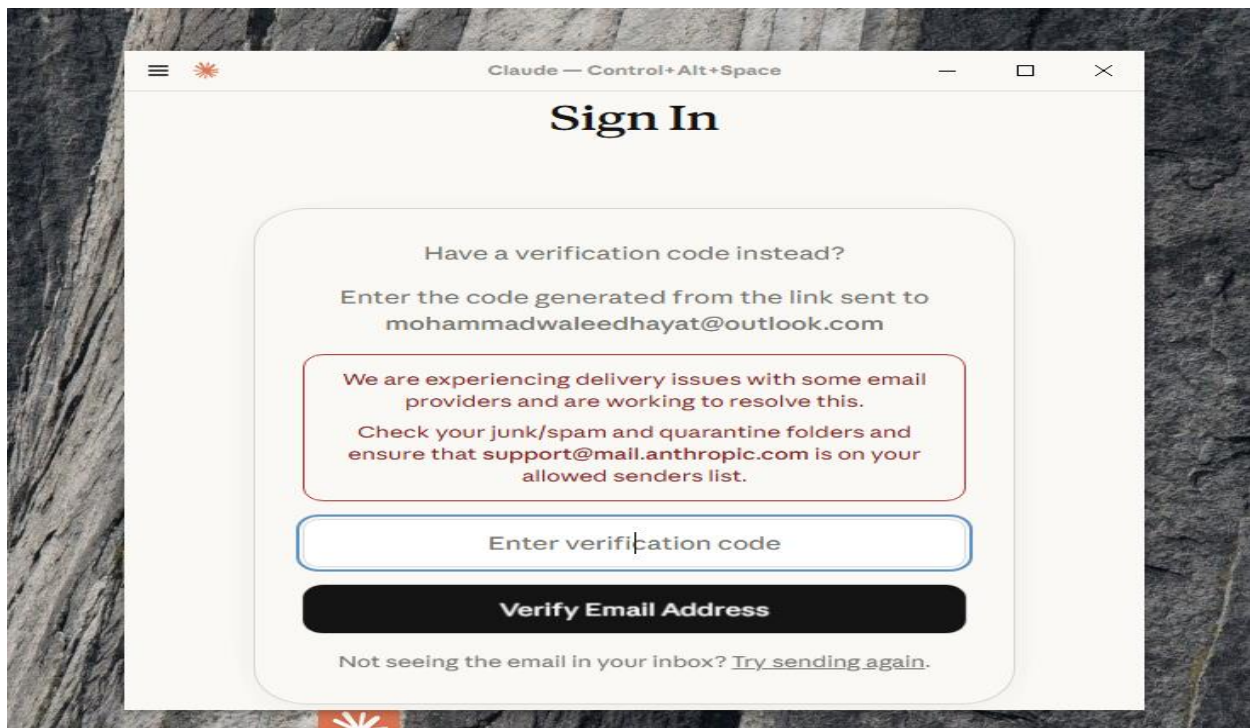


Figure 79-Accessing the Claude Assistant 2

Monitoring Alerts in Slack

Users can monitor real-time security alerts through the designated Slack channel once they are added to the alerts channel. To access alerts, users must join the Slack channel and subscribe to the alert channel where automated notifications are delivered.

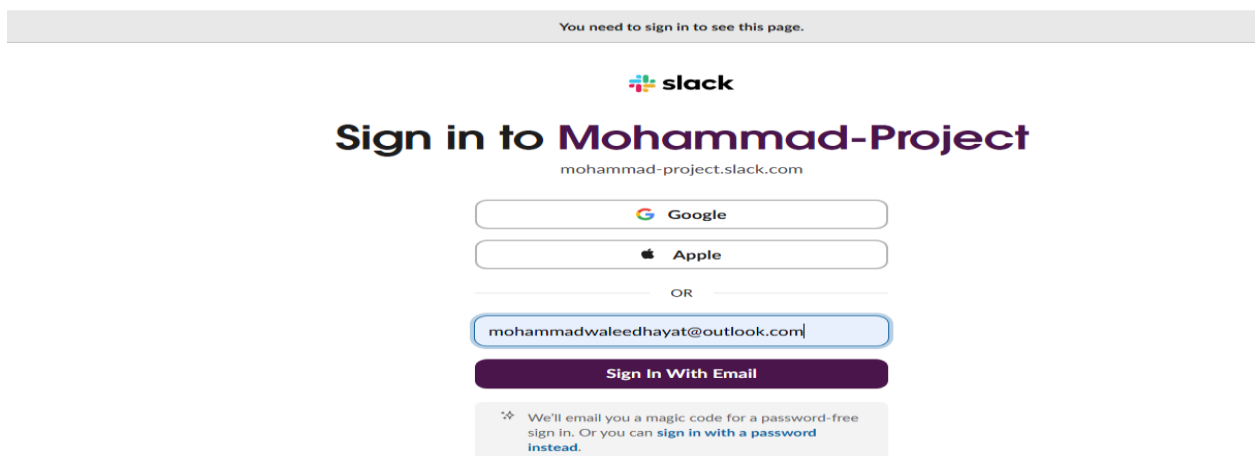


Figure 80- Accessing alerts 1

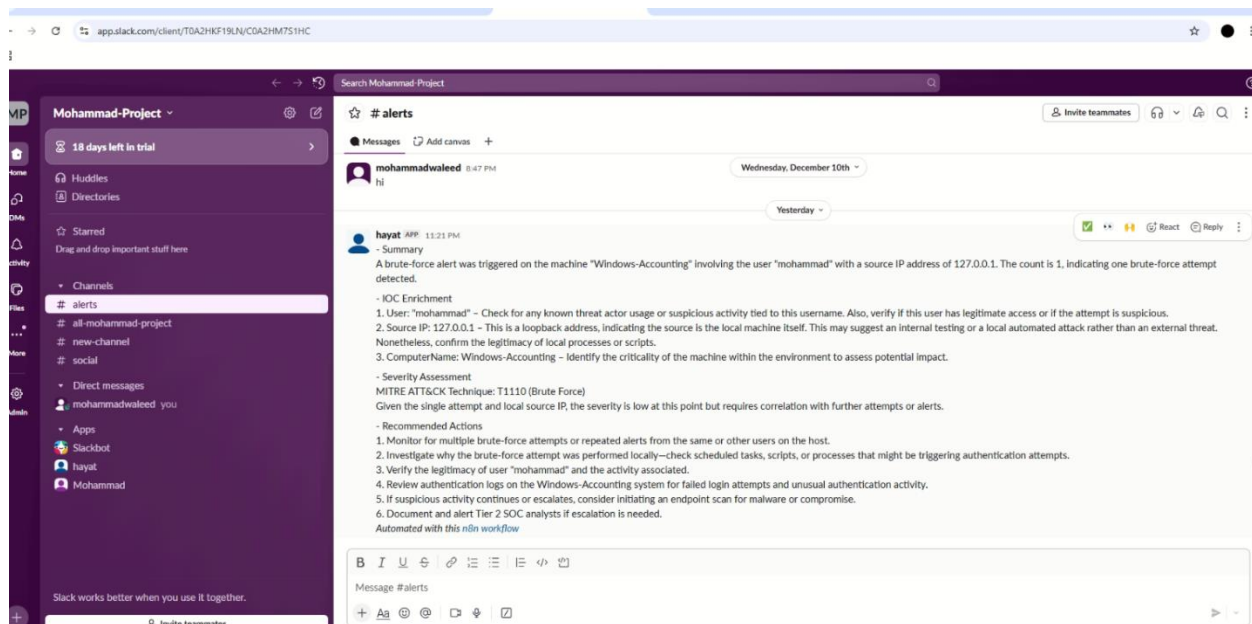


Figure 81- Accessing Alerts 2

Querying Logs Using Claude AI Assistant

Once our user has logged into Claude account and MCP server is running he can query about recent activity and triggered alerts,

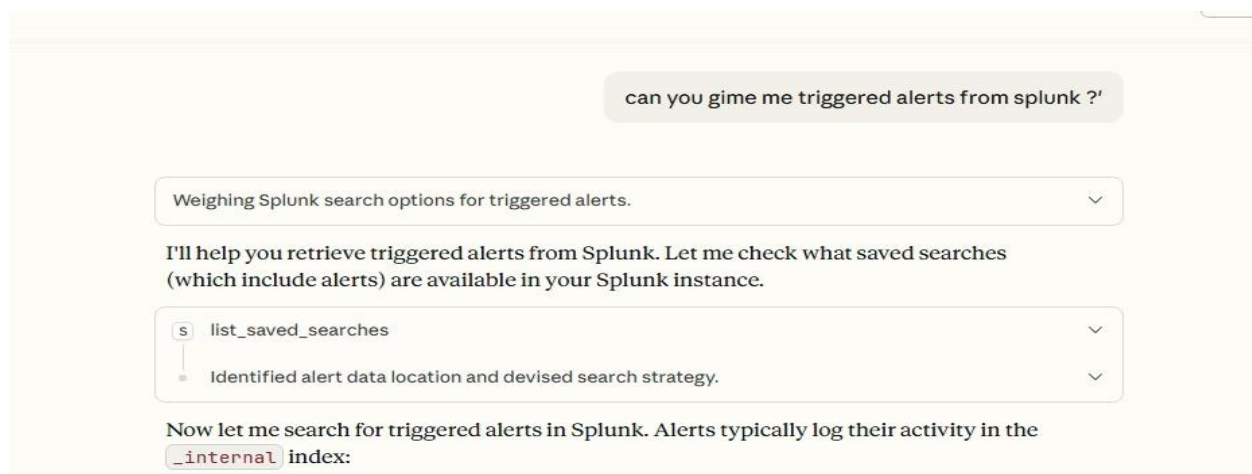


Figure 82- Querying into Assistant

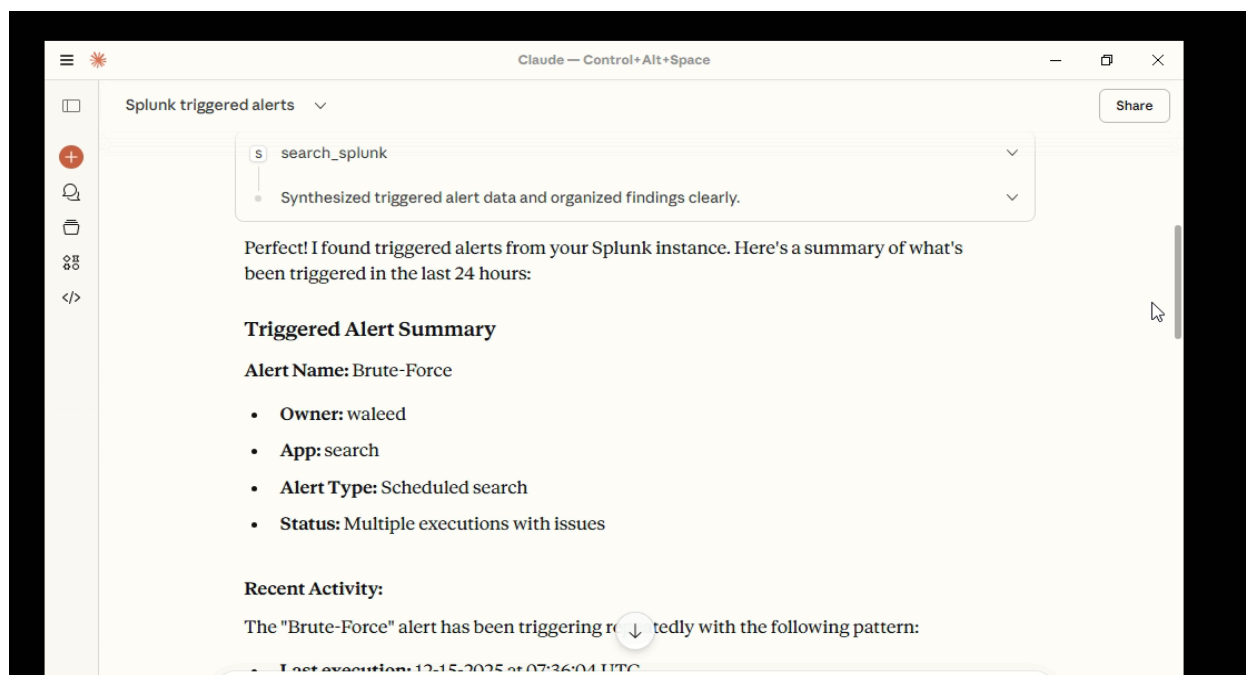


Figure 83- Receiving assistance with response measures

Interpreting and Responding to Alerts

- Severity Assessment

Given the single login attempt from localhost, this event has low immediate risk but could indicate the start of a brute-force attack or a misconfigured script. MITRE ATT&CK: T1110 (Brute Force) – initial access or credential access technique.

- Recommended Actions

1. Verify if the user "mohammad" is authorized and if this login attempt was expected.
2. Check logs for repeated failed login attempts from this or other accounts to identify a potential brute-force attempt.
3. Confirm there are no automated scripts or services causing repeated local login attempts.
4. If suspicious activity is confirmed, increase monitoring on the "Windows-Accounting" system for further authentication failures or anomalies.
5. Review account lockout policies and ensure they are configured to mitigate brute-force attacks.

Automated with this [n8n workflow](#)

Figure 84- Intercepting alerts with measures

Troubleshooting Common Issues

If the assistant has given any suspicious activity without a triggered alert, Our user can access splunk dashboard to check that activity and ask assistant about which precautions measures to take.

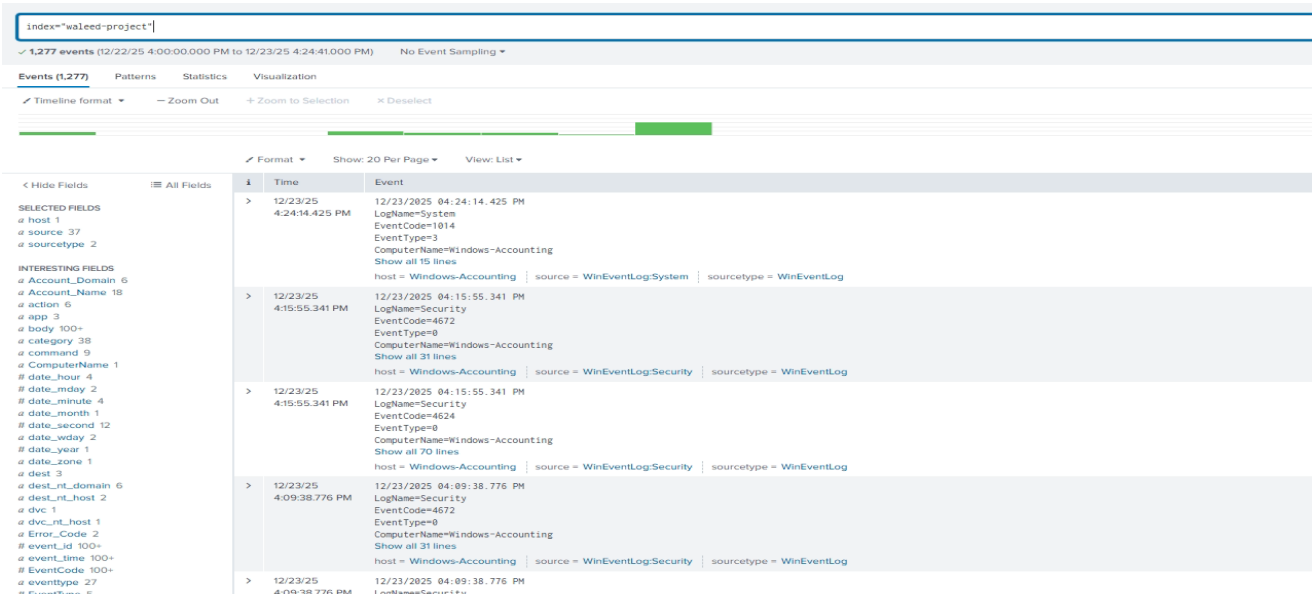


Figure 85- Searching for alerts manually.

System administrator manual

In this system and administrator manual we will demonstrate how an system administrator add an endpoint and starts receiving or configuring alerts.

System Installation and Configuration

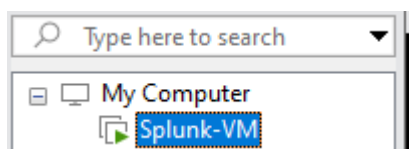


Figure 86- Start Splunk VM

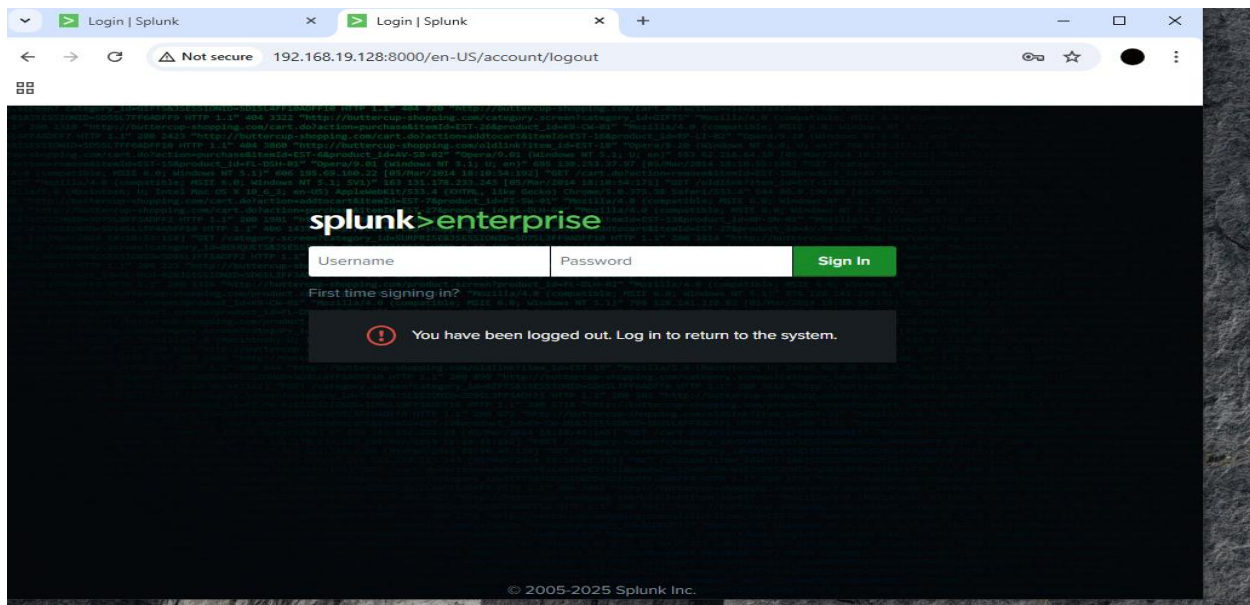


Figure 87- Login with splunk cridentials with which you registered

Configuring Splunk and Endpoint Integration

To add an endpoint on splunk you first our administrator needs to install Microsoft ads on for windows or Linux system, after that we need to create a new index on which they need to receive telemetry,

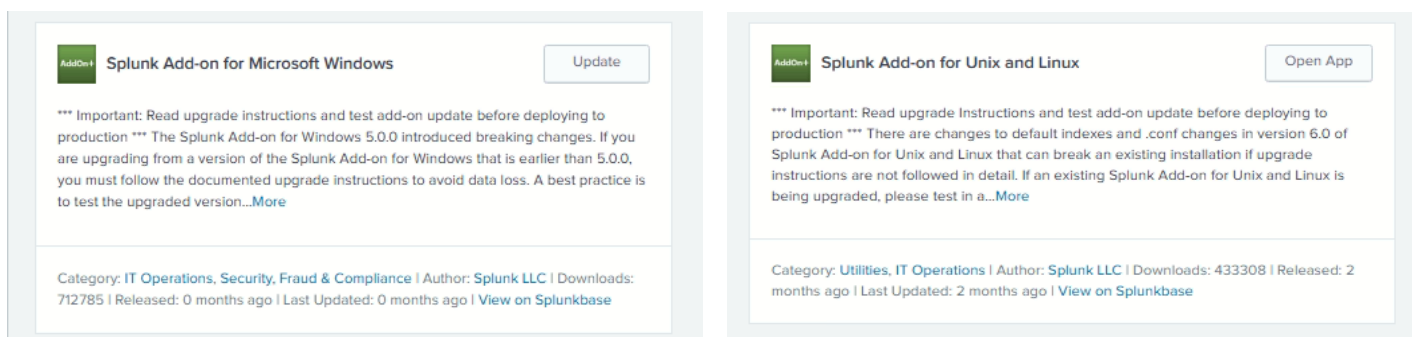


Figure 88- Installing Splunk Add-on for windows and Linux

After that administrator needs to install forwarder on both devices, after that and configure .conf file to see logs and activity forwarding.

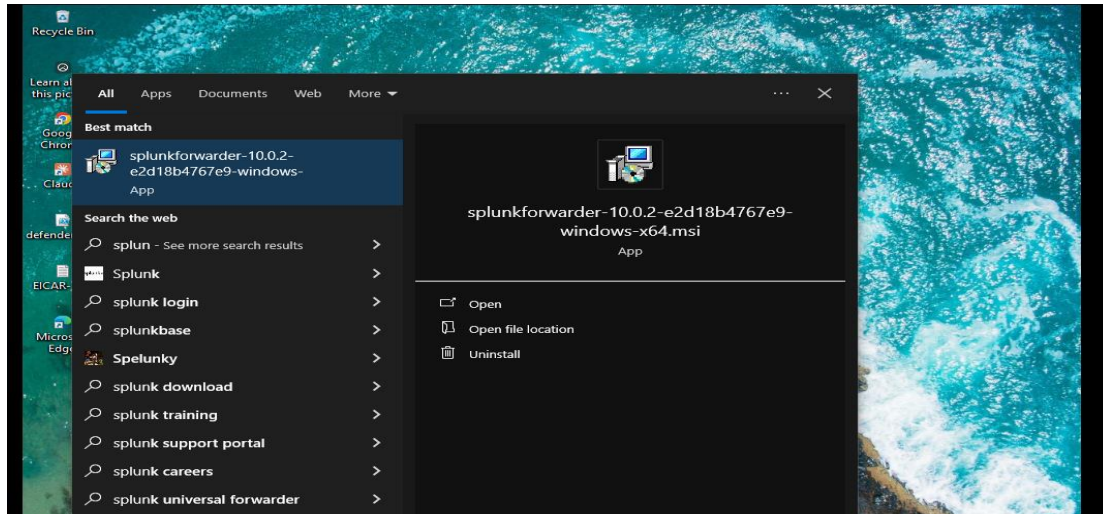


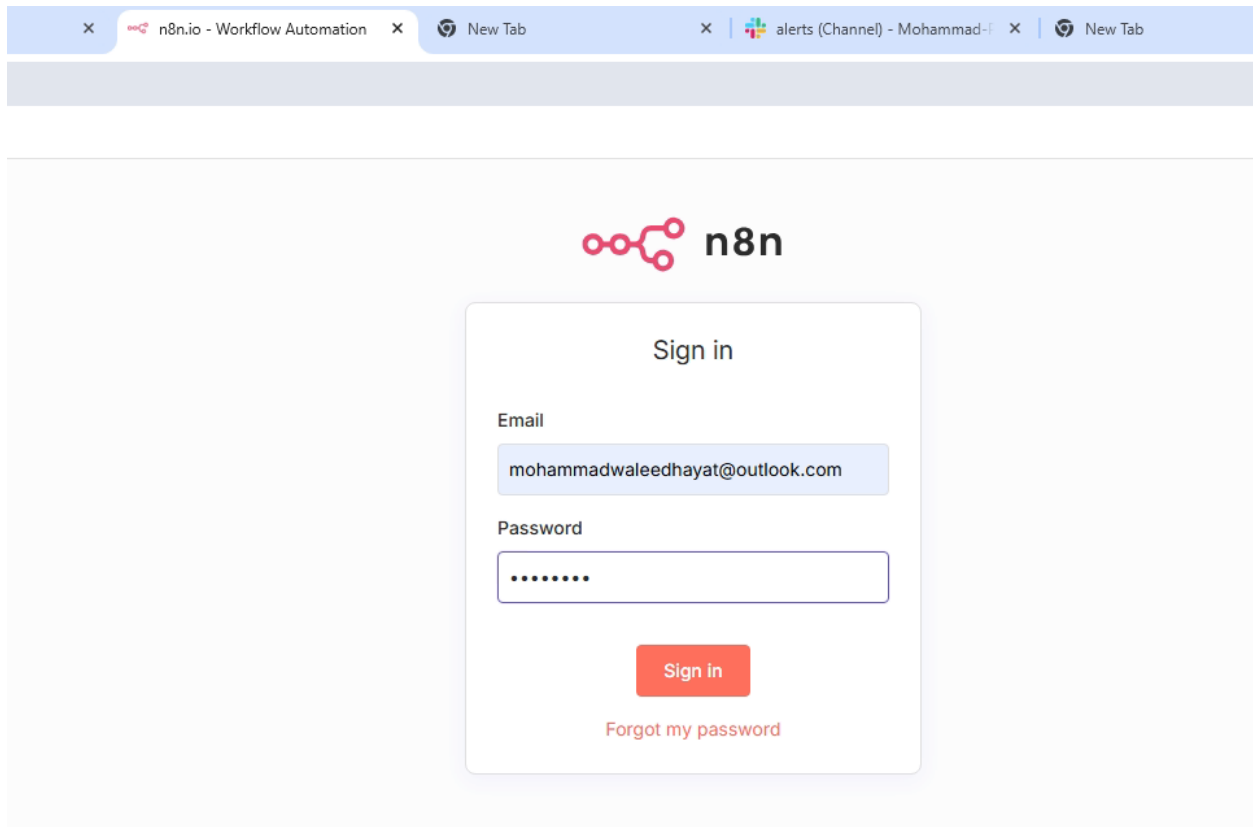
Figure 89- Installing forwarder on Windows

After that system administrator has installed forwarder, he needs

Using N8N automation For Alerts

To start the automation to work you need to turn on n8n VM first, once you have started that you can access n8n.

To Access use VMs IP at port 5678



The screenshot shows a web browser window with four tabs: 'n8n.io - Workflow Automation', 'New Tab', 'alerts (Channel) - Mohammad-', and another 'New Tab'. The main content area displays the n8n logo at the top, followed by a 'Sign in' form. The form includes an 'Email' field with the value 'mohammadwaleedhayat@outlook.com', a 'Password' field with masked characters, a red 'Sign in' button, and a link for 'Forgot my password'.

After accessing UI Insert your credentials. As the purpose of this workflow is just to automate your alerts from a webhook to Open Ai model and then to Slack.

So to get the Alerts on a webhook copy the production link of a webhook and paste it in the triggered action of send to a webhook under triggered alerts you need to see in slack.

Trigger Actions

+ Add Actions ▾

When triggered

Webhook	Remove
URL https://your.server.com/foo/bar	
Specified URL to send JSON payload via HTTP POST (ex., https://your.server.com/api/v1/webhook).	
Learn More	
> Add to Triggered Alerts	Remove

Figure 90- paste your webhook

1. Webhook Trigger Setup

- Node Type: Webhook
- Method: POST
- Webhook URL: http:// N8N-IP :5678/webhook/splunk-alert

Webhook

Listen for test event

Parameters

Settings

Docs

Webhook URLs

Test URL

Production URL

POST

http://192.168.19.130:5678/webhook/e02bae01-3e22-45a0-95a0-37c4602dea5e

HTTP Method

POST

Path

e02bae01-3e22-45a0-95a0-37c4602dea5e

Authentication

None

Respond

Immediately

If you are sending back a response, add a "Content-Type" response header with the appropriate value to avoid unexpected behavior

Options

No properties

Add option

Figure 91 – configuring webhool

2. AI Processing Node

- Node Type: OpenAI
- Action: Message a model
- Model: GPT-4 or GPT-3.5
- Purpose: Analyzes alert and provides context

The screenshot shows the configuration interface for the 'Message a model' node. The interface is divided into several sections:

- Parameters:** This section contains the main configuration options.
 - Resource:** A dropdown menu set to 'Text'.
 - Operation:** A dropdown menu set to 'Message a Model'.
 - Model:** A dropdown menu set to 'GPT-4.1-MINI'.
 - Messages:** This section contains the message configuration.
 - Type:** A dropdown menu set to 'Text'.
 - Role:** A dropdown menu set to 'System'.
 - Prompt:** A text area containing the following text:

Act as a Tier 1 SOC analyst assistant. When provided with a security alert or incident details (including indicators of compromise, logs, or metadata), perform the following:

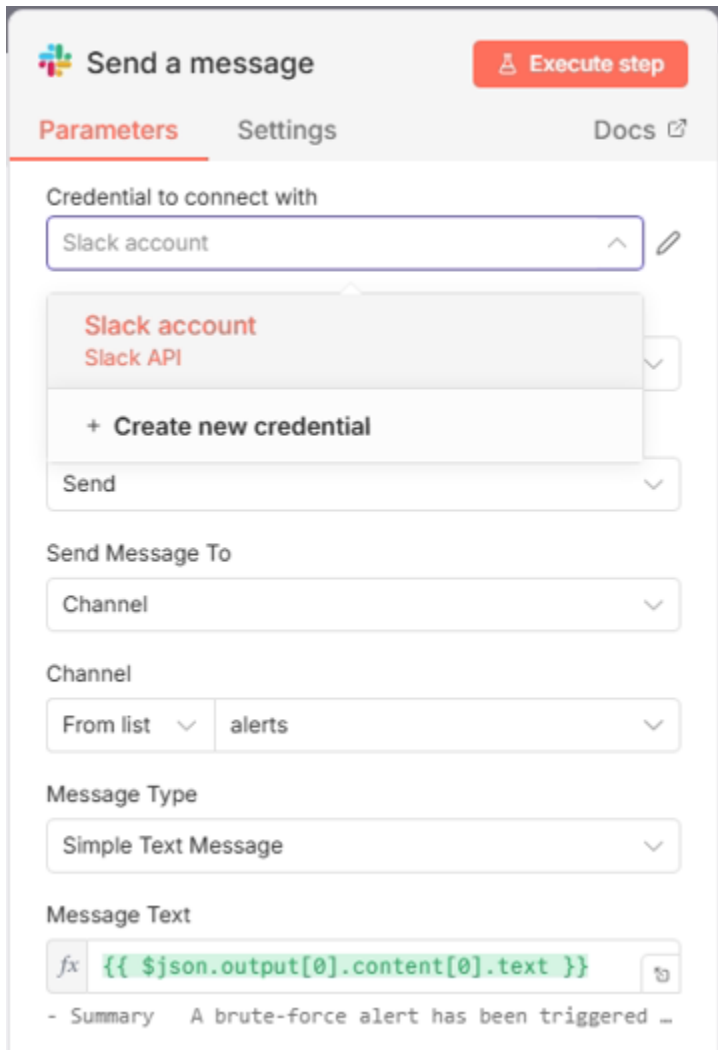
 1. Summary of the alert
 2. IOC enrichment
 3. Severity assessment (MITRE ATT&CK)
 4. Recommended next actions

Return output in this structure:

 - Summary
 - IOC Enrichment
 - Severity Assessment
 - Recommended Actions
- Settings:** A tab for additional settings.
- Docs:** A link to documentation.
- Execute step:** A red button to execute the step.

3. Slack Notification Node

- Node Type: Slack
- Action: Send message
- Channel: security-alerts
- Message includes Alert details + AI analysis



Send a message Execute step

Parameters Settings Docs

Credential to connect with

Slack account

Slack account
Slack API

+ Create new credential

Send

Send Message To

Channel

Channel

From list alerts

Message Type

Simple Text Message

Message Text

`{{ $json.output[0].content[0].text }}`

- Summary A brute-force alert has been triggered ...

Figure 92- Setting up slack channel

Activating a Workflow:

1. Make sure N8N VM is on.
2. Open the workflow
3. Click the toggle switch (top right)
4. Ensure status shows "Active"

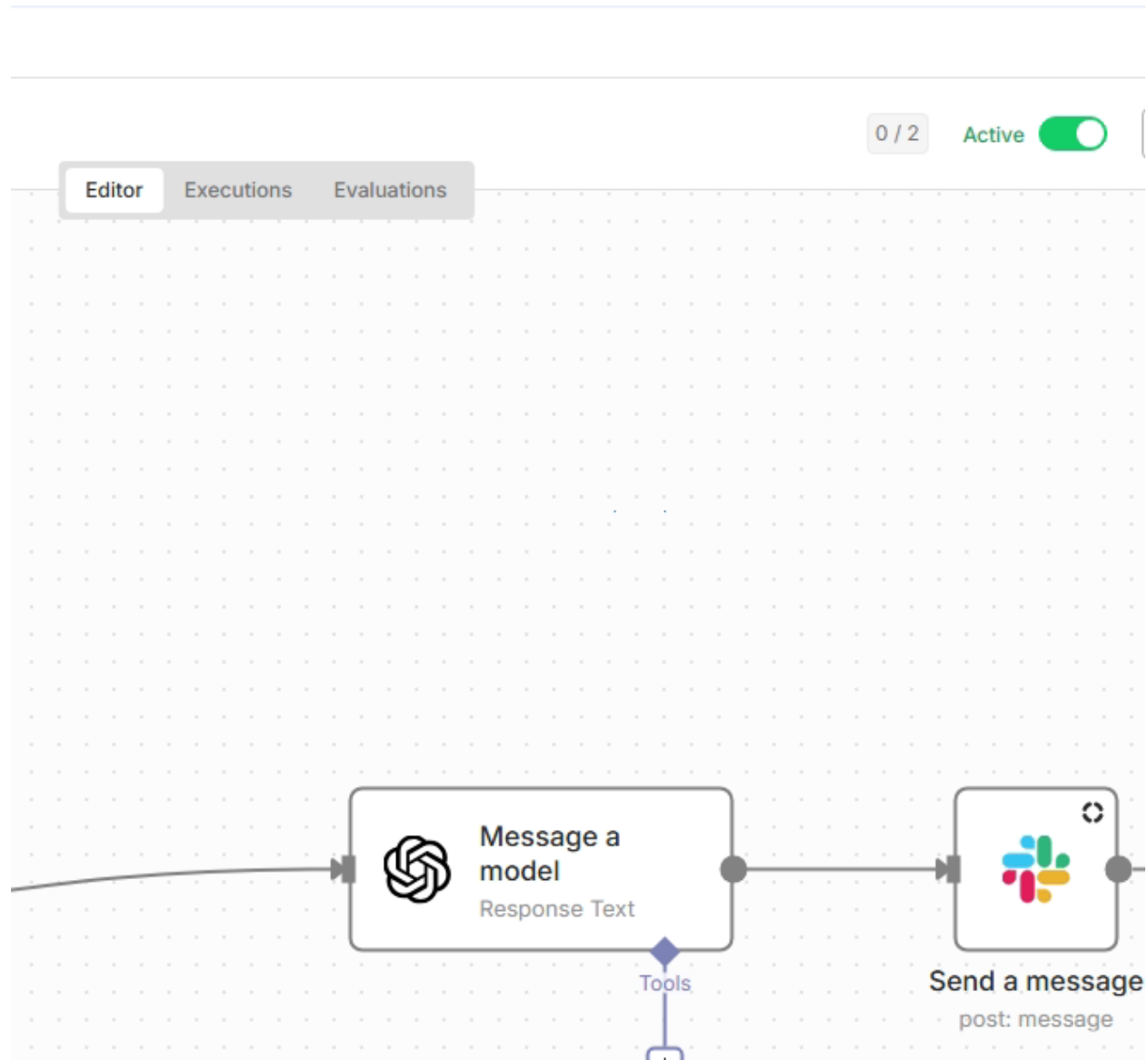


Figure 93- Turn the toggle on

Deploying and Connecting MCP Server

Installation and Setup

Prerequisites:

- Splunk Enterprise installed and running
- Claude Pro account with API access
- Python 3.8+ installed
- Splunk MCP Server
- Network connectivity between Splunk and MCP server

Installing MCP Server:

```
bash
```

```
# Install MCP via pip
```

```
pip install mcp
```

```
# Verify installation
```

```
mcp --version
```

Configure MCP for Splunk:

```
bash
```

```
# Create MCP configuration file
```

```
nano ~/.config/mcp/config.json
```

Configuration File Content:

```
json
```

```
{
```

```
  "splunk": {
```

```
    "host": "localhost ip ",
```

```
    "port": 8089,
```

```
    "token": "YOUR_SPLUNK_TOKEN"
```

{Your splunk user credentials}

“

{Splunk Username}

{Password}

},

}

Once you make sure Server is running you can start Querying about logs and event in claude AI .

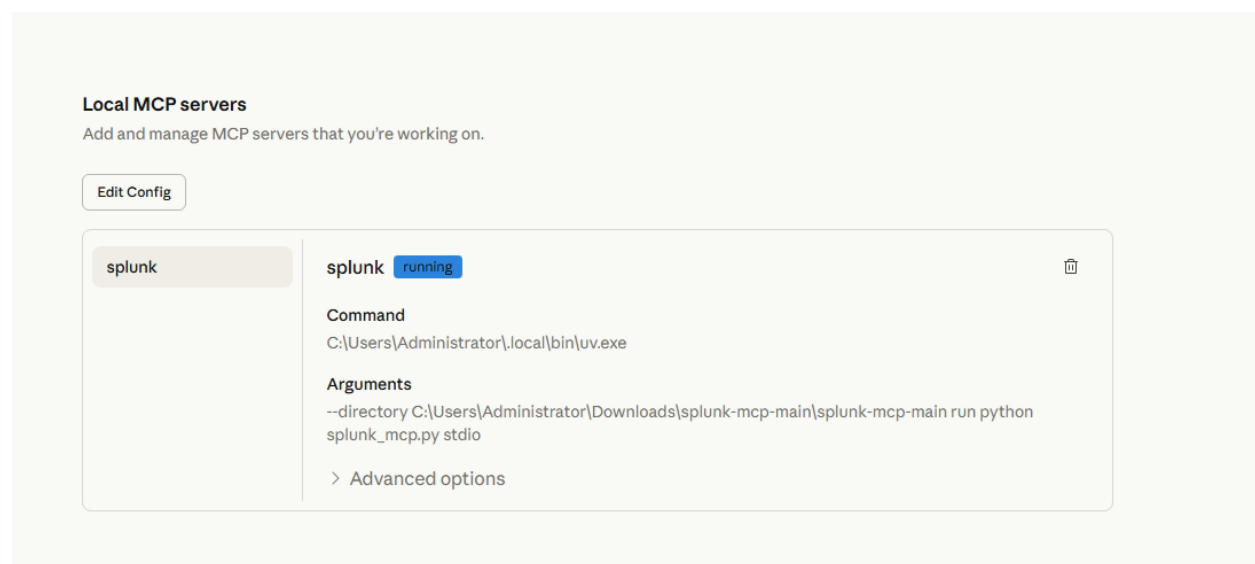


Figure 94 -Running MCP Server

System Maintenance and Monitoring

Querying Splunk Data Through Claude

Basic Query Process:

1. Open Claude interface (claude.ai)
2. Ensure MCP connection to Splunk is active (green indicator)
3. Ask Claude questions about your Splunk data in natural language

Example Queries:**Check Recent Security Events:**

"Show me all failed login attempts in the last 24 hours"

Analyze Specific Alerts:

"What privilege escalation attempts occurred today?"

Search for Patterns:

"Find any brute force attacks from the past week"

Get Event Details:

"Show details of the latest phishing detection alert"

Can you tell me mohammad's IP address ?

Let me search for network-related information about the mohammad user account.

 search_splunk 

Based on the logs, I found mohammad's IP address information:

Mohammad's IP Address: 192.168.19.1

This IP address appears in multiple logon events where mohammad connected to the Windows-Accounting system. Here's what I found:

Connection Details:

- **Remote connections from:** 192.168.19.1 (multiple Remote Desktop sessions)
- **Local connections from:** 127.0.0.1 (localhost - for local interactive logons)
- **Workstation Name:** DESKTOP-BSUB3KC (the computer mohammad connected from)

Connection Activity:

- Remote Desktop (RDP) connections from 192.168.19.1
- Multiple session reconnections and disconnections throughout the day
- Both network logons (Type 3) and interactive logons (Type 2 and 7) from this IP

Figure 95 - Querying About IP

Can you just give me any critical alerts i should be aware about in last 24 hours ?only just this.

I'll search for critical alerts in your Splunk instance from the last 24 hours.

 search_splunk 

 search_splunk 

No critical alerts found in the last 24 hours. Your systems appear to be running normally.

Figure 96 - Checking system health

Appendix II: Detailed Design

Explaining Design Through Sequence diagram

In this section we will cover detailed design of how our system will function. We will also include diagrams of different working components of our system.

To describe the Design of our system in detailed way, I will start with sequence diagram demonstrating the sequence of our operations, since sequence diagram can be our best option for explaining the user interaction to the components of our project.

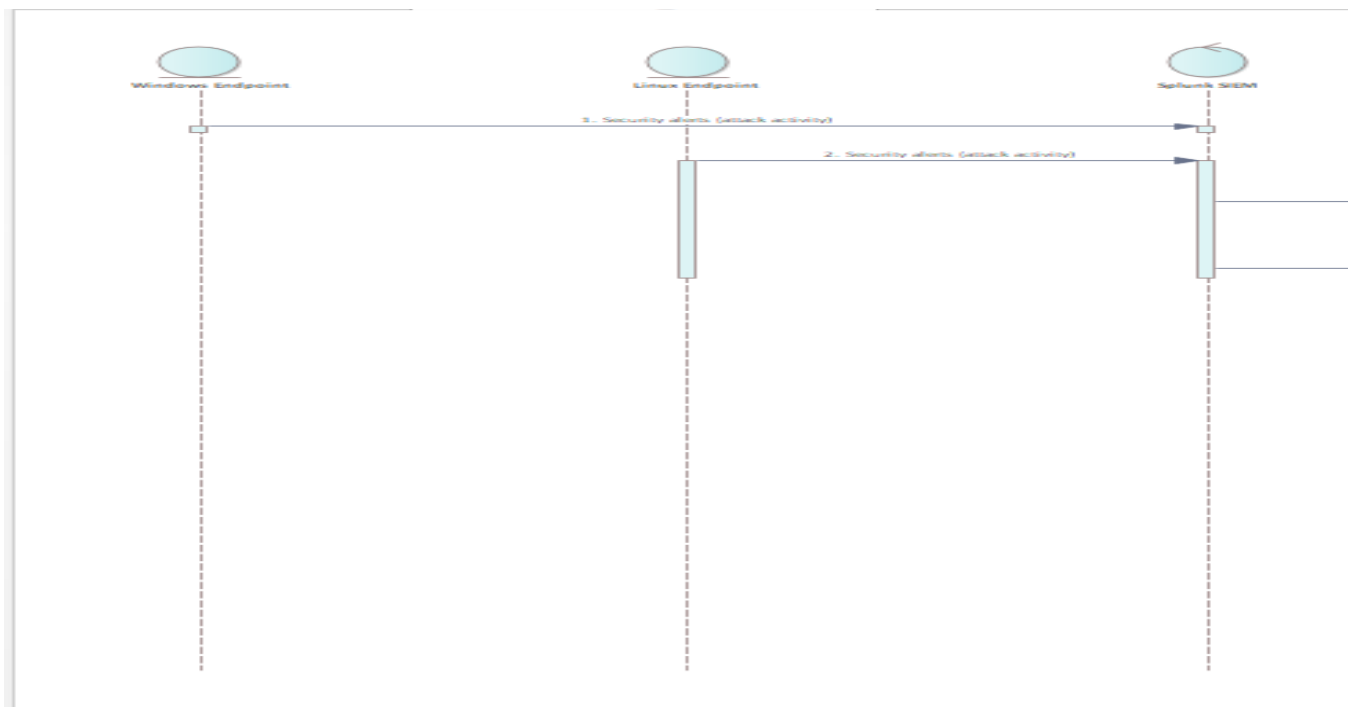


Figure 97- Sequence diagram part 1

The above figure demonstrates the initial phase of our product In which different operating systems such as windows and Linux are configured to send telemetry to splunk SIEM to continuously monitor and capture security events such as login attempts, file modifications, network connections, and suspicious activities. These telemetry data and security logs are then forwarded in real-time to Splunk SIEM for centralized collection, correlation, and analysis. This establishes the foundation for our automated threat detection pipeline, where raw security events are transformed into actionable intelligence through AI-powered analysis and automated response workflows.

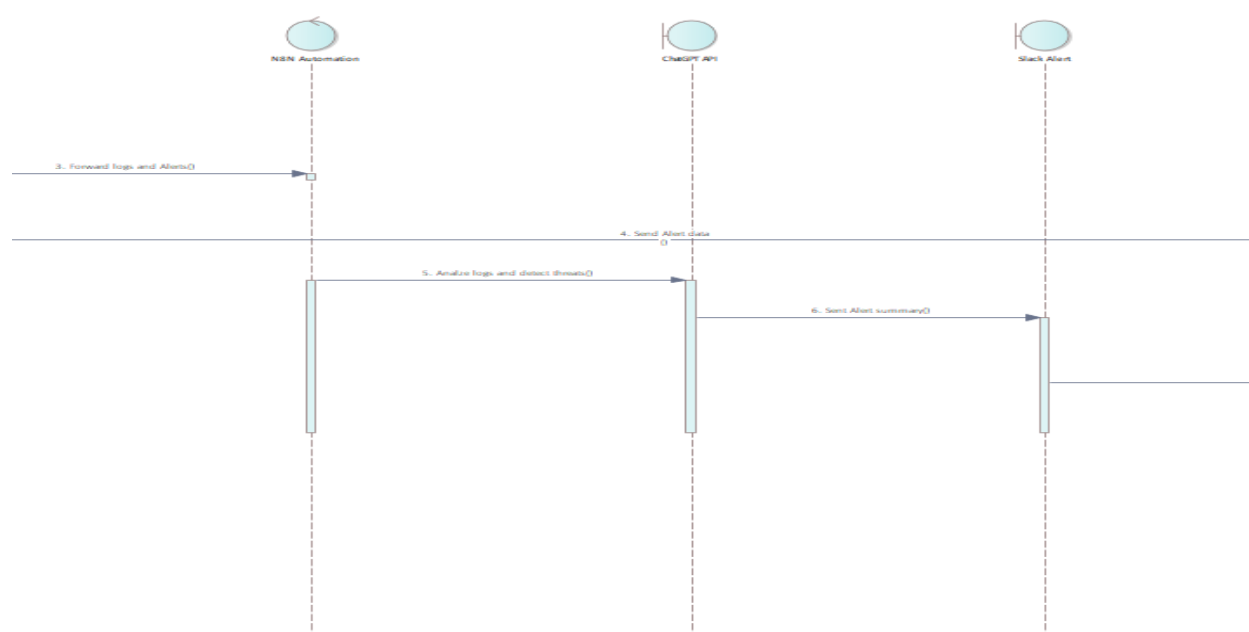


Figure 98- Sequence diagram part 2

In the above figure we can see that automation software is receiving triggered alerts and they are being processed by Open AI's API and send to slack with suitable response and what action can be taken again these alerts. The processed information, including alert summaries and suggested remediation actions, is automatically sent to Slack, enabling analysts to quickly understand the threat and take appropriate response measures without manual triage.

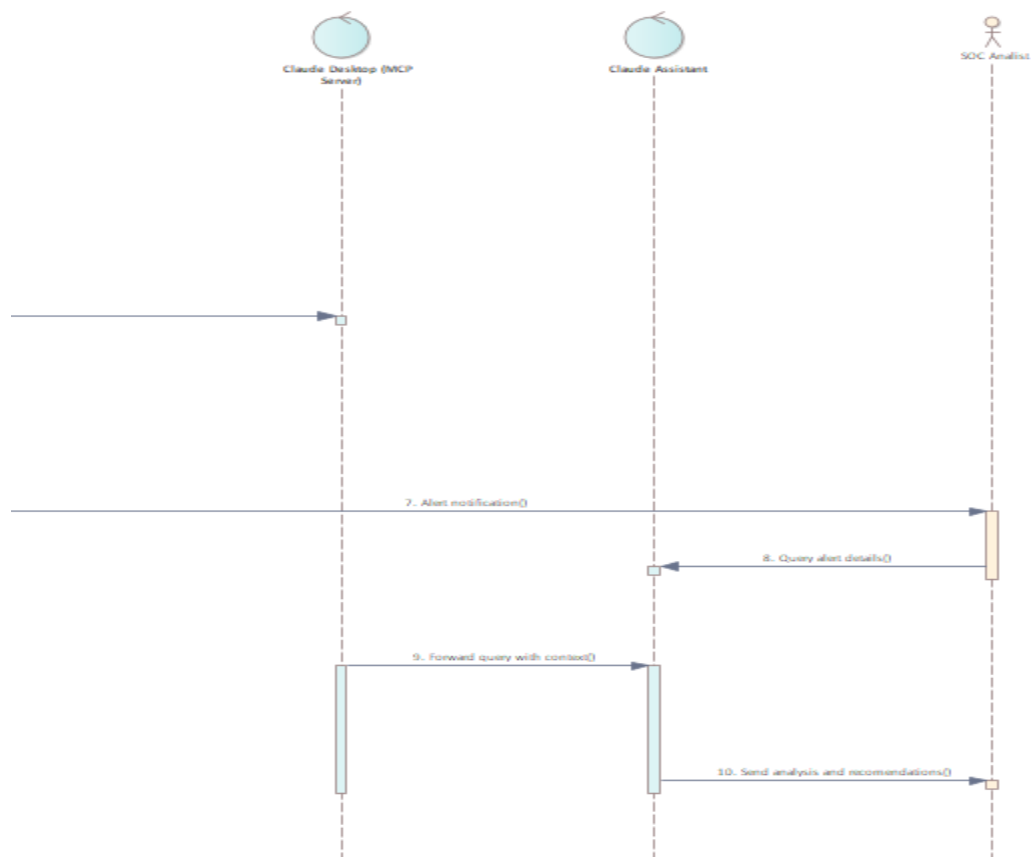


Figure 99- Sequence diagram part 3

In the above figure, we can see that our MCP server which is connected straight to SIEM is getting activity analysis and sending it to Claude assistant. we can also see our analyst getting straight alerts from Slack. When our analyst will send query to Claude assistant it grabs reply from mcp sever connected to SIEM. Claude retrieves the necessary context and data from the MCP server, which pulls information directly from SIEM. This ensures that all AI-generated insights are based on current, accurate security data.

Attack Surface Diagram

We have another attack surface diagram, this diagram shows what the surface of attacks is and what are the operations getting performed once our network has been attacked. An attack surface diagram visualizes all the potential entry points and vulnerabilities that attackers could exploit within a system or network. It maps out the exposed components, services, and interfaces that are accessible to potential threats, helping security teams understand where attacks might originate and how they propagate through the infrastructure. This diagram demonstrates the various attack vectors targeting our environment and shows how our LLM-powered SOC assistant detects, analyzes, and responds to these threats through integrated security tools and automated workflows.

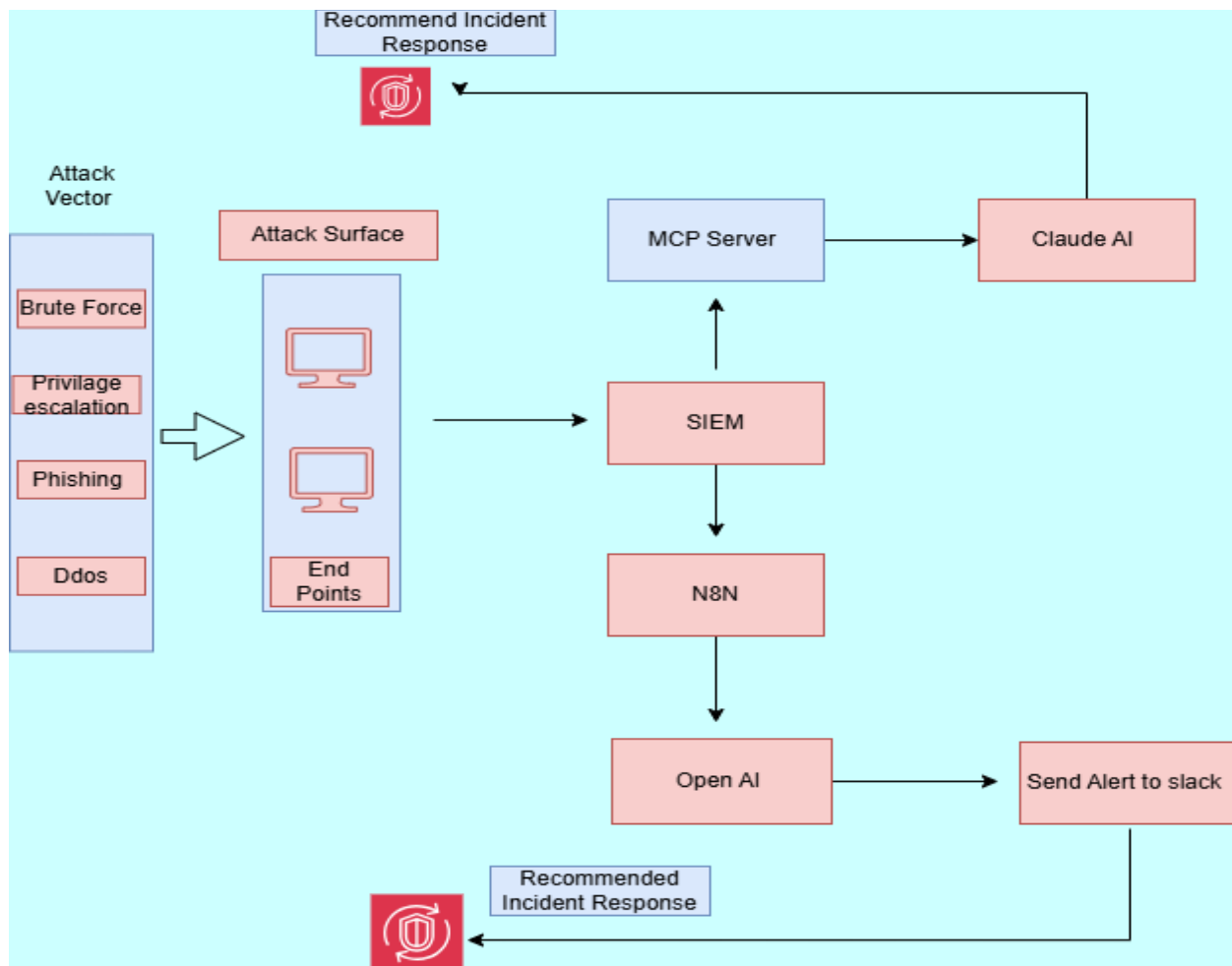


Figure 100- Attack Surface diagram

Operational Flow:

Attack Detection Phase: When attacks or suspicious activities occur on the monitored infrastructure, security events are immediately captured and forwarded to the analysis pipeline in SIEM.

Data Aggregation & Analysis:

- Events are ingested into the **SIEM** system for centralization and correlation
- Data flows through **MCP (Model Context Protocol)** to provide context to the AI assistant.
- **Claude LLM** analyzes events, identifying patterns, severity, and potential threats.

Intelligent Processing: Claude evaluates alerts against historical data, threat intelligence, and user behavior patterns to determine legitimacy, priority, and recommended actions.

Automated Response:

- **N8N** orchestrates automated workflows based on Claude's recommendations
- **Slack** notifications alert analysts with contextualized summaries and suggested actions
- Predefined measures and playbooks are suggested for known threat patterns.

Deployment Diagram:

A deployment diagram describes the physical architecture of a system, showing how software components are deployed across hardware infrastructure. It illustrates the hardware nodes (servers, devices, machines), the software artifacts running on them, and the communication pathways between these components. This diagram demonstrates how our LLM-powered SOC assistant is deployed across various platforms including endpoint machines, Automation software, SIEM infrastructure, automation servers, and AI assistants, showing the physical distribution and interaction of all system components in the production environment.

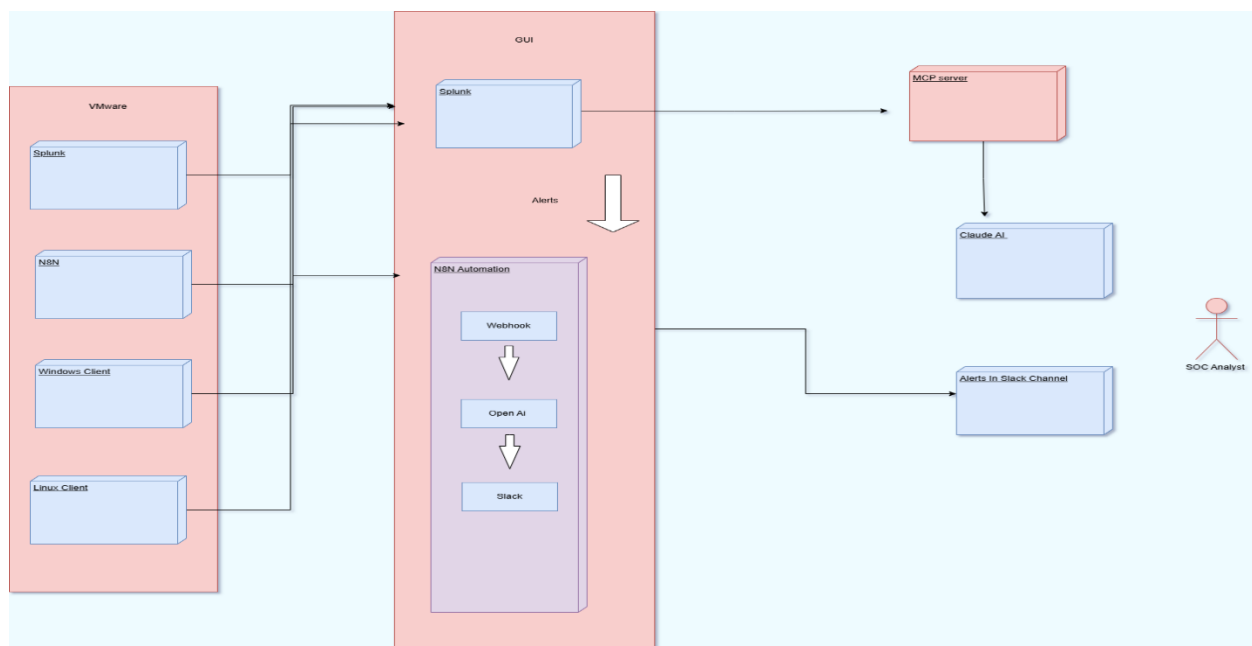


Figure 101 - Deployment Diagram

The System is deployed on VMware infrastructure with four virtual machines: a Splunk SIEM server for centralized log management, an N8N automation server for workflow orchestration, and Windows and Linux endpoint machines that generate security events and forward telemetry data to Splunk for real-time monitoring and analysis. After running Splunk and N8N they are accessed from GUI interfaces through their designated ports.

The N8N automation server operates as the orchestration layer, receiving real-time security alerts from Splunk SIEM through webhook triggers. Upon receiving an alert, N8N forwards the event data to OpenAI's API for intelligent analysis, where the LLM evaluates the threat severity, identifies attack patterns, and generates actionable recommendations. The processed results, including alert summaries and suggested response actions, are then automatically sent to Slack channels, allowing our analysts to receive contextualized threat intelligence and remediation guidance instantly.

At the last we have our MCP (model context protocol) server which maintains a direct connection to Splunk SIEM, continuously querying and retrieving security activity data, alert logs, and telemetry information. This real-time integration allows MCP to collect comprehensive insights about ongoing security events, user behaviors, and threat patterns detected within Splunk. When analysts interact with Claude Assistant to investigate alerts or query security events, Claude leverages the MCP server to access current SIEM data, providing context-aware analysis and recommendations based on live security rather than static information.

Appendix III: Detailed Implementation

In this section I will provide the detailed Implementation of my project, Due to the large number of elements in implementation phase I will try to divide into multiple sections, such as Setting up SIEM, configuration attacks, building automation and configuring MCP server to cloud, I will demonstrate the each as much possible.

Building SIEM:

To build SIEM we needed a SIEM solution. We had multiple options such as wazuh, Microsoft sentinel and IBM Qradar but I decided to choose Splunk because of ability to configure with multiple components and easy to operate.

To build SIEM we need multiple elements such as Virtualization software, server to run SIEM on and endpoints to configure SIEM on them. For virtualization we utilized VMware.

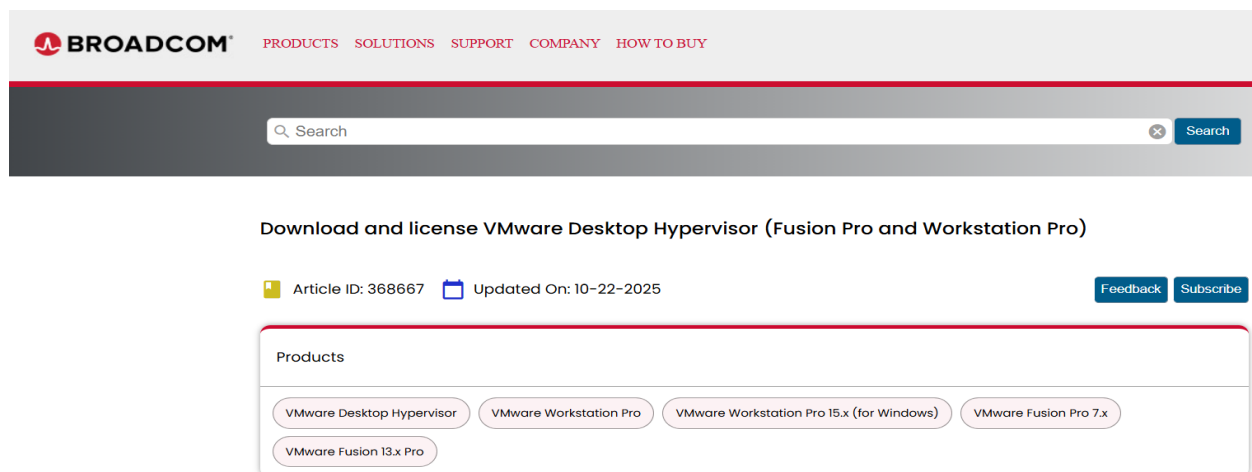


Figure 102- Installing VMware from Broadcom



Figure 103- VMware Workstation

After this we will install Ubuntu Servers to RUN Splunk on, we need to make sure we install the licensed version, so we don't face any issue later.

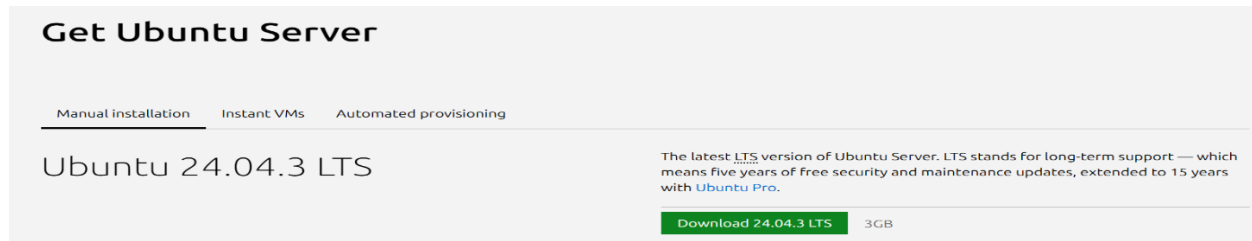


Figure 104- Installing Ubuntu server

After that you need to download ubuntu server. You need to make splunk account for getting the link download splunk enterprise.

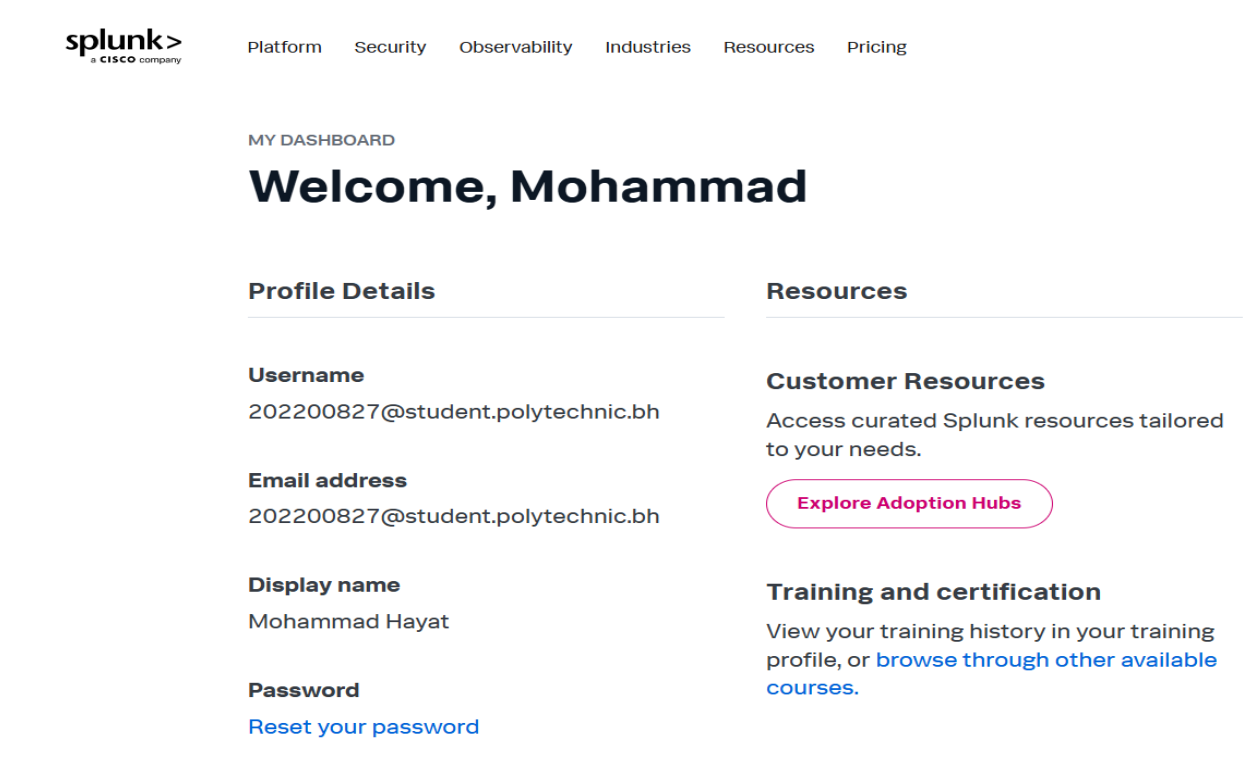


Figure 105- Creating Splunk Account

```

splunk:~$ wget -O splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.0.0/linux/splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb"
7:37:45-- https://download.splunk.com/products/splunk/releases/10.0.0/linux/splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb
load.splunk.com (download.splunk.com)... 18.161.21.49, 18.161.21.112, 18.161.21.84, ...
download.splunk.com (download.splunk.com)|18.161.21.49|:443... connected.
ent, awaiting response... 200 OK
2564 (1.3G) [binary/octet-stream]
splunk-10.0.0-e8eb0c4654f8-linux-amd64.deb'

8-linux-amd64.deb      11%[=====>] 146.14M  25.9MB/s  eta 54s

```

Figure 106- Installing Splunk

Commands for installing Splunk

Download Splunk Enterprise

```

wget -O splunk-9.1.2-b6b9c8185839-linux-2.6-amd64.deb
"https://download.splunk.com/products/splunk/releases/9.1.2/linux/splunk-9.1.2-
b6b9c8185839-linux-2.6-amd64.deb"

```

```

# Install Splunk sudo dpkg -i splunk-9.1.2-b6b9c8185839-linux-2.6-amd64.deb

```

```

# Start Splunk and accept license sudo /opt/splunk/bin/splunk start --accept-license

```

```

# Enable Splunk to start at boot sudo /opt/splunk/bin/splunk enable boot-start -user
splunk # Create splunk user if not exists sudo useradd -r -m -U -d /opt/splunk -s
/bin/bash splunk

```

```

# Change ownership to splunk user sudo chown -R splunk:splunk /opt/splunk

```

```

# Start Splunk as splunk user sudo -u splunk /opt/splunk/bin/splunk start --accept-
license

```

After that You need to Access GUI at port 8000 with the Ip of your Splunk Instance

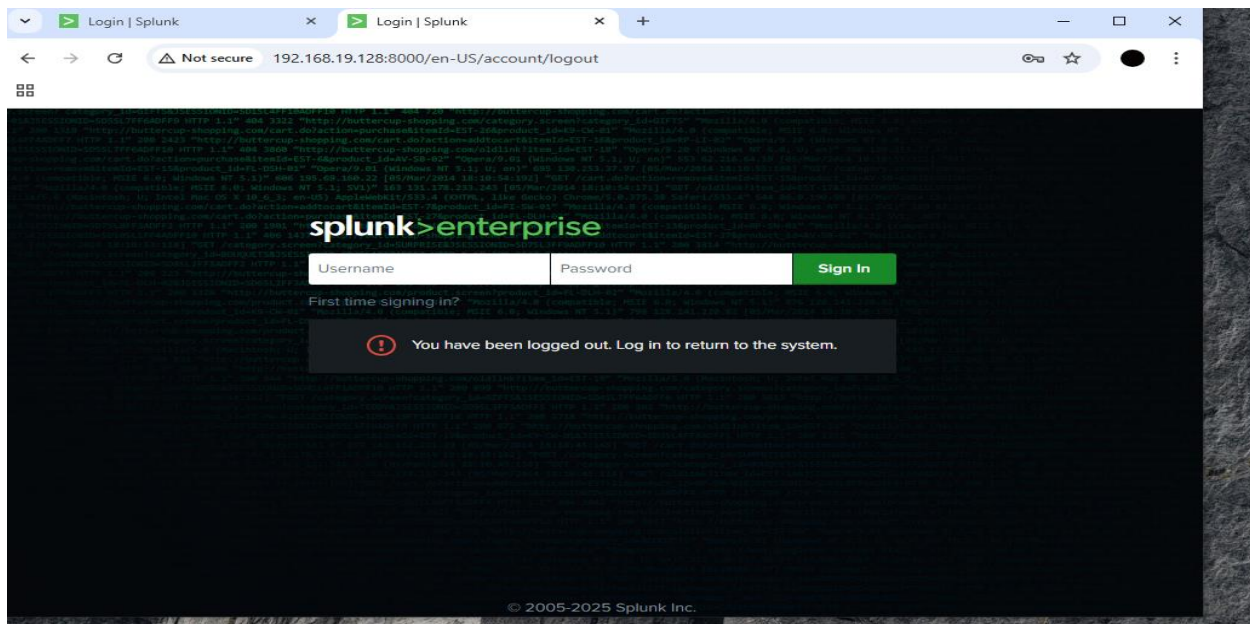


Figure 107- Accessing Splunk

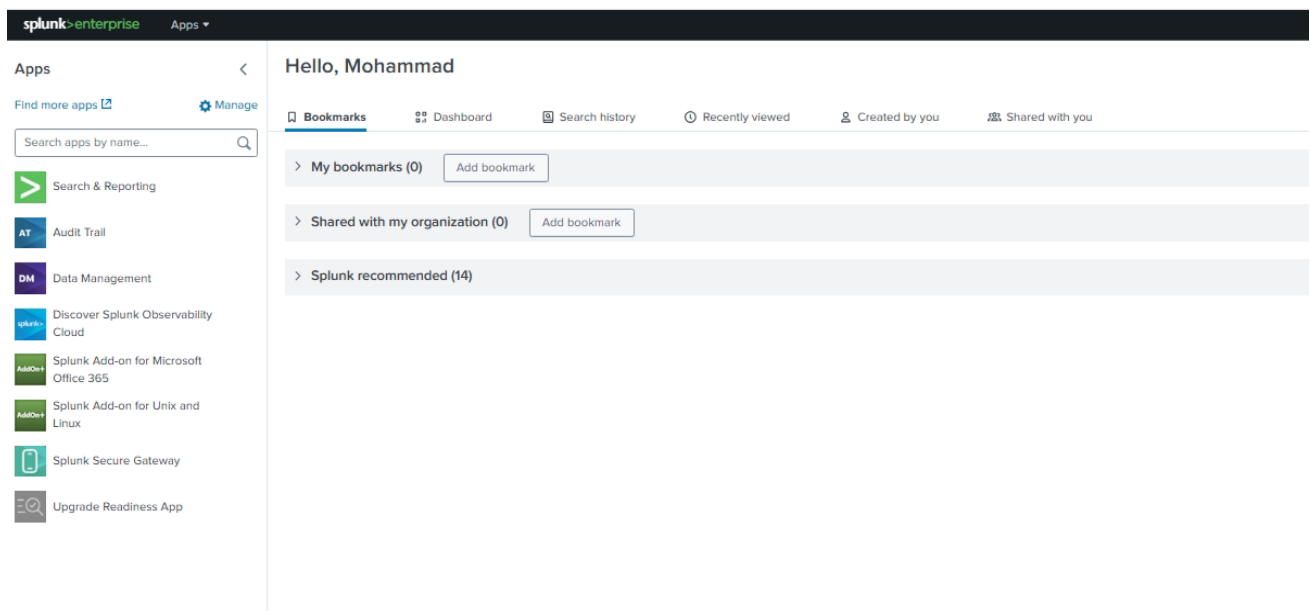


Figure 108- Splunk Dashboard

Splunk stores its captured activity in indexes, we need to create an index on which we will send activity and splunk will receive.

Figure 109- Creating new Index

Just before we move to our endpoints the last thing we need to do is to install apps for receiving logs from different operating systems.

Figure 110- Splunk Ads on for Windows and Linux

To Start Configuring Endpoints we will install endpoints virtual machines.

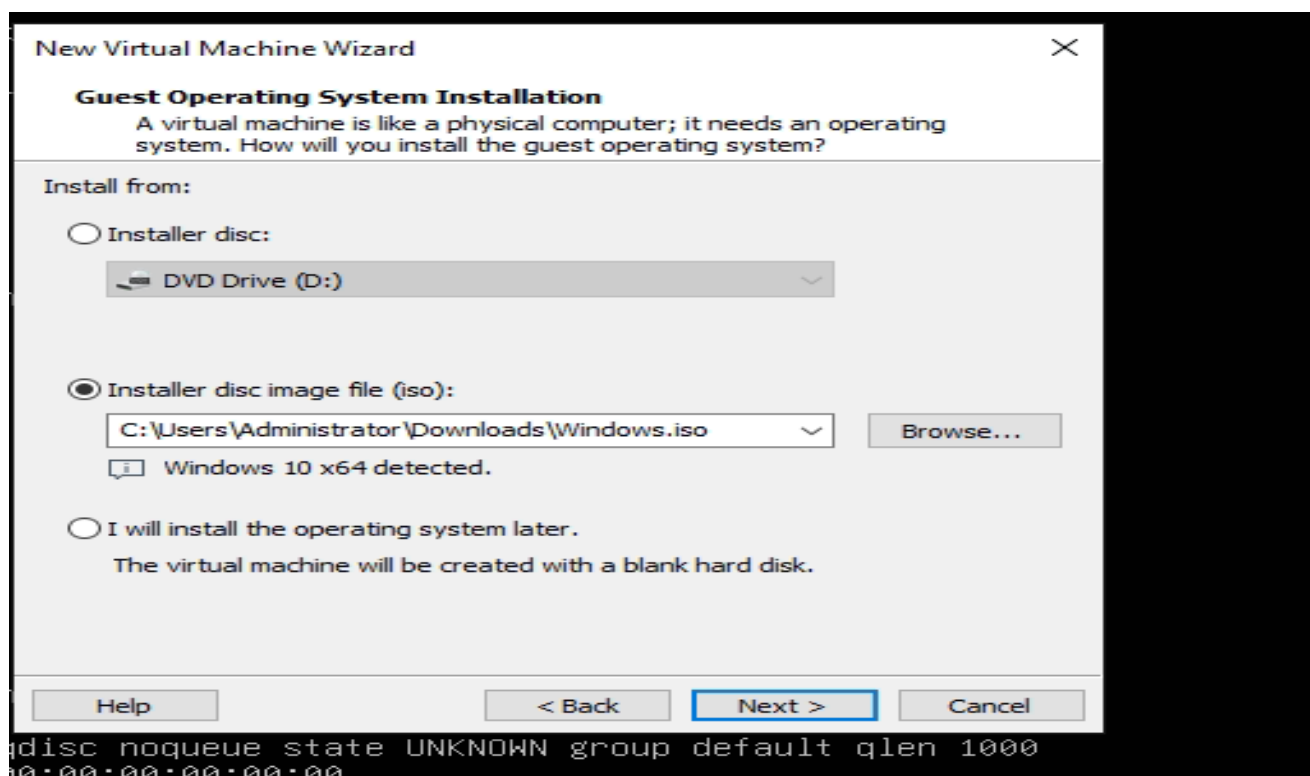


Figure 111-Installing Windows VM

After that we have installed windows, we need to install splunk forwarder on it.

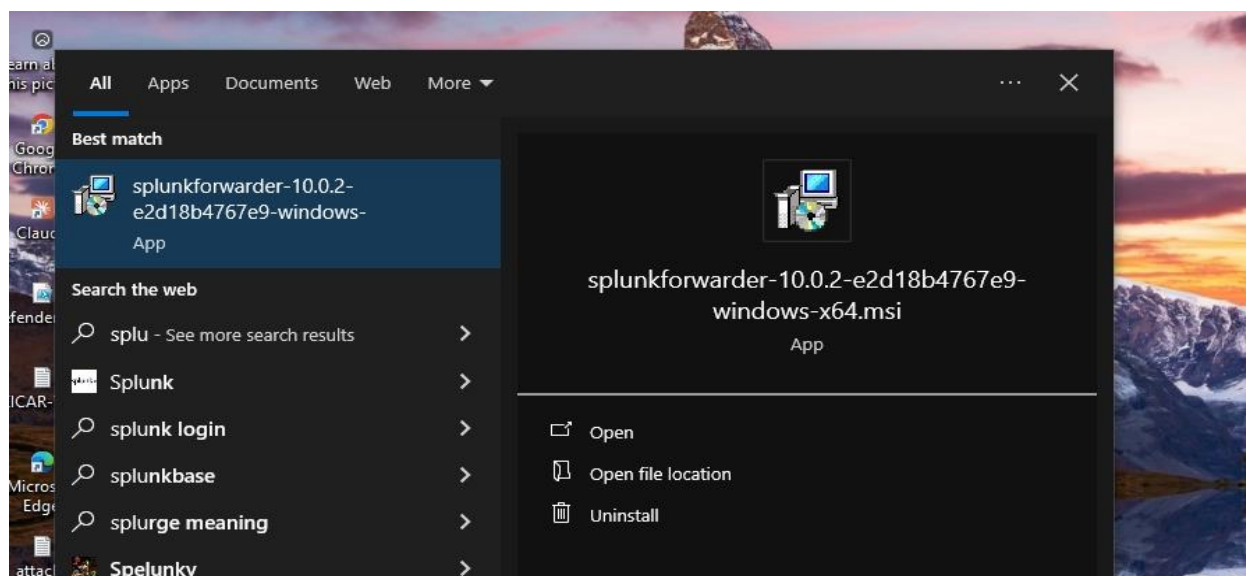


Figure 112 - Installing Forwarder

After installing forwarder, we need to give credentials for our splunk instance.

These are the commands for installing Forwarder in Linux.

```
# Download Splunk Universal Forwarder wget -O splunkforwarder-9.1.2-b6b9c8185839-linux-2.6-amd64.deb
```

```
"https://download.splunk.com/products/universalforwarder/releases/9.1.2/linux/splunkforwarder-9.1.2-b6b9c8185839-Linux-x86_64.tgz"
```

```
# For .deb package:
```

```
sudo dpkg -i splunkforwarder-9.1.2-b6b9c8185839-linux-2.6-amd64.deb
```

```
# Start the forwarder and accept license
```

```
sudo /opt/splunkforwarder/bin/splunk start --accept-license
```

```
# Set admin password
```

```
# Or set it non-interactively:
```

```
sudo /opt/splunkforwarder/bin/splunk edit user admin -password 'mohammad' -auth admin:mohammad
```

```
# Add forward server
```

```
sudo /opt/splunkforwarder/bin/splunk add forward-server <192.168.0.0>:9997 -auth admin:mohammad
```

```
# Enable boot start
```

```
sudo /opt/splunkforwarder/bin/splunk enable boot-start -user splunk
```

```
# Create splunk user if needed
```

```
sudo useradd -r -m -U -d /opt/splunkforwarder -s /bin/bash splunk
```

```
# Change ownership
```

```
sudo chown -R splunk:splunk /opt/splunkforwarder
```

```
# Restart forwarder
```

```
sudo /opt/splunkforwarder/bin/splunk restart
```

We will do the same on our Linux Vm.

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

You have new mail.
mohammad@mohammad:~$ ll
total 69740
drwxr-x--- 7 mohammad mohammad 4096 Dec  9 13:03 ./
drwxr-xr-x 4 root      root      4096 Dec  8 13:27 ../
-rw----- 1 mohammad mohammad 2552 Dec  8 21:05 .bash_history
-rw-r--r-- 1 mohammad mohammad 220  Mar 31 2024 .bash_logout
-rw-r--r-- 1 mohammad mohammad 3771 Mar 31 2024 .bashrc
drwx----- 3 mohammad mohammad 4096 Dec  9 07:30 .cache/
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:30 go/
-rw----- 1 mohammad mohammad 20  Dec  8 19:22 .lessht
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:49 .local/
drwxrwxr-x 2 mohammad mohammad 4096 Dec  9 13:03 MailHogData/
-rw-rw-r-- 1 mohammad mohammad 29122 Dec  9 13:03 .mailhog.log
-rw----- 1 mohammad mohammad 135  Dec  9 10:26 .msmtprc
-rw-rw-r-- 1 mohammad mohammad 27  Dec  7 18:52 passwords.txt
-rw-r--r-- 1 mohammad mohammad 807  Mar 31 2024 .profile
-rw-rw-r-- 1 mohammad mohammad 71311634 Nov 13 23:43 splunkforwarder-10.0.2-e2d18b4767e9-linux-and64.deb
drwx----- 2 mohammad mohammad 4096 Dec  1 20:24 .ssh/
-rw-r--r-- 1 mohammad mohammad 0  Dec  1 20:36 sudo_as_admin_successful
```

Figure 113- Installing Forwarder on Linux

Setting Alerts

To start configuring attacks we needed both systems running with Universal forwarder on them. Once we have received the activity we can start saving alert of suspicious activity, since we are using windows and Linux system, we will require their logs under their indexes. First, we will set both the system to send logs to Our splunk server for that we will need to configure their output.config file to our Splunk servers IP and other needed information.

These files are located in C:\programfiles\SplunkUniversalForwarder\etc\system\local

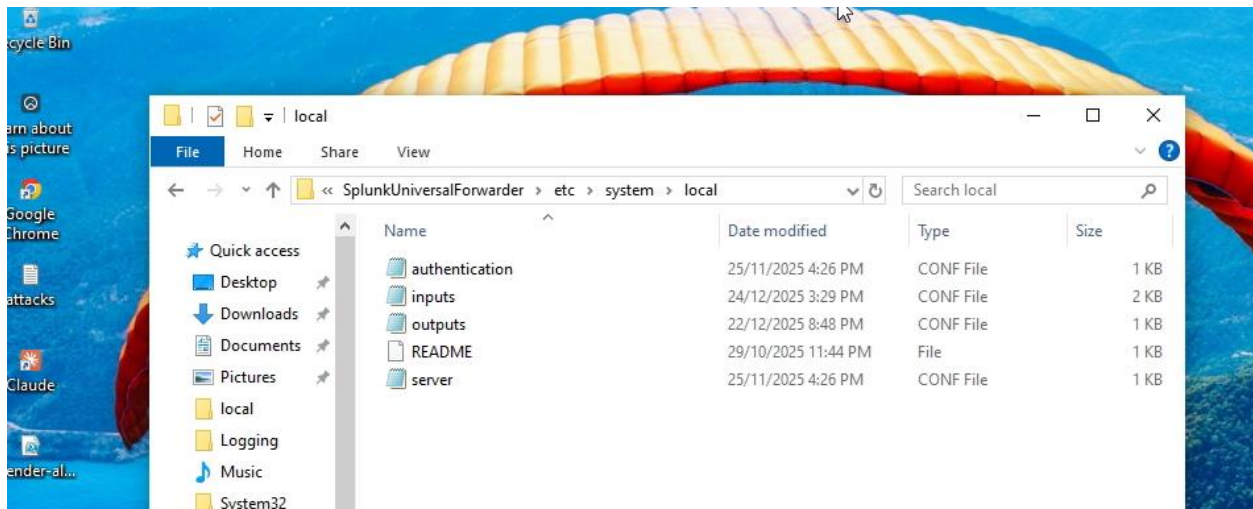


Figure 114 - input & output files of windows splunk forwarder

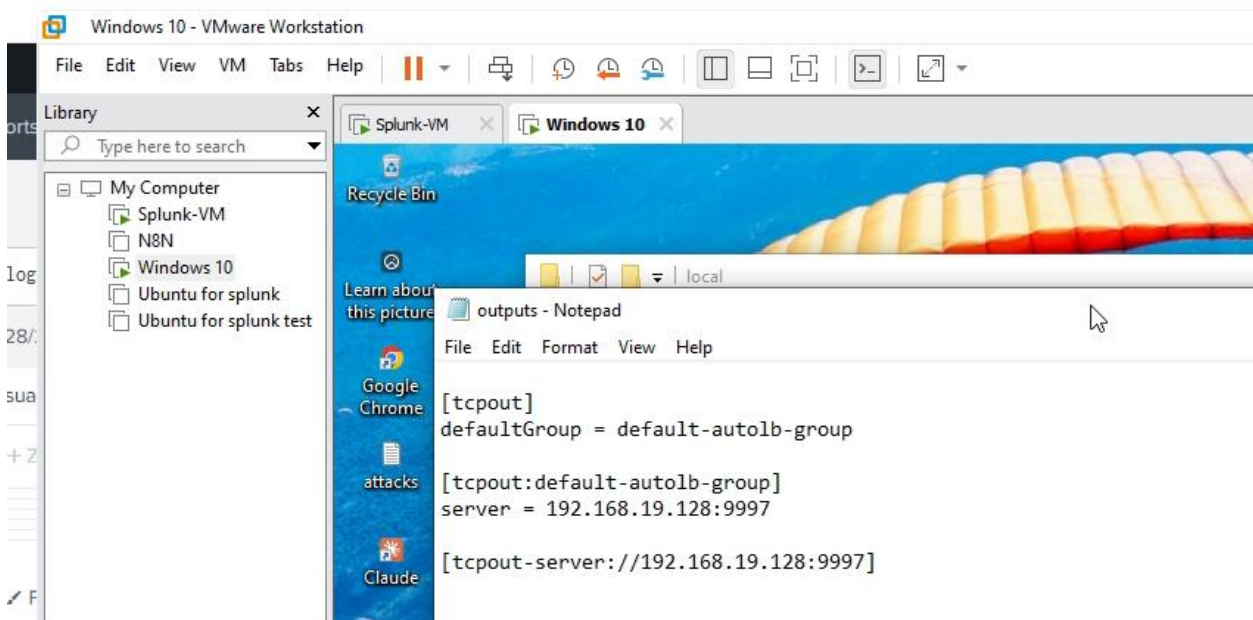


Figure 115- splunk server ip

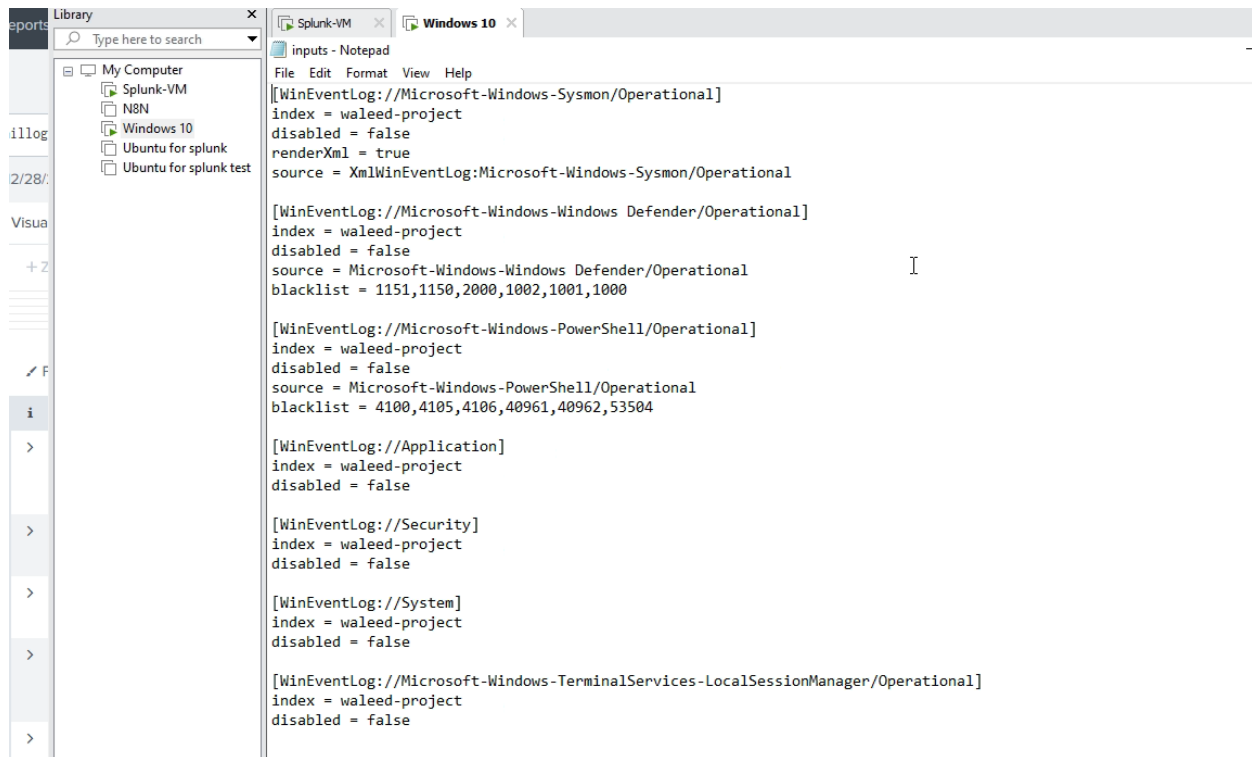
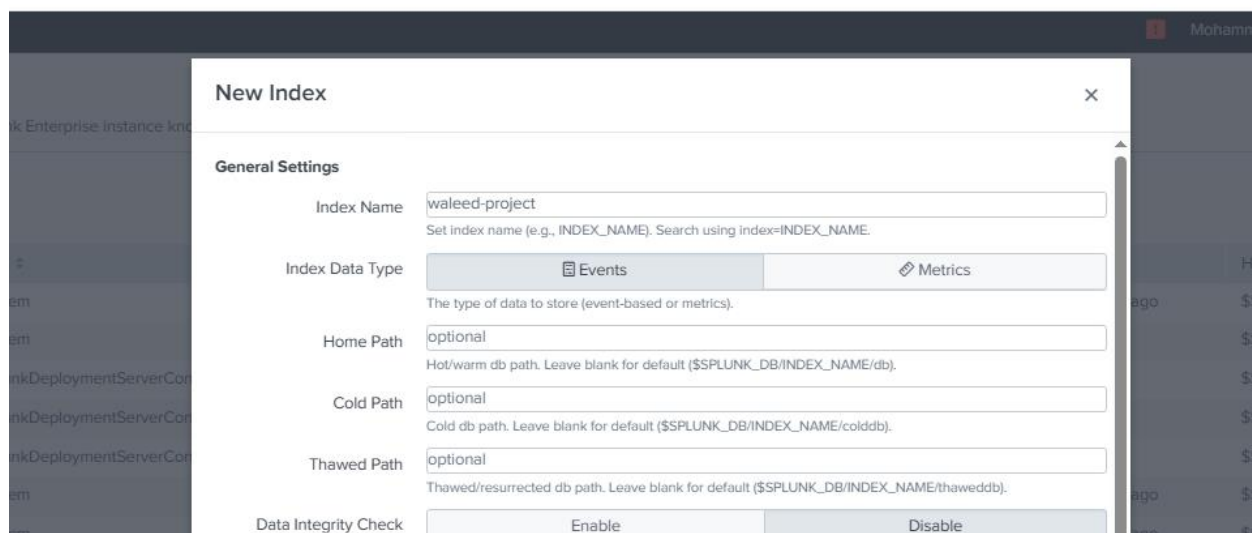


Figure 116 Inputs logs to be monitored

Attacks on windows



Brute Force Attack:

h page

Search) ▾

Edit Alert

×

Settings

Alert Brute-Force

Description

Optional

Search

index="waleed-project" EventCode=4625

|stats count by _time, ComputerName, user, src_ip

Alert type

Scheduled

Real-time

Expires

2 day(s) ▾

Trigger Conditions

Trigger alert when

Per-Result ▾

Throttle ?

☐

Trigger Actions

+ Add Actions ▾

When triggered

▼

🔔 Add to Triggered Alerts

Severity

Medium ▾

Remove

>

🔗 Webhook

Remove

Cancel

Save

Phishing Attack

For phishing attack we need to install an email server first

These were the commands for Installing SMTP server locally,

Install SMTP Server

Install-WindowsFeature SMTP-Server -IncludeManagementTools

Start SMTP Service:

cmd

net start smtpsvc

Configure SMTP Server:

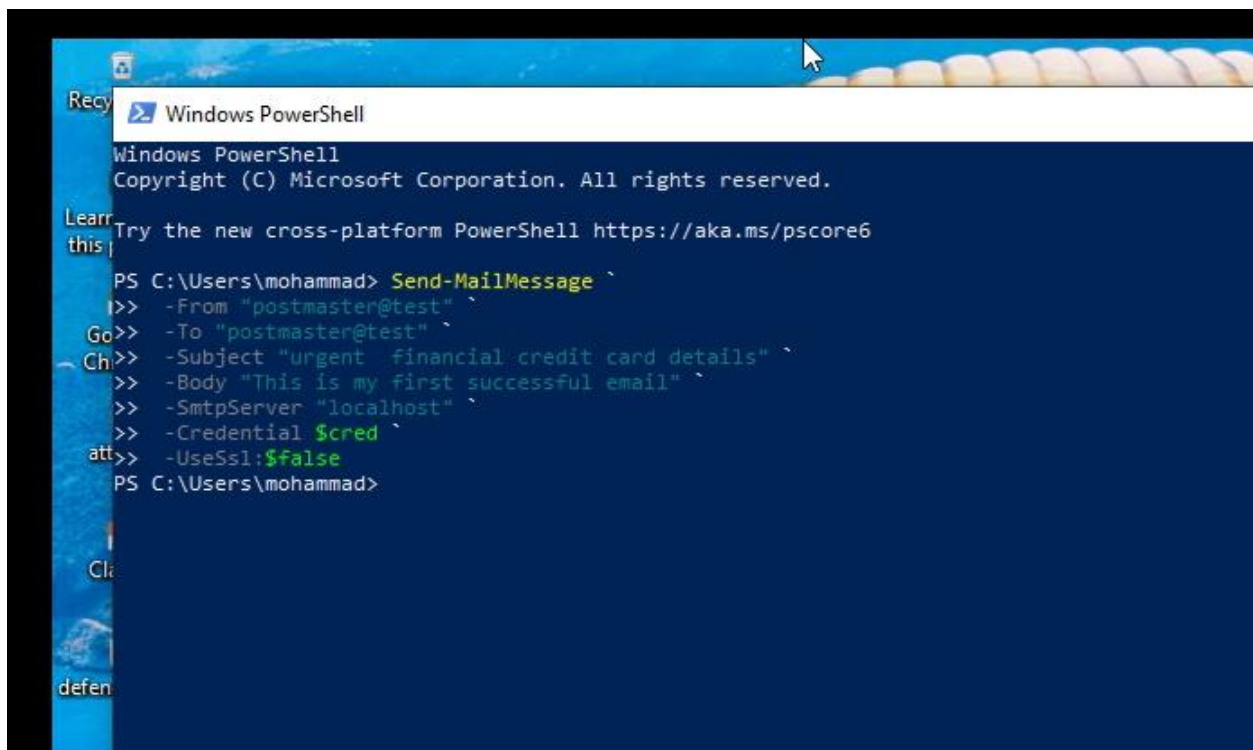
Set SMTP to start automatically

sc config smtpsvc start= auto

142 | Page

Check SMTP service status

sc query smtpsvc

A screenshot of a Windows PowerShell window. The title bar reads "Windows PowerShell". The window content shows the following text: "Windows PowerShell", "Copyright (C) Microsoft Corporation. All rights reserved.", "Try the new cross-platform PowerShell https://aka.ms/pscore6", "PS C:\Users\mohammad> Send-MailMessage -From 'postmaster@test' -To 'postmaster@test' -Subject 'urgent financial credit card details' -Body 'This is my first successful email' -SmtpServer 'localhost' -Credential \$cred -UseSsl:\$false", and "PS C:\Users\mohammad>". On the left side of the window, there is a vertical sidebar with several icons and labels: "Recy", "Learn this", "Go", "Ch", "att", "Cl", and "defen".

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\mohammad> Send-MailMessage -From 'postmaster@test' -To 'postmaster@test' -Subject 'urgent financial credit card details' -Body 'This is my first successful email' -SmtpServer 'localhost' -Credential $cred -UseSsl:$false
PS C:\Users\mohammad>
```

Figure 117 - sending mail locally

Privilege Escalation

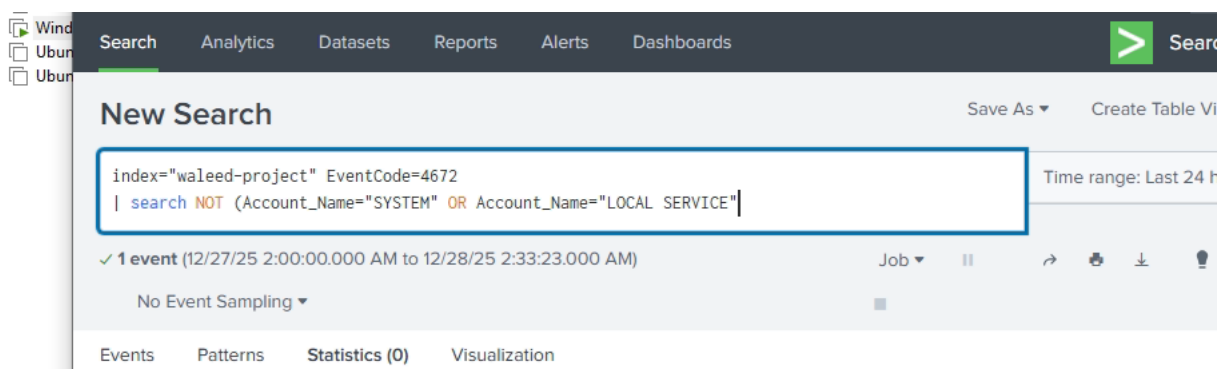


Figure 119 - Permissions Granted Event code

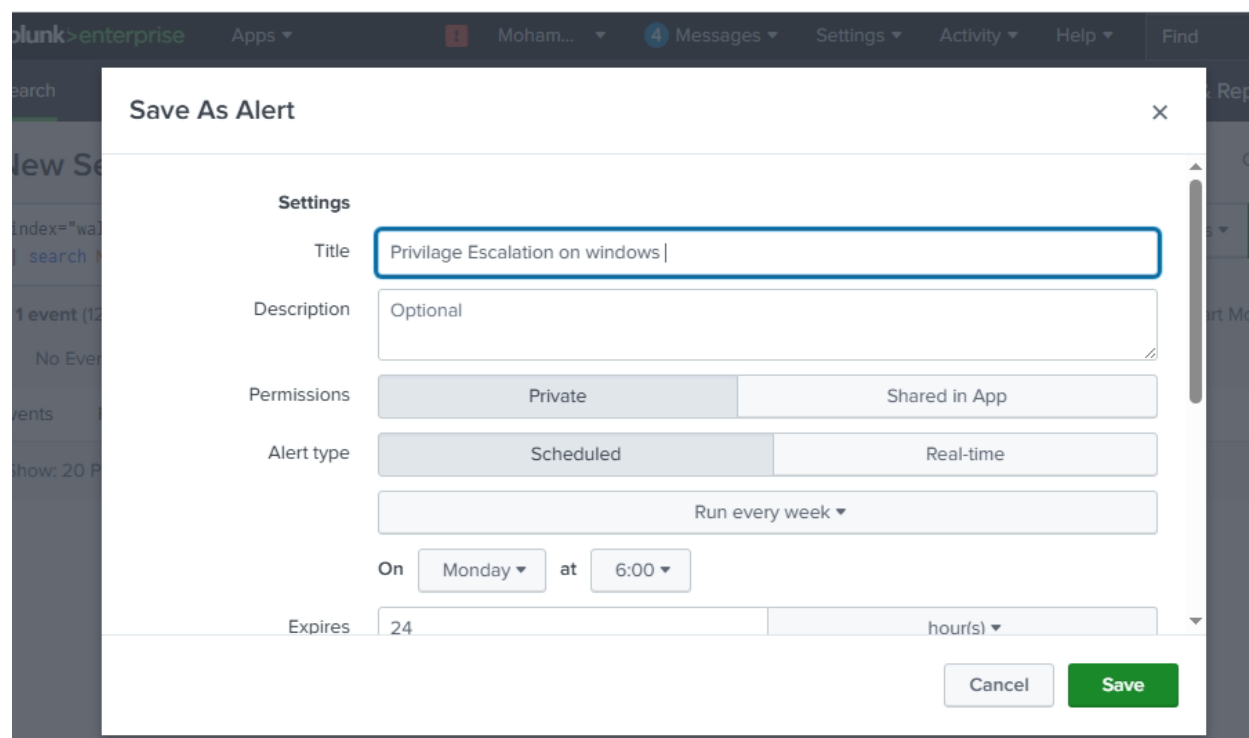


Figure 120 - Saving privilege escalation alert

Attacks On Linux

We will create an index for Linux, so we get all user activity on that

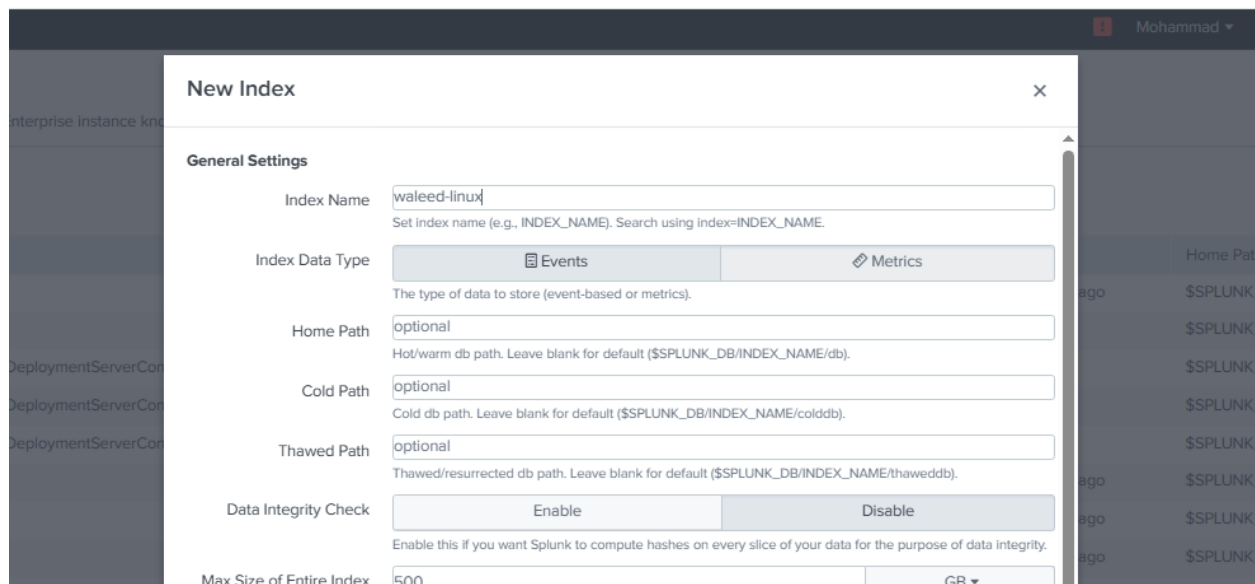


Figure 121 - Creating Linux index

Download Splunk Universal Forwarder

bash

```
wget -O splunkforwarder.tgz
```

```
"https://download.splunk.com/products/universalforwarder/releases/9.1.2/linux/splunkforwarder-9.1.2-b6436b649711-Linux-x86_64.deb"
```

Step 2: Extract the Package

bash

```
# Extract to /opt directory
```

```
sudo tar xvzf splunkforwarder.deb -C /opt
```

Step 3: Start Forwarder and Accept License

bash

```
sudo /opt/splunkforwarder/bin/splunk start --accept-license
```

```
# Set admin username and password when prompted
```

Step 4: Configure Forwarder to Start on Boot

bash

```
# Enable boot start
```

```
sudo /opt/splunkforwarder/bin/splunk enable boot-start
```

As you can see, we have 2 files in our Linux Forwarders directory,

```
mohammad@mohammad-linux:~$ cd /opt/splunkforwarder/etc/system/local/
mohammad@mohammad-linux:/opt/splunkforwarder/etc/system/local$ ll
total 24
drwxr-xr-x 2 splunkfwd splunkfwd 4096 Dec 25 23:13 ./
drwxr-xr-x 8 splunkfwd splunkfwd 4096 Dec  1 21:01 ../
-rw-r--r-- 1 splunkfwd splunkfwd  604 Dec 25 23:13 inputs.conf
-rw----- 1 splunkfwd splunkfwd  144 Dec 25 22:51 outputs.conf
-r--r--r-- 1 splunkfwd splunkfwd  265 Oct 29 23:44 README
-rw----- 1 splunkfwd splunkfwd  433 Dec  1 21:05 server.conf
mohammad@mohammad-linux:/opt/splunkforwarder/etc/system/local$
```

Figure 122 - Linux Forwarders files to be monitored

We will configure output.conf file to send all the alerts& activity to Our Splunk server.

We will use command `sudo nano output.conf`

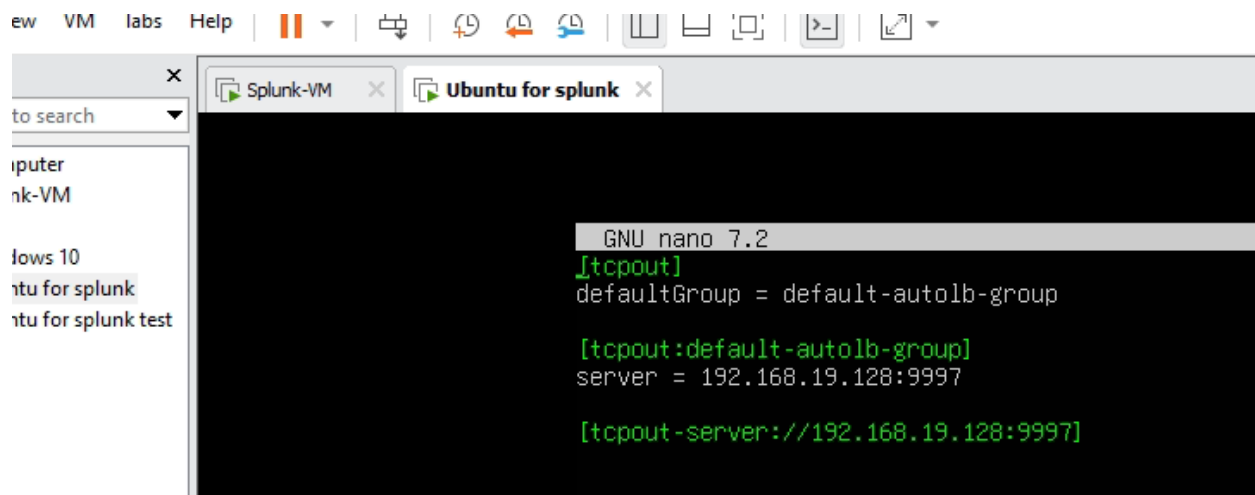
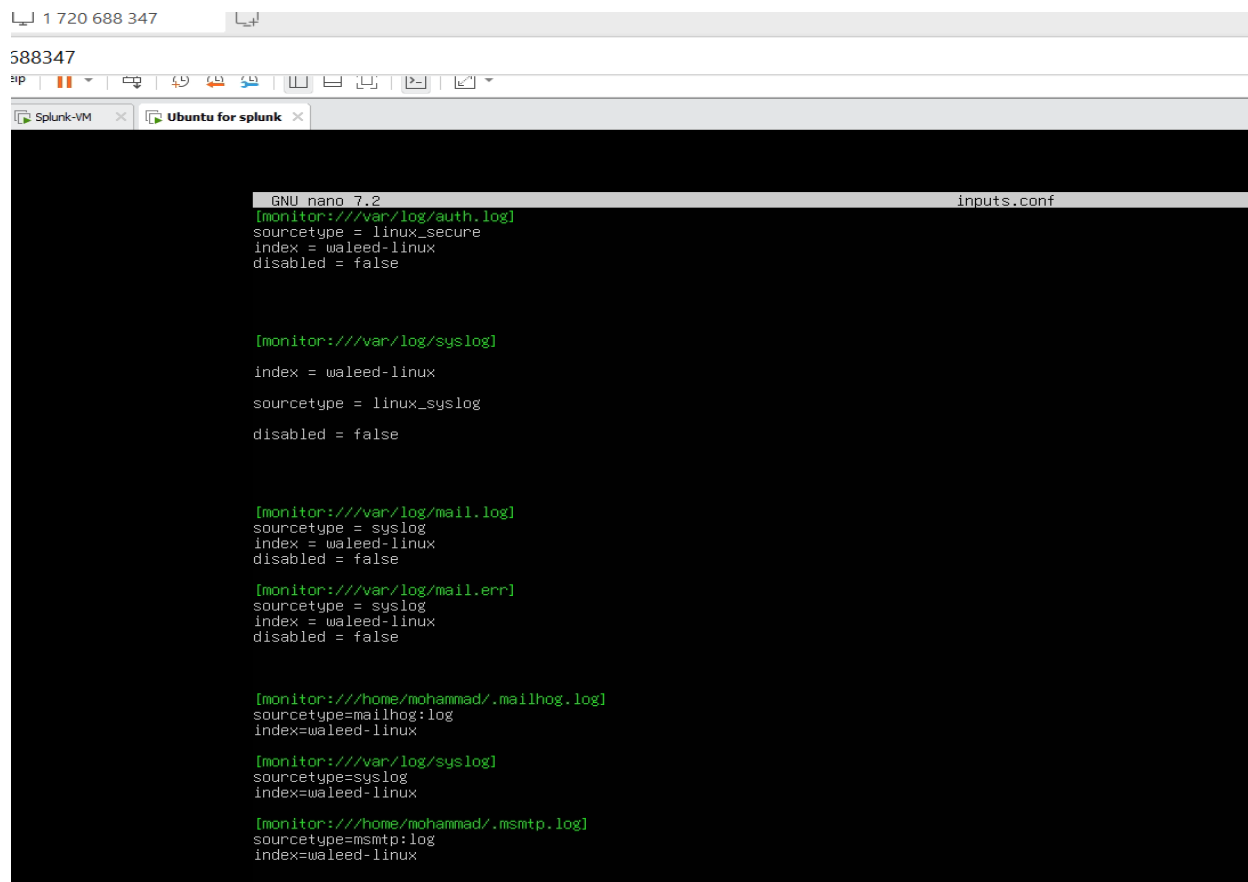


Figure 123 - setting splunk servers ip

Sending required logs for saving alerts from a received events and activity.



The screenshot shows a terminal window with the title bar "588347" and a tab labeled "Ubuntu for splunk". The terminal is running the nano text editor, editing the file "inputs.conf". The content of the file is as follows:

```
GNU nano 7.2 inputs.conf
[monitor:///var/log/auth.log]
sourcetype = linux_secure
index = waleed-linux
disabled = false

[monitor:///var/log/syslog]
index = waleed-linux
sourcetype = linux_syslog
disabled = false

[monitor:///var/log/mail.log]
sourcetype = syslog
index = waleed-linux
disabled = false

[monitor:///var/log/mail.err]
sourcetype = syslog
index = waleed-linux
disabled = false

[monitor:///home/mohammad/.mailhog.log]
sourcetype=mailhog:log
index=waleed-linux

[monitor:///var/log/syslog]
sourcetype=syslog
index=waleed-linux

[monitor:///home/mohammad/.msmtp.log]
sourcetype=msmtp:log
index=waleed-linux
```

Figure 124- Setting up log files needed for saving alerts and events.

After that you need to restart your splunk forwarder,

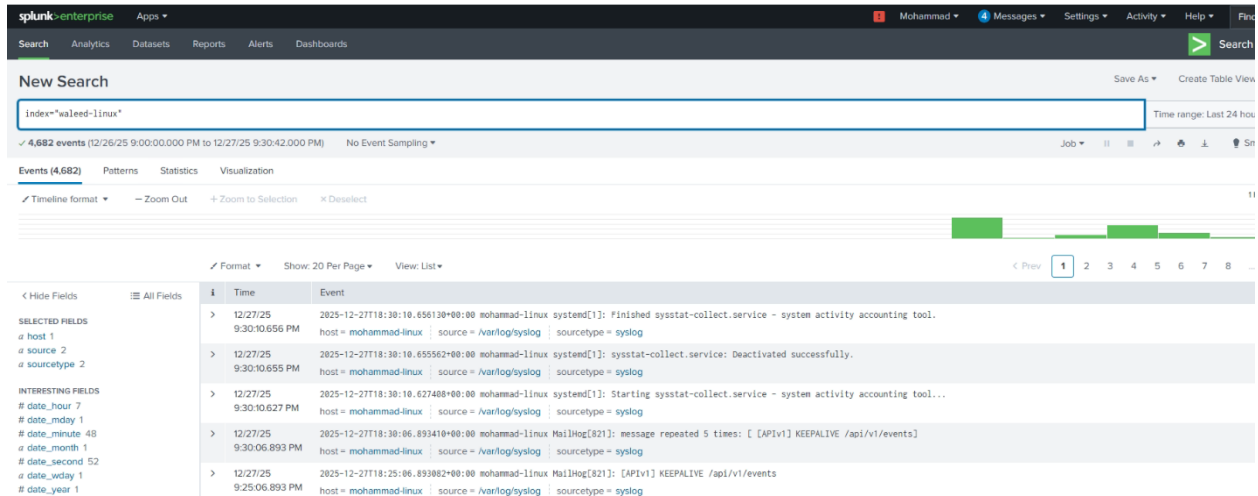


The screenshot shows a terminal window with the following commands and output:

```
drwxr-xr-x 12 root root      4096 Aug  5 16:54 usr/
drwxr-xr-x 13 root root      4096 Dec  1 20:24 var/
mohammad@mohammad-linux:/$ cd home
mohammad@mohammad-linux:/home$ ll
total 20
drwxr-xr-x  5 root    root    4096 Dec 27 15:17 ./
drwxr-xr-x 23 root    root    4096 Dec  1 20:19 ../
drwxr-x---  2 hacker  hacker  4096 Dec  8 13:27 hacker/
drwxr-x---  7 mohammad mohammad 4096 Dec  9 13:03 mohammad/
drwxr-x---  2 testuser testuser 4096 Dec 27 15:17 testuser/
mohammad@mohammad-linux:/home$ cd mohammad/
mohammad@mohammad-linux:~$ sudo /opt/splunkforwarder/bin/splunk restart_
```

After that we need to perform activity for creating specific alerts with which our system can correlate to later and create an alert.

As we can see there are lot of events and logs under our Linux index.



Brute Force Attack

A brute force attack is generated if there is a failed password alert.

First, we will create Some failed password logs .

```
drwxr-xr-x 23 root root 4096 Dec 1 20:19 ../
drwxr-x--- 2 hacker hacker 4096 Dec 8 13:27 hacker/
drwxr-x--- 7 mohammad mohammad 4096 Dec 9 13:03 mohammad/
drwxr-x--- 2 testuser testuser 4096 Dec 27 15:17 testuser/
mohammad@mohammad-linux:/home$ cd mohammad/
mohammad@mohammad-linux:~$ ssh whatever@localhost
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
whatever@localhost: Permission denied (publickey,password).
mohammad@mohammad-linux:~$
```

Figure 125- Generating Failed Password logs

Then we will go in search and report and check for failed password activity.

New Search

Save As Create

index=waleed-linux sourcetype=linux_secure "Failed password"

Time range:

10 events (12/26/25 9:00:00.000 PM to 12/27/25 9:37:16.000 PM) No Event Sampling

Job

Events (10) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

	Time	Event
Hide Fields All Fields SELECTED FIELDS a host 1 a source 1 a sourcetype 1 INTERESTING FIELDS a action 2 a app 1 # date_hour 4 # date_mday 1 # date_minute 5 # date_month 1 # date_second 10 a date_wday 1 # date_year 1 # date_zone 1 a eventtype 6 a index 1 # linecount 3 # pid 6	12/27/25 9:34:59.792 PM	2025-12-27T18:34:59.792649+00:00 mohammad-linux sshd[2972]: Failed password for invalid user whatever from 127.0.0.1 port 51032 ssh2 2025-12-27T18:35:00.792035+00:00 mohammad-linux sshd[2972]: Connection closed by invalid user whatever 127.0.0.1 port 51032 [preauth] 2025-12-27T18:35:00.792163+00:00 mohammad-linux sshd[2972]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 2025-12-27T18:35:01.050689+00:00 mohammad-linux CRON[2974]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) 2025-12-27T18:35:01.065232+00:00 mohammad-linux CRON[2974]: pam_unix(cron:session): session closed for user root
	12/27/25 9:34:54.306 PM	2025-12-27T18:34:54.306590+00:00 mohammad-linux sshd[2972]: Failed password for invalid user whatever from 127.0.0.1 port 51032 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
	12/27/25 9:34:45.953 PM	2025-12-27T18:34:45.953042+00:00 mohammad-linux sshd[2972]: Failed password for invalid user whatever from 127.0.0.1 port 51032 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
	12/27/25 8:18:37.371 PM	2025-12-27T17:18:37.371608+00:00 mohammad-linux sshd[2811]: Failed password for invalid user fakeuser from 127.0.0.1 port 56372 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure
12/27/25 8:18:23.158 PM	2025-12-27T17:18:23.158505+00:00 mohammad-linux sshd[2807]: Failed password for invalid user fakeuser from 127.0.0.1 port 33128 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure	
12/27/25 6:59:22.956 PM	2025-12-27T15:59:22.956308+00:00 mohammad-linux sshd[2234]: Failed password for invalid user fakeuser from 127.0.0.1 port 43074 ssh2 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure	

Figure 126- Search for a failed password

Saving a Brute Force alert.

Edit Alert

×

Settings

Alert Brute Force On linux testing

Description Triggered when ever a Failed password is detected

Search index=waleed-linux sourcetype=linux_secure "Failed password"

Alert type Scheduled Real-time

Expires 24 hour(s) ▼

Trigger Conditions

Trigger alert when Per-Result ▼

Throttle ? ☐

Trigger Actions

+ Add Actions ▼

When triggered ▼

🔔 Add to Triggered Alerts

Remove

Severity Medium ▼

Figure 127- Saving a brute force alert

Privilege Escalation

Privilege escalation alert is generated when someone tries to perform any activity which he has not any privileges of so in Linux “not in sudoers” alert is generated.

1.create user

```
sudo useradd -m testuser
```

```
sudo passwd testuser
```

2. Switch to that user:

```
bash
```

```
su testuser
```

3. Try sudo commands (this will trigger the alert):

```
bash
```

```
sudo apt update
```

```
or
```

```
bash
```

```
sudo systemctl restart ssh
```

```
or
```

```
bash
```

```
sudo nano /etc/passwd
```

First, we will generate a Not in Sudoers Alert

```

mohammad@mohammad-linux:~$ ssh whatever@localhost
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
Permission denied, please try again.
whatever@localhost's password:
whatever@localhost: Permission denied (publickey,password).
mohammad@mohammad-linux:~$ sudo su - testuser
[sudo] password for mohammad:
testuser@mohammad-linux:~$ sudo ls
[sudo] password for testuser:
testuser is not in the sudoers file.
This incident has been reported to the administrator.
testuser@mohammad-linux:~$

```

Figure 128- Generating not in Sudoers Alert

New Search

✓ 2 events (12/26/25 9:00:00.000 PM to 12/27/25 9:52:19.000 PM)
No Event Sampling ▼

Events (2)

Patterns

Statistics

Visualization

✓ Timeline format ▼
— Zoom Out
+ Zoom to Selection
× Deselect

✓ Format ▼
Show: 20 Per Page ▼
View: List ▼

		i	Time	Event
		>	12/27/25 9:50:37.554 PM	2025-12-27T18:50:37.554629+00:00 mohammad-linux sudo: testuser : user NOT in sudoers ; TTY=pts/0 host = mohammad-linux source = /var/log/auth.log sourcetype = linux_secure

< Hide Fields

≡ All Fields

SELECTED FIELDS

a host 1

Figure 129- checking for not in sudoers alert

Edit Alert

Settings

AlertPrivilege Escalation on Linux

DescriptionOptional

Searchindex="waleed-linux" not in sudoers

Alert type

Scheduled

Real-time

Expires

24

hour(s)

Trigger Conditions

Trigger alert whenPer-Result

Throttle?

Trigger Actions

+ Add Actions

When triggered

SeverityHigh

Cancel

Save

Figure 130 - Saving privilege escalation alert on Linux

Phishing Email

To configure phishing email first we will install mail server.

```
5 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

You have new mail.
mohammad@mohammad-linux:~$ ll
total 69740
drwxr-xr-x 7 mohammad mohammad 4096 Dec  9 13:03 ./
drwxr-xr-x 5 root      root      4096 Dec 27 15:17 ../
-rw-r--r-- 1 mohammad mohammad 3568 Dec 25 23:56 .bash_history
-rw-r--r-- 1 mohammad mohammad 220  Mar 31 2024 .bash_logout
-rw-r--r-- 1 mohammad mohammad 3771 Mar 31 2024 .bashrc
drwxr-xr-x 3 mohammad mohammad 4096 Dec  9 07:30 .cache/
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:30 go/
-rw-r--r-- 1 mohammad mohammad 20  Dec  8 19:22 .lessht
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:49 .local/
drwxrwxr-x 2 mohammad mohammad 4096 Dec  9 13:03 MailHogData/
-rw-rw-r-- 1 mohammad mohammad 29122 Dec  9 13:03 .mailhog.log
-rw-r--r-- 1 mohammad mohammad 135  Dec  9 10:26 .msmtprc
-rw-rw-r-- 1 mohammad mohammad 27  Dec  7 18:52 passwords.txt
-rw-r--r-- 1 mohammad mohammad 807  Mar 31 2024 .profile
-rw-rw-r-- 1 mohammad mohammad 71311634 Nov 13 23:43 splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64.deb
drwxr-xr-x 2 mohammad mohammad 4096 Dec 26 01:01 .ssh/
-rw-r--r-- 1 mohammad mohammad 0  Dec  1 20:36 .sudo_as_admin_successful
-rw-rw-r-- 1 mohammad mohammad 165  Dec  9 09:50 .wget-hsts
mohammad@mohammad-linux:~$
```

Figure 131- MailHog Email server

```
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:30 go/
-rw-r--r-- 1 mohammad mohammad 20  Dec  8 19:22 .lessht
drwxrwxr-x 3 mohammad mohammad 4096 Dec  9 07:49 .local/
drwxrwxr-x 2 mohammad mohammad 4096 Dec  9 13:03 MailHogData/
-rw-rw-r-- 1 mohammad mohammad 29122 Dec  9 13:03 .mailhog.log
-rw-r--r-- 1 mohammad mohammad 135  Dec  9 10:26 .msmtprc
-rw-rw-r-- 1 mohammad mohammad 27  Dec  7 18:52 passwords.txt
-rw-r--r-- 1 mohammad mohammad 807  Mar 31 2024 .profile
-rw-rw-r-- 1 mohammad mohammad 71311634 Nov 13 23:43 splunkforwarder-10.0.2-e2d18b4767e9-linux-amd64.deb
drwxr-xr-x 2 mohammad mohammad 4096 Dec 26 01:01 .ssh/
-rw-r--r-- 1 mohammad mohammad 0  Dec  1 20:36 .sudo_as_admin_successful
-rw-rw-r-- 1 mohammad mohammad 165  Dec  9 09:50 .wget-hsts
mohammad@mohammad-linux:~$ echo "Subject: Urgent Credit card Information Cridentials" | sendmail mohammad@localdomain
mohammad@mohammad-linux:~$
```

Figure 132- Sending Email Locally on Linux

Edit Alert

Settings

Alert: **Phishing On Linux**

Description: Original Alert

Search: index="waleed-linux" Urgent Credit card cridentials

Alert type: ☒ Scheduled ☐ Real-time

Run on Cron Schedule ▼

Time Range: Last 24 hours ▶

Cron Expression: *****
e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires: 24 hour(s) ▼

Trigger Conditions

Figure 133 - Saving a Phishing Email Alert

Triggered Alerts

Filter App Search & Report... Owner All owners ▼ Severity All severity ▼ Alert name All alerts ▼ Showing 1 - 20 of 138 results Show: 20 per page ▼

☐	Time ▼	Alert name ↕	App ↕	Type ↕	Severity ↕	Mode ↕	Actions
☐	2025-12-28 03:11:01 +03	Phishing On Linux	search	Scheduled	Medium	Digest	View Results Edit Search Delete
☐	2025-12-28 03:05:01 +03	Suspicious User created	search	Scheduled	Medium	Digest	View Results Edit Search Delete
☐	2025-12-28 03:00:05 +03	Phishing On Linux	search	Scheduled	Medium	Digest	View Results Edit Search Delete

Figure 134- Phishing alert Triggered

Suspicious user Alert

Suspicious alert is generated when a new user is added to the users group.

Adding new user

sudo useradd -m testuser

sudo passwd testuser

```
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
1025/tcp  open  NFS-or-IIS

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
mohammad@mohammad-linux:~$ sudo useradd 123
[sudo] password for mohammad:
useradd: invalid user name '123': use --badname to ignore
mohammad@mohammad-linux:~$ sudo useradd usertest
mohammad@mohammad-linux:~$
```

Figure 135- Adding a new user to user group

Suspicious User created

index="waheed-linux" "useradd" OR "new user"

✓ 5 events (12/26/25 10:00:00.000 PM to 12/27/25 10:31:23.000 PM) No Event Sampling ▼

Events (5) Patterns Statistics Visualization

✓ Timeline format ▼ — Zoom Out + Zoom to Selection × Deselect

Format ▼ Show: 20 Per Page ▼ View: List ▼

< Hide Fields	≡ All Fields	i	Time	Event
>				
			12/27/25	2025-12-27T19:25:13.928507+00:00 mohammad-linux sudo: mohammad : TTY=pts/0 ; PWD=/home/mohammad ; USER=root ; COMMAND
			10:25:13.928 PM	2025-12-27T19:25:13.931845+00:00 mohammad-linux sudo: pam_unix(sudo:session): session opened for user root(uid=0) by
				2025-12-27T19:25:13.973724+00:00 mohammad-linux useradd[3114]: new group: name=usertest, GID=1004
				2025-12-27T19:25:13.974360+00:00 mohammad-linux useradd[3114]: new user: name=usertest, UID=1004, GID=1004, home=/hom
				2025-12-27T19:25:14.013825+00:00 mohammad-linux sudo: pam_unix(sudo:session): session closed for user root

Figure 136- Looking for useradd Activity

Save As Alert

Settings

Title

Suspicious User created

Description

testing

Permissions

Private

Shared in App

Alert type

Scheduled

Real-time

Run on Cron Schedule

Time Range

Last 24 hours

Cron Expression

e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires

24

hour(s)

Trigger Conditions

Trigger alert when

Number of Results

is greater than

0

Trigger

Once

For each result

Throttle ?

☒

Suppress triggering for

10

minute(s)

Figure 137- Saving Suspicious User Alert

Setting Up Automation

To Set Automation we will start from Download our automation software on Ubuntu Server.

Ater installing the Server we will install docker and compose .yaml file for installing N8N for eliminating dependencies problem.

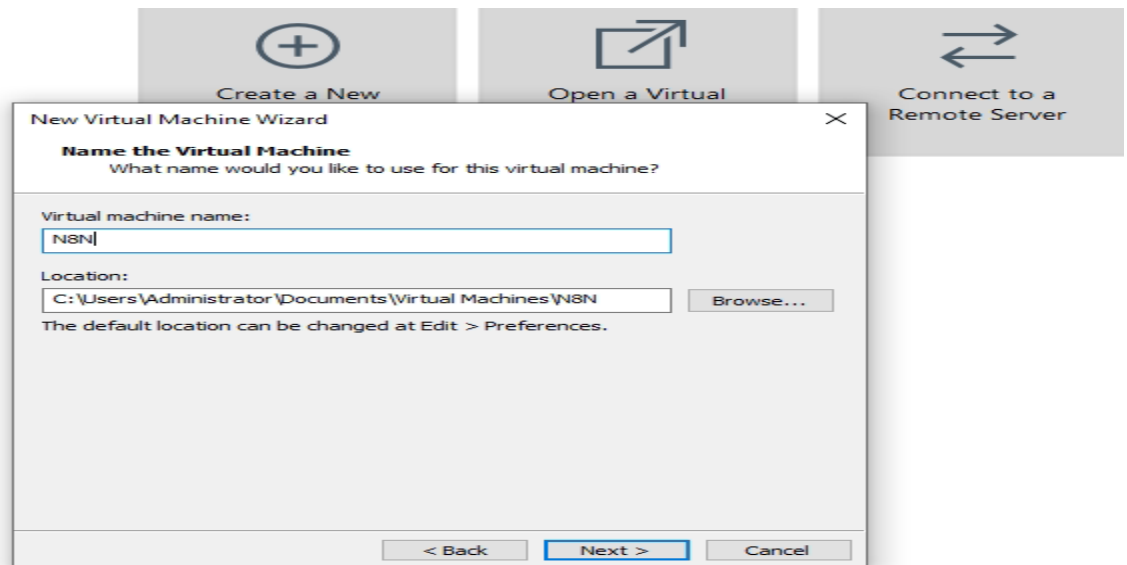


Figure 138- Creating N8N server

```
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

n8n@n8n:~$ ll
total 32
drwxr-x--- 5 n8n  n8n  4096 Nov 25 19:19 ./
drwxr-xr-x 3 root root  4096 Nov 25 10:26 ../
-rw-r--r-- 1 n8n  n8n   220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 n8n  n8n  3771 Mar 31  2024 .bashrc
drwx----- 2 n8n  n8n  4096 Nov 25 10:27 .cache/
drwxrwxr-x 3 n8n  n8n  4096 Nov 25 19:58 n8n-compose/
-rw-r--r-- 1 n8n  n8n   807 Mar 31  2024 .profile
drwx----- 2 n8n  n8n  4096 Nov 25 10:26 .ssh/
-rw-r--r-- 1 n8n  n8n     0 Nov 25 10:27 .sudo_as_admin_successful
n8n@n8n:~$
```

First we will create a Directory n8n-Compose for installing N8n in to avoid dependencies.

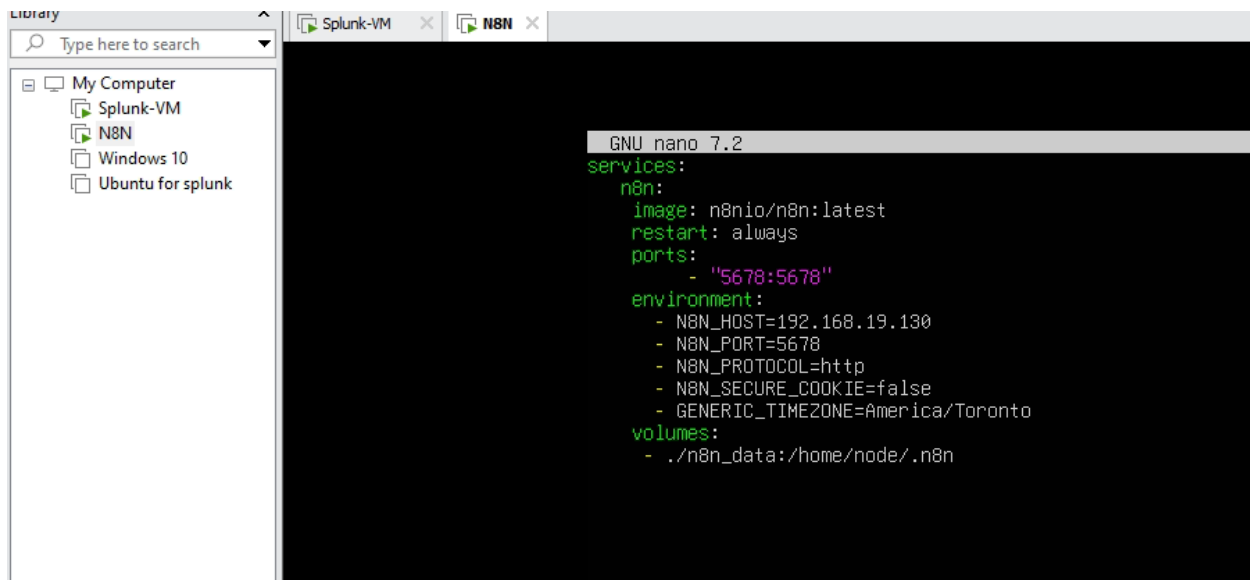


Figure 139- Docker Compose. Yaml for installing N8N

After that it is Installed, we can access the GUI at port 5678.

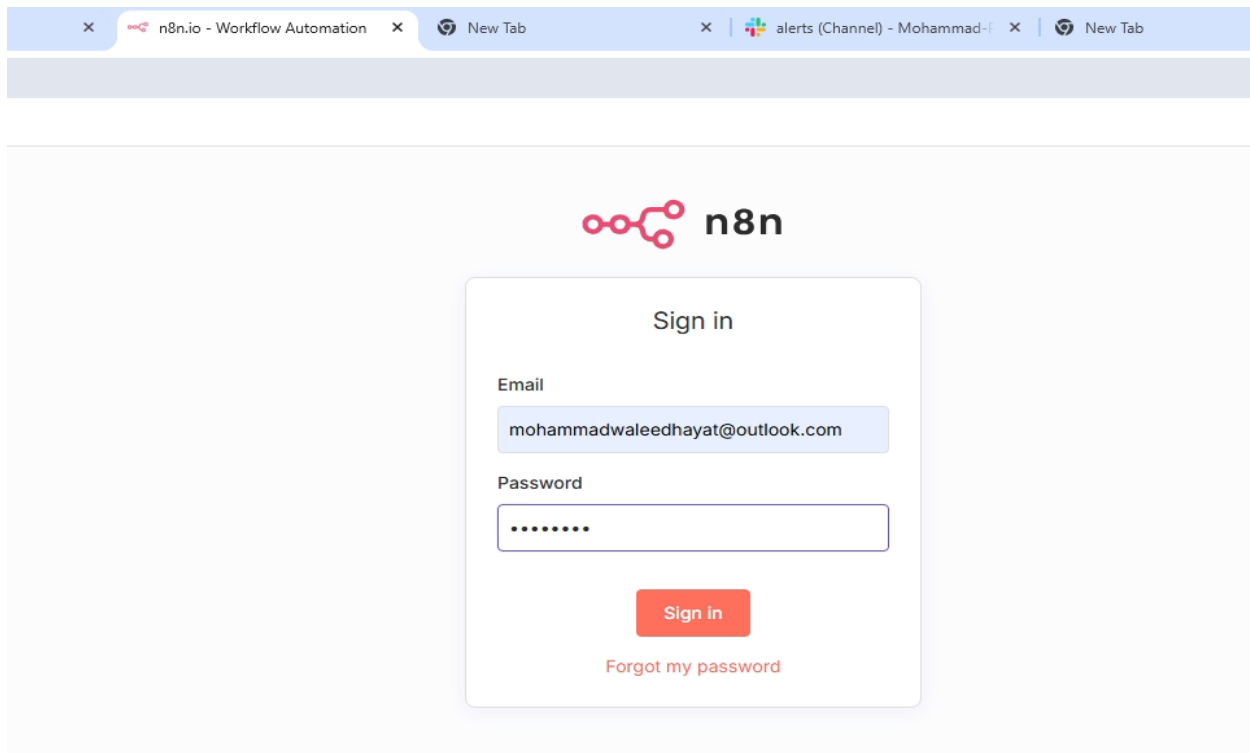


Figure 140- Accessing N8N

To start making Alert notification we will create a Webhook for catching alerts in N8N, and select the post method and we will copy its API and add it to send to a Webhook option of our triggered Alerts in Splunk, so each time an Alert is triggered it sends its result to this webhook.

Webhook Listen for test event

Parameters Settings Docs

Webhook URLs

Test URL Production URL

POST http://192.168.19.130:5678/webhook-test/e02bae01-3e22-45a0-95a0-37c4602dea5e

HTTP Method

POST

Path

e02bae01-3e22-45a0-95a0-37c4602dea5e

Authentication

None

OUTPUT

1 item

headers
accept-encoding : identity
content-length : 397
host : 192.168.19.130:5678
content-type : application/json
user-agent : Splunk/FF289EB5-D1CA-42E4-9F9E-30B0D80471AB
connection : close

Figure 141- - Setting Up webhook for catching Alerts from Splunk

The screenshot displays the Splunk Alert configuration page. At the top, there's a 'Description' field with the placeholder 'Optional'. Below it is a 'Search' field containing the query: `index="waleed-project" EventCode=4625` and a sub-query: `| stats count by _time, ComputerName, user, src_ip`. The 'Alert type' is set to 'Real-time' (selected over 'Scheduled'). The 'Expires' field is set to '2' days. Under 'Trigger Conditions', 'Trigger alert when' is set to 'Per-Result'. The 'Throttle' checkbox is unchecked. In the 'Trigger Actions' section, a '+ Add Actions' button is visible. Below it, a table lists the actions. The first action is 'Add to Triggered Alerts'. The second action is 'Webhook', which is expanded to show a 'URL' field with the value `http://192.168.19.130:5678/webhook-test/e`. Below the URL, there is explanatory text: 'Specified URL to send JSON payload via HTTP POST (ex., https://your.server.com/api/v1/webhook).' and a 'Learn More' link. At the bottom right, there are 'Cancel' and 'Save' buttons.

Figure 142- Saving Webhook Api in Splunk's Alert

Now we will Connect Open AI Chatbot Api to Our webhook to create Customized incident Response recommendations according to each type of alert.

We will select the method Send a message give a prompt to act the way we want it to.

We have created 2 roles, 1st role is System which is for setting the AI's identity and behavior. We have given instructions to act like an SOC tier 1 analyst and has been instructed to evaluate incoming alerts based on the MITRE ATT&CK framework, identifying tactics, techniques, and procedures (TTPs) associated with potential threats. Upon receiving alert data, the AI provides initial triage recommendations, severity assessments, and suggests appropriate response actions aligned with industry's best practices. It correlates indicators of compromise (IOCs) such as suspicious IP addresses, user behaviors, and system anomalies to determine if an alert represents a genuine security incident or a false positive.

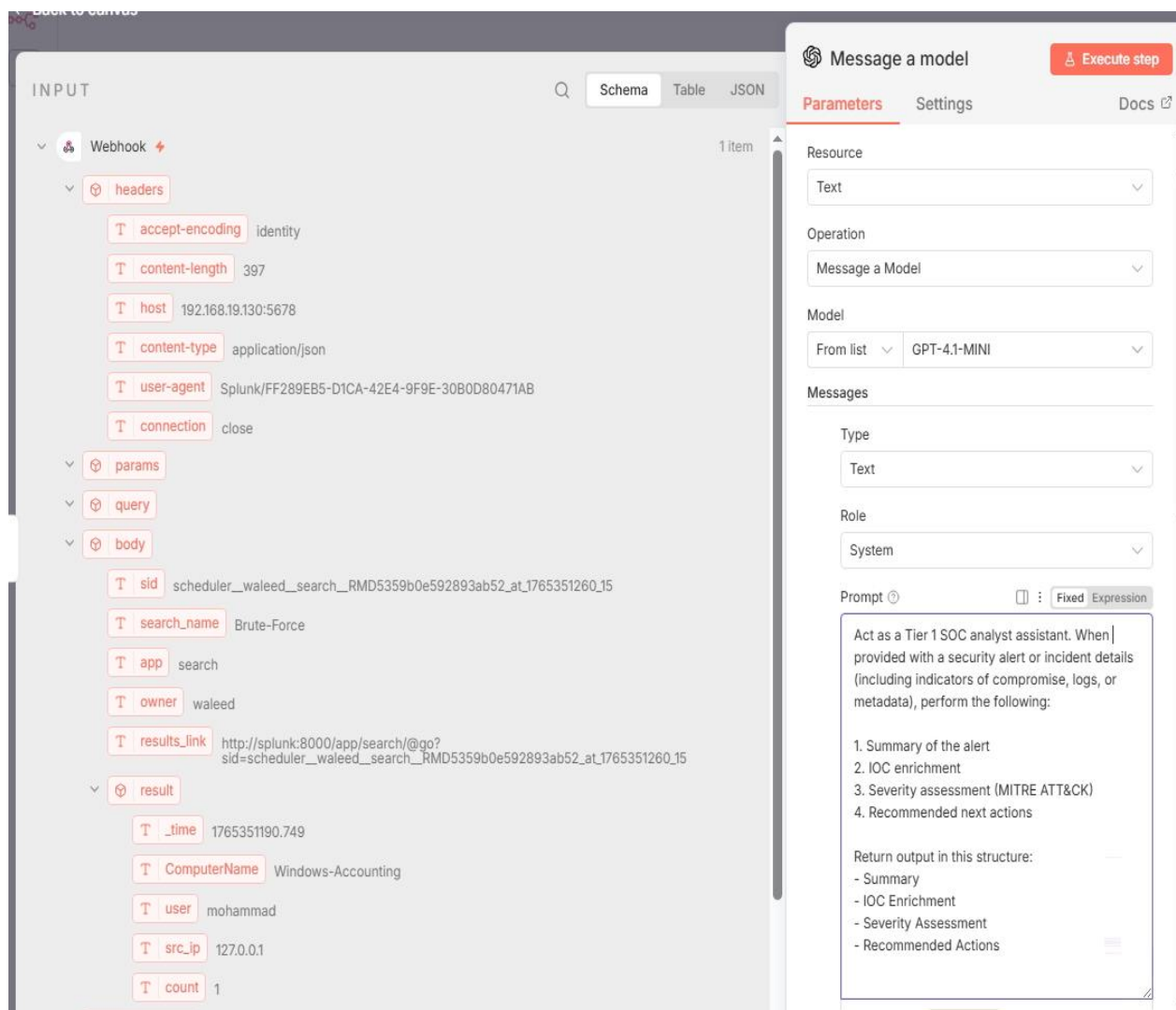


Figure 143- Open AI's 1st role to Act like an Analyst

The 2nd role is for System where we are training system that what to do with received data from webhook. We will prompt a code in .json to train system We trained n8n to catch incoming alert data from a webhook and format it for OpenAI. When the webhook receives a security alert (like a brute-force attack, Phishing, Privilege escalation), n8n extracts the alert name from \$json.body.search_name and the detailed results from \$json.body.result. It then structures this information into a readable message format using JSON.stringify to properly format the alert details with indentation. This formatted alert message is then sent to OpenAI's API, which can analyze the security event, provide context about the threat, suggest remediation steps, or generate human-

readable summaries of the technical alert data for security teams to review and act upon.

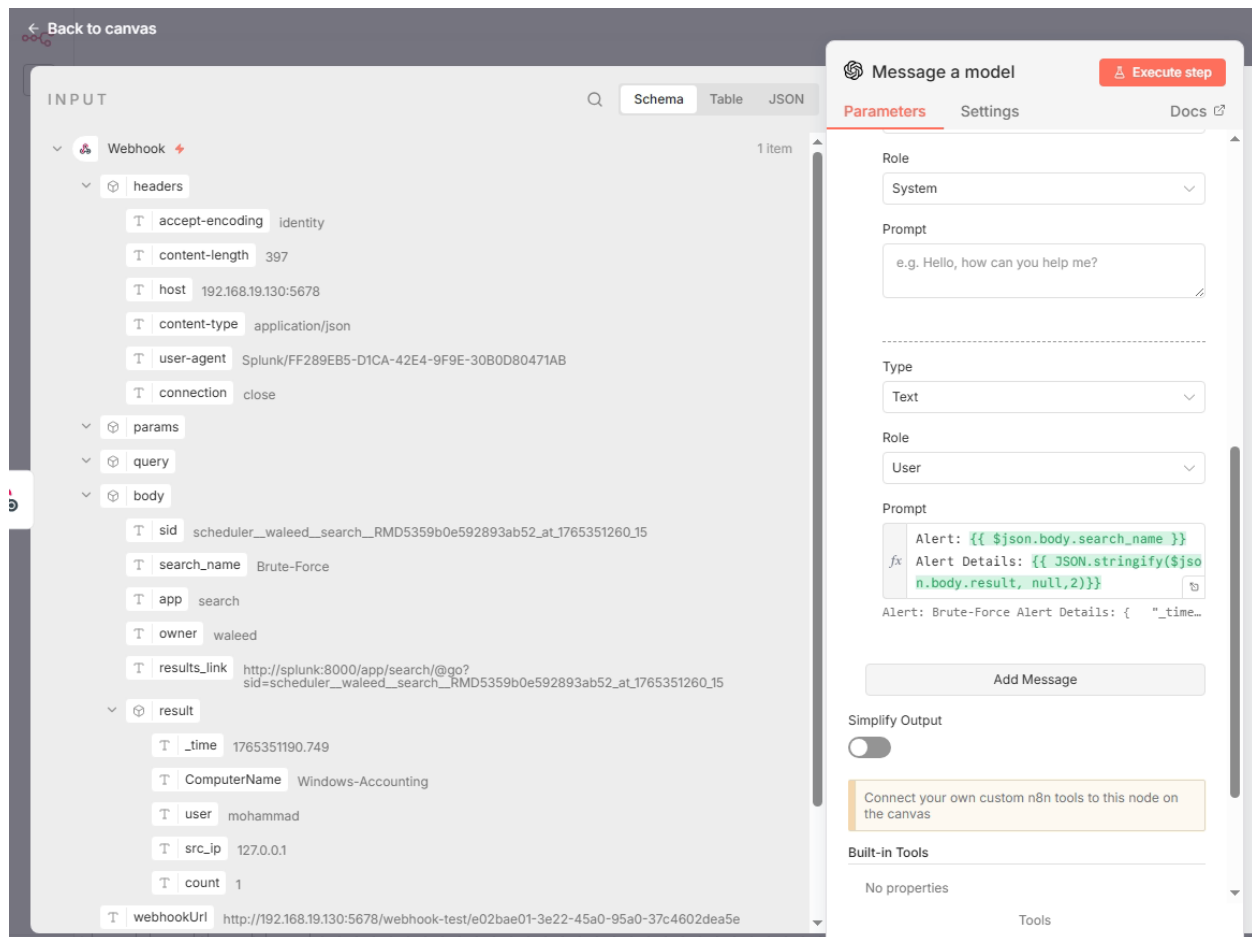


Figure 144- Training OpenAI to send Alerts Acting as an Analyst

Configuring Slack Alerts

Now to receive Alerts in slack you need to create a designated Slack App.

For creating A slack app and

Mohammad

Settings

Basic Information

Collaborators

Socket Mode

Install App

Manage Distribution

Features

App Home

Agents & AI Apps NEW

Work Object Previews NEW

Workflow Steps NEW

Org Level Apps

Incoming Webhooks

Interactivity & Shortcuts

Slash Commands

Steps from Apps LEGACY

OAuth & Permissions

Event Subscriptions

User ID Translation

App Manifest NEW

Data Feature

OAuth & Permissions

Advanced token security via token rotation

Recommended for developers building on or for security-minded organizations – opting into token rotation allows app tokens to automatically expire after they're issued within your app code. [View documentation.](#)

At least one redirect URL needs to be set below before this app can be opted into token rotation

Opt In

OAuth Tokens

These tokens were automatically generated when you installed the app to **Mohammad-Project**. You can use these to authenticate your app. [Learn more.](#)

Bot User OAuth Token

xoxb-10085661043702-10081468322371-T3WnTLPcjo2aOvw7RWdRI9qM

Copy

Access Level: Workspace

Reinstall to Mohammad-Project

Schema

Table

JSON

1 item

id

resp_0ad7578f5065875200694c583dca08190a7d7e53b893f88

object

response

created_at

1766611005

status

completed

background

false

billing

T payer

developer

completed_at

1766611011

error

[null]

incomplete_details

[null]

instructions

[null]

max_output_tokens

[null]

max_tool_calls

[null]

model

gpt-4.1-mini-2025-04-14

Send a message

Execute step

Parameters

Settings

Docs

Credential to connect with

Slack account

Stack account

Stack API

+ Create new credential

Send

Send Message To

Channel

From list

alerts

Message Type

Simple Text Message

Message Text

```

{{ $json.output[0].content[0].text }}

```

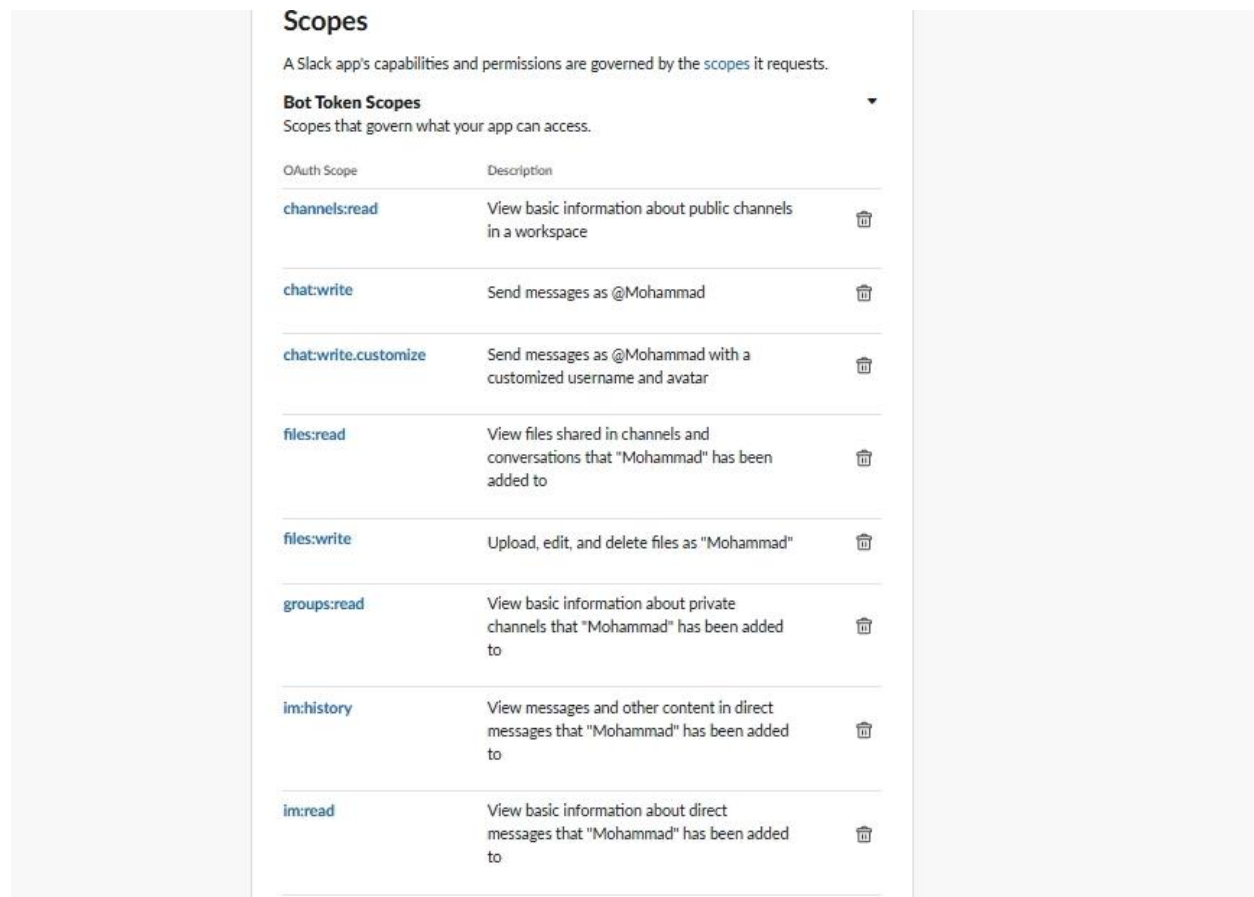
Summary A brute-force alert has been triggered

OUTPUT

This node will output an empty item if nothing would normally be returned

1 item

ok	channel	message
true	C0A2HM7SHHC	user : U0A2GJ59MCM type : message ts : 1766611012.438319 bot_id : B0A2P1PL3CL app_id : A0A2KV5KREW text : - Summary \n A brute-force alert has been triggered on the host "Windows-Accounting" involving the user "mohammad" with a source IP of 127.0.0.1. The alert count is 1, indicating a single detected brute-force attempt.\n - IOC Enrichment \n 1. User: "mohammad" — verify if this is a valid user and check for any recent suspicious activities associated with this account. \n 2. Source IP: 127.0.0.1 — This IP is the localhost. A brute-force attempt originating from localhost may indicate either a misconfigured monitoring tool, a script running on the machine, or potentially a malicious insider or malware operating locally. Investigate local processes and services. \n 3. Host: "Windows-



Configuring Splunk MCP with Claude AI:

We need to install Claude Desktop and Splunk MCP server.

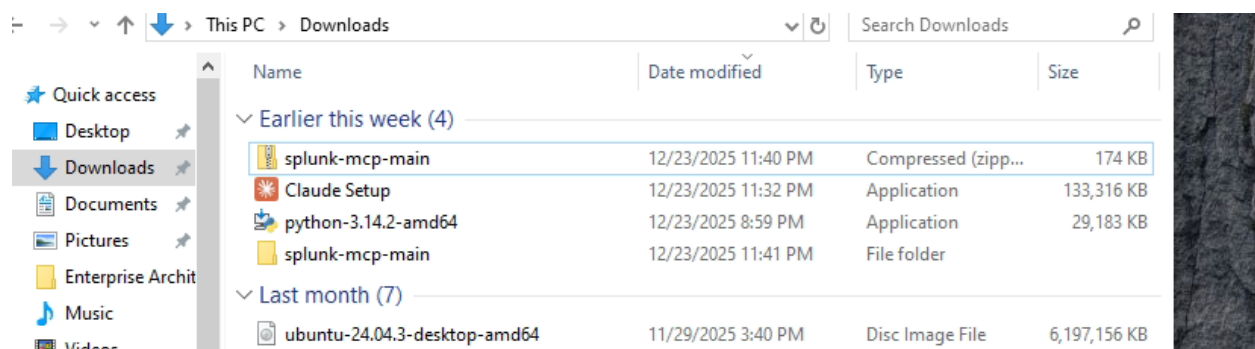


Figure 145- Downloading Claude Desktop

After that to configure MCP server first we need to install Splunk's MCP Server.

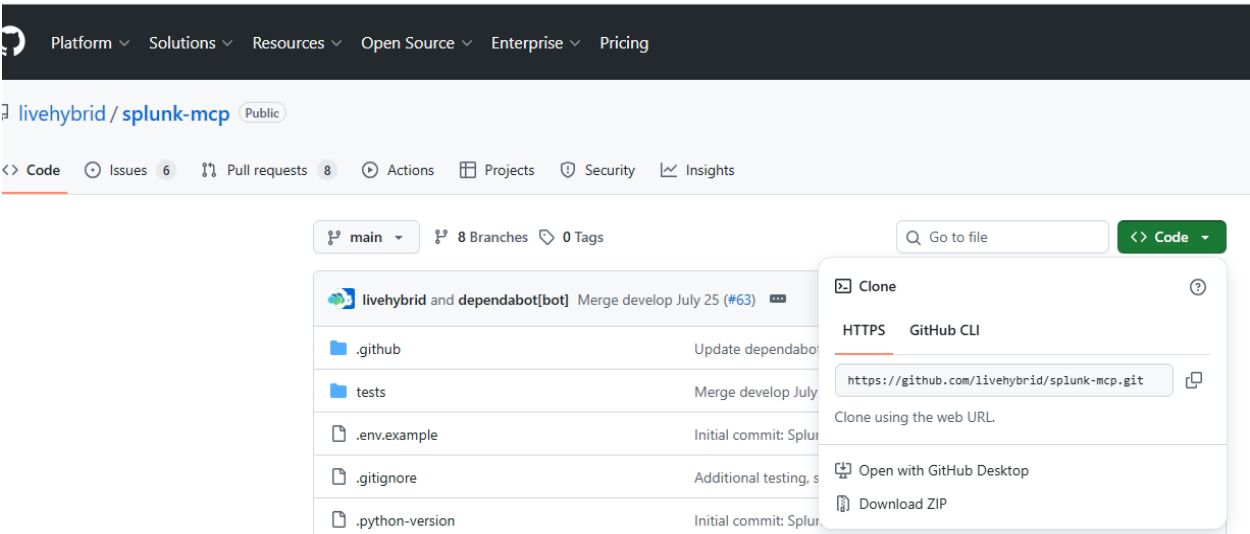


Figure 146- Installing Splunk MCP

python-3.14.2-amd64	12/23/2025 8:59 PM	Application	29,183
splunk-mcp-main	12/23/2025 11:41 PM	File folder	

After that you have installed MCP server you need to configure it with Claude Desktop. First, we will install Claude desktop. After you have installed you need to go on developer option to start configuring the desktop_configure file to add MCP server.

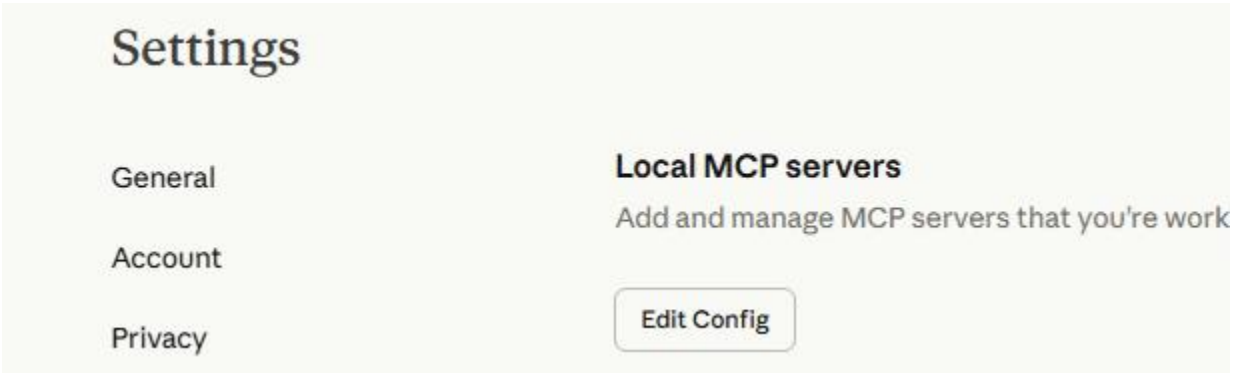


Figure 147- Configuring Claude Desktop

webstorage	12/9/2025 5:21 PM	File Folder	
ant-did	12/9/2025 9:21 PM	File	1 KB
claude_desktop_config	12/15/2025 5:57 AM	JSON File	1 KB
config	12/23/2025 9:03 PM	JSON File	1 KB
DIPS	12/14/2025 2:05 PM	File	36 KB

Now before configuring this File, we need to create a new user Account in our SIEM Splunk and give permission. After adding new user, you need to remember credentials of new user.

🔍
☆
●
⋮

⚠️
Administrator ▾
4 Messages ▾
Settings ▾
Activity ▾
Help ▾
Find 🔍

New User

	Default app ▾	Default app inherited from ▾	Roles ▾	Last login ▾	Status ⓘ ▾	⚙️
launcher	system		admin user	12/24/2025, 12:51:26 AM	● Active	⋮
launcher	system		admin	12/24/2025, 12:38:58 AM	● Active	⋮

splunk>enterprise Apps ▾

Users / Edit user

Name mohammad

Full name

Optional

Email address

Optional

Set password

New password

Confirm password

Confirm new password

Password requirements ⓘ
 ⓘ Must contain at least 8 character(s)

Time zone ⓘ

Default app ⓘ

Assign roles ⓘ

Available item(s)
0/7 selected

☐ can_delete
☐ data_management_admin
☐ data_management_agent
☐ data_management_agent_spl2
☐ power
☐ splunk-system-role
☐ splunk_system_upgrader

>

<

Selected item(s)
0/2 selected

☐ admin
☐ user

After creating new user you need to go back to Claude desktop file and configure its file located in %APPDATA%\Claude\claude_desktop_config.json.

2025 11:33 PM File f
 2025 12:08 AM File f
 2025 11:33 PM File f
 2025 11:33 PM File f
 2025 11:33 PM File f
 2025 11:33 PM File f
 2025 11:58 PM File f
 2025 12:08 AM File f
 2025 11:58 PM File f
 2025 11:58 PM File f
 2025 11:33 PM File f
 2025 11:33 PM File f
 2025 11:33 PM File f
 2025 11:56 PM JSON
 2025 12:01 AM JSON
 2025 11:33 PM File
 2025 11:56 PM File
 2025 11:58 PM JSON
 2025 11:33 PM JSON
 2025 11:33 PM File
 2025 11:58 PM File
 2025 12:01 AM File
 2025 11:35 PM File
 2025 11:35 PM File
 2025 11:57 PM JSON

*claude_desktop_config - Notepad
 File Edit Format View Help

```

{
  "mcpServers": {
    "splunk": {
      "command": "uv",
      "env": {
        "SPLUNK_HOST": "192.168.19.128",
        "SPLUNK_PORT": "8089",
        "SPLUNK_USERNAME": "Mohammad",
        "SPLUNK_PASSWORD": "mohammad123",
        "SPLUNK_SCHEME": "https",
        "VERIFY_SSL": "false"
      },
      "args": [
        "--directory",
        "C:\\Users\\Administrator\\Downloads\\splunk-mcp-main\\splunk-mcp-main",
        "run",
        "python",
        "splunk_mcp.py",
        "stdio"
      ]
    }
  }
}

```

You need to add an MCP server's .json code.

At Splunk-Host you need to give your Splunk's instance IP, At port at which port Splunk's operating. For username and password, you need to give the User and password You created for MCP client and for directory you need to give the address of splunk server you just downloaded.

To install MCP server we need to install UV because the Splunk MCP server operates as an external Python-based service that communicates with Claude Desktop through the Model Context Protocol (MCP). Claude Desktop itself is an Electron application built on JavaScript and does not execute Python code directly. Instead, it launches the MCP server using the uv tool, which provides a lightweight and efficient way to create isolated Python environments, install dependencies, and run Python scripts without manual virtual environment management. This eliminates the need for heavier tools such as Poetry.

```
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Users\Administrator> uv
An extremely fast Python package manager.

Usage: uv.exe [OPTIONS] <COMMAND>

Commands:
  auth      Manage authentication
  run        Run a command or script
  init       Create a new project
  add        Add dependencies to the project
  remove     Remove dependencies from the project
  version    Read or update the project's version
  sync       Update the project's environment
  lock       Update the project's lockfile
  export     Export the project's lockfile to an alternate format
  tree       Display the project's dependency tree
  format     Format Python code in the project
  tool       Run and install commands provided by Python packages
  python     Manage Python versions and installations
  pip        Manage Python packages with a pip-compatible interface
  venv       Create a virtual environment
  build      Build Python packages into source distributions and wheels
  publish    Upload distributions to an index
  cache      Manage uv's cache
  self       Manage the uv executable
  help       Display documentation for a command

Cache options:
  -n, --no-cache          Avoid reading from or writing to the cache, instead using a temporary directory for the
                           duration of the operation [env: UV_NO_CACHE=]
  --cache-dir <CACHE_DIR> Path to the cache directory [env: UV_CACHE_DIR=]

Python options:
  --managed-python      Require use of uv-managed Python versions [env: UV_MANAGED_PYTHON=]
  --no-managed-python   Disable use of uv-managed Python versions [env: UV_NO_MANAGED_PYTHON=]
  --no-python-downloads Disable automatic downloads of Python. [env: "UV_PYTHON_DOWNLOADS=never"]

Global options:
  -q, --quiet...          Use quiet output
  -v, --verbose...        Use verbose output
  --color <COLOR_CHOICE>
```

We needed to Python install and correctly configure Python on the system before configuring the MCP server, as Claude relies on it to start the server process. A modern Python version (3.10 or higher) is recommended to ensure compatibility with contemporary libraries and asynchronous features used by MCP servers.



We need to check use admin privileges and add python to path before installing.



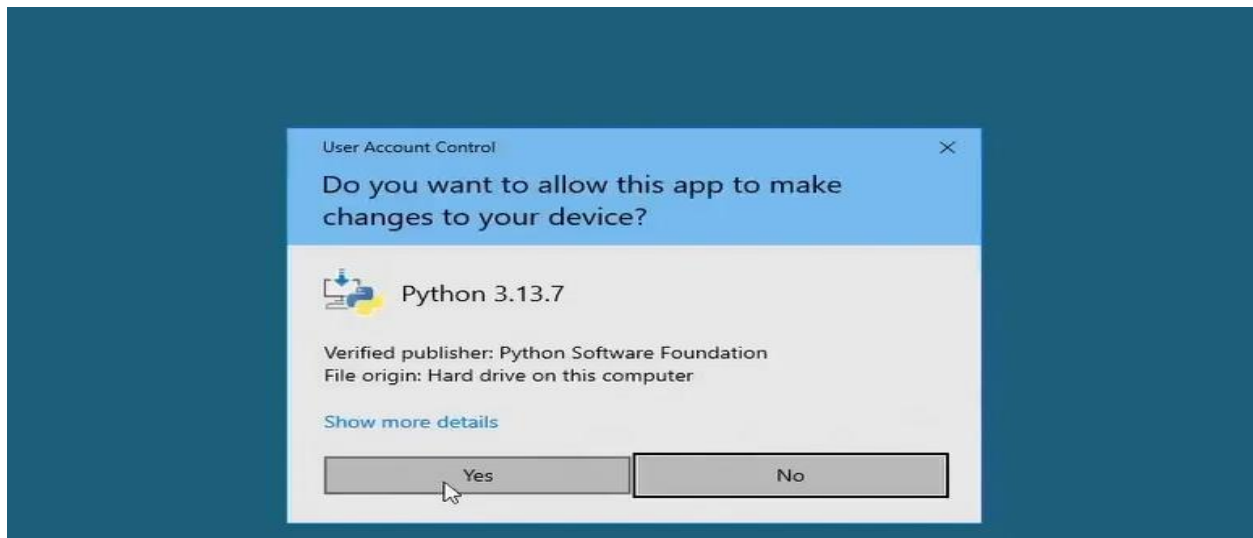
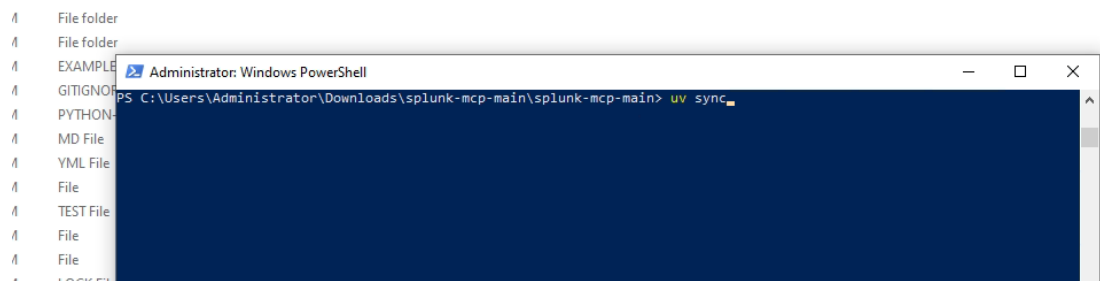


Figure 148 - Install python

After that we Synced the MCP server with UV.



After configuring this we updated the .conf file with Updated UV version.

```
5 11:33 PM File f *claude_desktop_config - Notepad
5 12:08 AM File f
5 11:33 PM File f
5 11:33 PM File f
5 11:33 PM File f
5 11:33 PM File f
5 11:58 PM File f
5 12:08 AM File f
5 11:58 PM File f
5 11:58 PM File f
5 11:33 PM File f
5 11:33 PM File f
5 11:33 PM File f
5 11:56 PM JSON
5 12:01 AM JSON
5 11:33 PM File
5 11:56 PM File
5 11:58 PM JSON
5 11:33 PM JSON
5 11:33 PM File
5 11:58 PM File
5 12:01 AM File
5 11:35 PM File
5 11:35 PM File

{
  "mcpServers": {
    "splunk": {
      "command": "uv",
      "env": {
        "SPLUNK_HOST": "192.168.19.128",
        "SPLUNK_PORT": "8089",
        "SPLUNK_USERNAME": "Mohammad",
        "SPLUNK_PASSWORD": "mohammad123",
        "SPLUNK_SCHEME": "https",
        "VERIFY_SSL": "false"
      },
      "args": [
        "--directory",
        "C:\\Users\\Administrator\\Downloads\\splunk-mcp-main\\splunk-mcp-main",
        "run",
        "python",
        "splunk_mcp.py",
        "stdio"
      ]
    }
  }
}
```

```
{
  "mcpServers": {
    "splunk": {
      "command": "C:\\Users\\Administrator\\local\\bin\\uv.exe",
      "env": {
        "SPLUNK_HOST": "192.168.19.128",
        "SPLUNK_PORT": "8089",
        "SPLUNK_USERNAME": "admin",
        "SPLUNK_PASSWORD": "your_password_here",
        "SPLUNK_SCHEME": "https",
        "VERIFY_SSL": false
      },
    },
  },
}
```

After that, once your MCP server is running successfully you can start querying commands for alerts assistance.

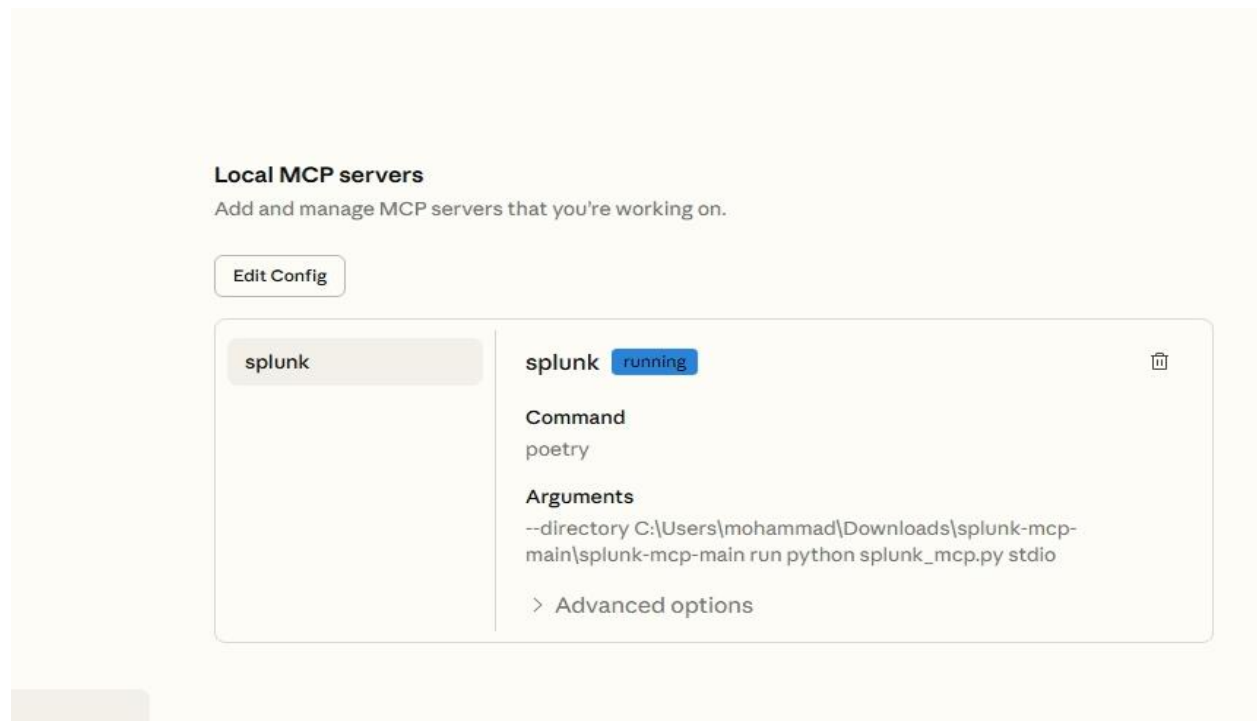


Figure 149 - MCP server running